

42. I²C-bus Interface (RIICa)

This MCU has a three-channel I²C-bus interface (RIIC0, RIIC1, RIIC2).

The RIIC module conforms with the NXP I²C-bus (Inter-IC bus) interface and provides a subset of its functions.

In this section, “PCLK” is used to refer to PCLKB.

42.1 Overview

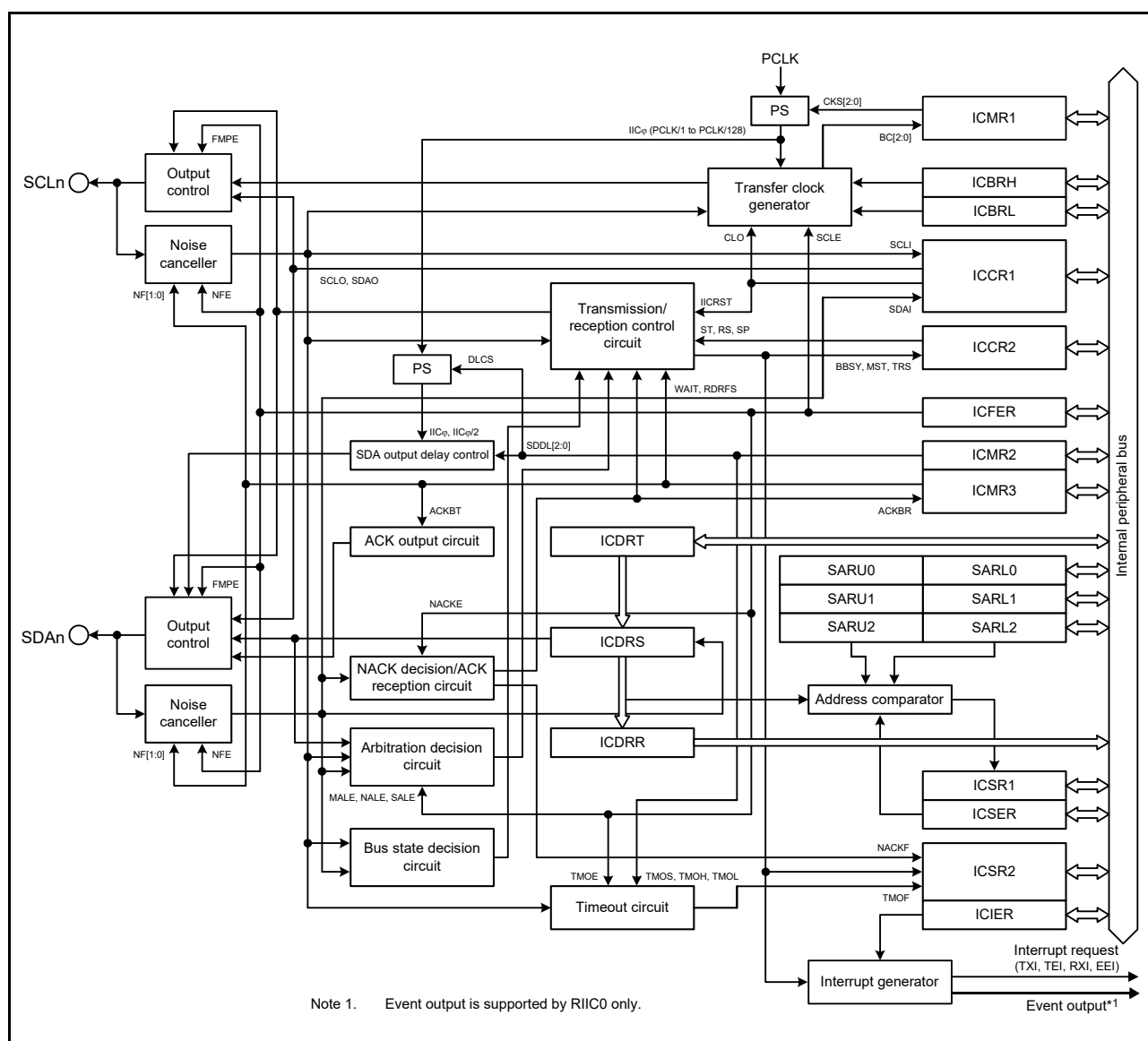
Table 42.1 lists the specifications of the RIIC, Figure 42.1 shows a block diagram of the RIIC. Table 42.2 lists the I/O pins of the RIIC.

Table 42.1 RIIC Specifications (1/2)

Item	Description
Communications format	- I²C-bus format or SMBus format - Master mode or slave mode selectable - Automatic securing of the various setup times, hold times, and bus-free times for the transfer rate
Transfer rate	Fast-mode Plus is supported (up to 1 Mbps)
Serial clock (SCL)	For master operation, the duty cycle of the SCL is selectable in the range from 4 to 96%.
Generating and detecting conditions	Start, restart, and stop conditions are automatically generated. Start conditions (including restart conditions) and stop conditions are detectable.
Slave address	- Up to three different slave addresses can be set. 7-bit and 10-bit address formats are supported (along with the use of both at once). - General call addresses, device ID addresses, and SMBus host addresses are detectable.
Acknowledgment	- For transmission, the acknowledgment bit is automatically loaded. Transfer of the next data for transmission can be automatically suspended on detection of a not-acknowledge signal. - For reception, the acknowledgment bit is automatically transmitted. If a wait between the eighth and ninth clock pulses has been selected, software control of the acknowledgment in response to the received data is possible.
Wait function	- During reception, cycles of waiting by holding the SCL line low can be inserted at the following two types of timing: - Waiting between the eighth and ninth clock pulses - Waiting between the ninth clock pulse and the first clock pulse of the next byte
SDA output delay function	Changes in the timing of the output of data bits for transmission, and of the acknowledgment bit, can be delayed relative to the falling edge of SCL.
Arbitration	- For multi-master operation - Clock synchronization for the SCL line in cases of conflict with the SCL signal from another master is possible. - When generating the start condition would create conflict on the bus, loss in arbitration is detected by testing for non-matching between the internal data level and the actual level on the SDA line. - During master operation, loss in arbitration is detected by testing for non-matching between the actual level on the SDA line and the internal data level. - Loss in arbitration due to detection of the start condition while the bus is busy is detectable (to prevent the generating of double start conditions). - Loss in arbitration in transfer of a not-acknowledge signal due to the internal signal (NACK) and the actual level on the SDA line not matching is detectable. - Loss in arbitration due to non-matching of internal data level and the actual level on the SDA line is detectable in slave transmission.
Timeout function	The internal timeout function is capable of detecting long-interval stop of the SCL.
Noise cancellation	The interface incorporates digital noise filters for both the SCL and SDA signals, and the width for noise cancellation by the filters is adjustable by software.
Interrupt sources	Four sources: - Communication error/communication event - Detection of arbitration-lost, NACK, timeout, a start condition including a restart condition, or a stop condition - Receive data full (including matching with a slave address) - Transmit data empty (including matching with a slave address) - Transmission end

Table 42.1 RIIC Specifications (2/2)

Item	Description
Low power consumption function	Module stop state can be set.
RIIC operating modes	- Four - Master transmit mode, master receive mode, slave transmit mode, and slave receive mode
Event link function (output)	Four sources (RIIC0): - Communication error/communication event - Detection of arbitration-lost, NACK, timeout, a start condition including a restart condition, or a stop condition - Receive data full (including matching with a slave address) - Transmit data empty (including matching with a slave address) - Transmission end

**Figure 42.1 RIIC Block Diagram (n = 0 to 2)**

The logic levels of the input signals for RIIC are CMOS when the I²C-bus is selected (ICMR3.SMBS bit is 0), or TTL when the SMBus is selected (ICMR3.SMBS bit is 1).

Table 42.2 RIIC Pin Configuration

Channel	Pin Name	I/O	Function
RIIC0	SCL0	I/O	RIIC0 serial clock I/O pin
	SDA0	I/O	RIIC0 serial data I/O pin
RIIC1	SCL1	I/O	RIIC1 serial clock I/O pin
	SDA1	I/O	RIIC1 serial data I/O pin
RIIC2	SCL2	I/O	RIIC2 serial clock I/O pin
	SDA2	I/O	RIIC2 serial data I/O pin

42.2 Register Descriptions

42.2.1 I²C-bus Control Register 1 (ICCR1)

Address(es): RIIC0.ICCR1 0008 8300h, RIIC1.ICCR1 0008 8320h, RIIC2.ICCR1 0008 8340h

b7	b6	b5	b4	b3	b2	b1	b0
ICE	IICRST	CLO	SOWP	SCLO	SDAO	SCLI	SDAI
Value after reset:	0	0	0	1	1	1	1

Bit	Symbol	Bit Name	Description	R/W
b0	SDAI	SDA Line Monitor	0: SDA _n line is low. 1: SDA _n line is high.	R
b1	SCLI	SCL Line Monitor	0: SCL _n line is low. 1: SCL _n line is high.	R
b2	SDAO	SDA Output Control/Monitor	- Read: 0: The RIIC has driven the SDA_n pin low. 1: The RIIC has released the SDA_n pin. - Write: 0: The RIIC drives the SDA_n pin low. 1: The RIIC releases the SDA_n pin. (High level output is achieved through an external pull-up resistor.)	R/W
b3	SCLO	SCL Output Control/Monitor	- Read: 0: The RIIC has driven the SCL_n pin low. 1: The RIIC has released the SCL_n pin. - Write: 0: The RIIC drives the SCL_n pin low. 1: The RIIC releases the SCL_n pin. (High level output is achieved through an external pull-up resistor.)	R/W
b4	SOWP	SCLO/SDAO Write Protect	0: SCLO and SDAO bits can be written. 1: SCLO and SDAO bits are protected. (This bit is read as 1.)	R/W
b5	CLO	Additional SCL Output	0: Does not output an additional SCL (default). 1: Outputs an additional SCL. (The CLO bit is cleared automatically after one clock pulse is output.)	R/W
b6	IICRST	I ² C-bus Interface Internal Reset	0: Releases the RIIC reset or internal reset. 1: Initiates the RIIC reset or internal reset. (Clears the bit counter and the SCL _n /SDA _n output latch)	R/W
b7	ICE	I ² C-bus Interface Enable	0: Disable (SCL _n and SDA _n pins in inactive state) 1: Enable (SCL _n and SDA _n pins in active state) (Combined with the IICRST bit to select either RIIC or internal reset.)	R/W

SDAO Bit (SDA Output Control/Monitor) and SCLO Bit (SCL Output Control/Monitor)

These bits are used to directly control the SDA_n and SCL_n signals output from the RIIC.

When writing to these bits, also write 0 to the SOWP bit.

The result of setting these bits is input to the RIIC via the input buffer. When slave mode is selected, a start condition may be detected and the bus may be released depending on the bit settings.

Do not rewrite these bits during a start condition, stop condition, restart condition, or during transmission or reception.

Operation after rewriting under the above conditions is not guaranteed.

When reading these bits, the state of signals output from the RIIC can be read.

CLO Bit (Additional SCL Output)

This bit is used to output an additional SCL for debugging or error processing.

Normally, set the bit to 0. Setting the bit to 1 in a normal communication state causes a communication error.

For details on this function, refer to section 42.11.2, Additional SCL Output Function.

IICRST Bit (I²C-bus Interface Internal Reset)

This bit is used to reset the internal states of the RIIC.

Setting this bit to 1 initiates an RIIC reset or internal reset.

Whether an RIIC reset or internal reset is initiated is determined according to the combination with the ICE bit. Table 42.3 lists the resets of the RIIC.

The RIIC reset initializes all registers and internal states of the RIIC, and the internal reset initializes the bit counter (ICMR1.BC[2:0] bits), the I²C-bus shift register (ICDRS), and the I²C-bus status registers (ICSR1 and ICSR2) as well as the internal states of the RIIC. For the reset conditions for each register, refer to section 42.14, Initialization of Registers and Functions When a Reset is Applied or a Condition is Detected.

An internal reset initiated with the IICRST bit set to 1 during operation (with the ICE bit set to 1) resets the internal states of the RIIC without initializing the port settings and the control and setting registers of the RIIC when the bus or RIIC hangs up due to a communication error.

If the RIIC hangs up in a low level output state, resetting the internal states cancels the low level output state and releases the bus with the SCLn pin and SDAn pin at a high impedance.

Note: If an internal reset is initiated using the IICRST bit for a bus hang-up occurred during communication with the master device in slave mode, the states may become different between the slave device and the master device (due to the difference in the bit counter information). For this reason, do not initiate an internal reset in slave mode, but initiate restoration processing from the master device. If it is necessary to perform an internal reset in slave mode, perform it during bus free state. If an internal reset is necessary because the RIIC has hung with the SCLn line in a low level output state in slave mode, initiate an internal reset and then generate a restart condition from the master device or resume communications from the start condition after having generated a stop condition. If communication is restarted by initiating a reset solely in the slave device without generating a start condition or restart condition from the master device, synchronization will be lost because the master and slave devices operate asynchronously.

Table 42.3 RIIC Resets

IICRST	ICE	State	Specifications
1	0	RIIC reset	Resets all registers and internal states of the RIIC.
	1	Internal reset	Resets the ICMR1.BC[2:0] bits, registers ICSR1, ICSR2, and ICDRS, and the internal states of the RIIC.

ICE Bit (I²C-bus Interface Enable)

This bit selects the active or inactive state of the SCLn and SDAn pins. It can also be combined with the IICRST bit to initiate two types of resets. See Table 42.3, RIIC Resets, for the types of resets.

Set the ICE bit to 1 when using the RIIC. The SCLn and SDAn pins are placed in the active state when the ICE bit is set to 1.

Set the ICE bit to 0 when the RIIC is not to be used. The SCLn and SDAn pins are placed in the inactive state when the ICE bit is set to 0. Do not assign the SCLn or SDAn pin to the RIIC when setting up the multi-function pin controller (MPC). Note that the slave address comparison operation is carried out if the pins are assigned to the RIIC.

42.2.2 I²C-bus Control Register 2 (ICCR2)

Address(es): RIIC0.ICCR2 0008 8301h, RIIC1.ICCR2 0008 8321h, RIIC2.ICCR2 0008 8341h

	b7	b6	b5	b4	b3	b2	b1	b0
	BBSY	MST	TRS	—	SP	RS	ST	—
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b1	ST	Start Condition Generation Request	0: Does not request to generate a start condition. 1: Requests to generate a start condition.	R/W
b2	RS	Restart Condition Generation Request	0: Does not request to generate a restart condition. 1: Requests to generate a restart condition.	R/W
b3	SP	Stop Condition Generation Request	0: Does not request to generate a stop condition. 1: Requests to generate a stop condition.	R/W
b4	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b5	TRS	Transmit/Receive Mode	0: Receive mode 1: Transmit mode	R/W*1
b6	MST	Master/Slave Mode	0: Slave mode 1: Master mode	R/W*1
b7	BBSY	Bus Busy Detection Flag	0: The I ² C-bus is released (bus free state). 1: The I ² C-bus is occupied (bus busy state).	R

Note 1. When the ICMR1.MTWP bit is set to 1, bits MST and TRS can be written to.

ST Bit (Start Condition Generation Request)

This bit is used to request transition to master mode and generation of a start condition.

When this bit is set to 1 to request to generate a start condition, a start condition is generated when the BBSY flag is set to 0 (bus free state).

For details on the start condition generation, refer to section 42.10, Start Condition/Restart Condition/Stop Condition Generating Function.

[Setting condition]

- When 1 is written to the ST bit

[Clearing conditions]

- When 0 is written to the ST bit
- When a start condition has been generated (a start condition is detected)
- When the ICSR2.AL (arbitration-lost) flag is set to 1
- When 1 is written to the ICCR1.IICRST bit to apply an RIIC reset or an internal reset

Note: Set the ST bit to 1 (requests to generate a start condition) when the BBSY flag is set to 0 (bus free state).
Note that arbitration may be lost due to a start condition generation error if the ST bit is set to 1 (requests to generate a start condition) when the BBSY flag is set to 1 (bus busy state).

RS Bit (Restart Condition Generation Request)

This bit is used to request that a restart condition be generated in master mode.

When this bit is set to 1 to request to generate a restart condition, a restart condition is generated when the BBSY flag is set to 1 (bus busy state) and the MST bit is set to 1 (master mode).

For details on the restart condition generation, refer to section 42.10, Start Condition/Restart Condition/Stop Condition Generating Function.

[Setting condition]

- When 1 is written to the RS bit with the ICCR2.BBSY flag set to 1

[Clearing conditions]

- When 0 is written to the RS bit
- When a restart condition has been generated (a start condition is detected)
- When the ICSR2.AL (arbitration-lost) flag is set to 1
- When 1 is written to the ICCR1.IICRST bit to apply an RIIC reset or an internal reset

Note: Do not set the RS bit to 1 while generating a stop condition.

Note: If 1 (requests to generate a restart condition) is written to the RS bit in slave mode, the restart condition is not generated but the RS bit remains set to 1. If the operating mode changes to master mode with the bit not being cleared, note that the restart condition may be generated.

SP Bit (Stop Condition Generation Request)

This bit is used to request that a stop condition be generated in master mode.

When this bit is set to 1 to request to generate a stop condition, a stop condition is generated when the BBSY flag is set to 1 (bus busy state) and the MST bit is set to 1 (master mode).

For details on the stop condition generation, refer to section 42.10, Start Condition/Restart Condition/Stop Condition Generating Function.

[Setting condition]

- When 1 is written to the SP bit with both the BBSY flag and the ICCR2.MST bit set to 1

[Clearing conditions]

- When 0 is written to the SP bit
- When a stop condition has been generated (a stop condition is detected)
- When the ICSR2.AL (arbitration-lost) flag is set to 1
- When a start condition and a restart condition are detected
- When 1 is written to the ICCR1.IICRST bit to apply an RIIC reset or an internal reset

Note: Writing to the SP bit is not possible while the setting of the BBSY flag is 0 (bus free state).

Note: Do not set the SP bit to 1 while a restart condition is being generated.

TRS Bit (Transmit/Receive Mode)

This bit indicates transmit or receive mode.

The RIIC is in receive mode when the TRS bit is set to 0 and is in transmit mode when the bit is set to 1. Combination of this bit and the MST bit indicates the operating mode of the RIIC.

The value of the TRS bit is automatically changed to 1 for transmission or 0 for reception in response to the generation or detection of a start condition and setting of the R/W# bit. Although writing to the TRS bit is possible when the ICMR1.MTWP bit is set to 1, writing to this bit is not necessary during normal usage.

[Setting conditions]

- When a start condition is generated normally according to the start condition generation request (when a start condition is detected with the ST bit set to 1)
- When a restart condition is generated normally according to the restart condition generation request (when a restart condition is detected with the RS bit set to 1)
- When the R/W# bit added to the slave address is set to 0 in master mode
- When the address received in slave mode matches the address enabled in the IC SER register, with the R/W# bit set to 1
- When 1 is written to the TRS bit with the ICMR1.MTWP bit set to 1

[Clearing conditions]

- When a stop condition is detected
- The ICSR2.AL (arbitration-lost) flag being set to 1
- In master mode, reception of a slave address to which an R/W# bit with the value 1 is appended
- In slave mode, a match between the received address and the address enabled in the IC SER register when the value of the received R/W# bit is 0 (including cases where the received address is the general call address)
- In slave mode, a restart condition is detected (that is, a start condition is detected while the ICCR2.BBSY flag is 1 and the ICCR2.MST bit is 0)
- When 0 is written to the TRS bit with the ICMR1.MTWP bit set to 1
- When 1 is written to the ICCR1.IICRST bit to apply an RIIC reset or an internal reset

MST Bit (Master/Slave Mode)

This bit indicates master or slave mode.

The RIIC is in slave mode when the MST bit is set to 0 and is in master mode when the bit is set to 1. Combination of this bit and the TRS bit indicates the operating mode of the RIIC.

The value of the MST bit is automatically changed to 1 for master mode or 0 for slave mode by generating of a start condition and generating or detection of a stop condition, etc. Although writing to the MST bit is possible when the ICMR1.MTWP bit is set to 1, writing to this bit is not necessary during normal usage.

[Setting conditions]

- When a start condition is generated normally according to the start condition generation request (when a start condition is detected with the ST bit set to 1)
- When 1 is written to the MST bit with the ICMR1.MTWP bit set to 1

[Clearing conditions]

- When a stop condition is detected
- When the ICSR2.AL (arbitration-lost) flag is set to 1
- When 0 is written to the MST bit with the ICMR1.MTWP bit set to 1
- When 1 is written to the ICCR1.IICRST bit to apply an RIIC reset or an internal reset

BBSY Flag (Bus Busy Detection Flag)

The BBSY flag indicates whether the I²C-bus is occupied (bus busy state) or released (bus free state).

This bit is set to 1 when the SDAn line changes from high to low under the condition of SCLn line = high, assuming that a start condition has been generated.

The RIIC recognizes the SDAn line changing from low to high while the SCLn line is high as generation of the stop condition. After that, this flag becomes 0 if the RIIC does not detect a start condition during the bus free time (the period set in the ICBRL register).

[Setting condition]

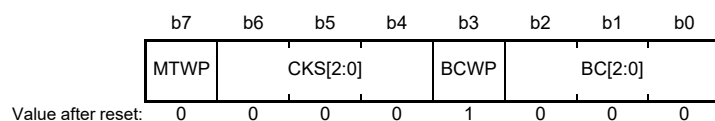
- When a start condition is detected

[Clearing conditions]

- When the bus free time (specified in the ICBRL register) start condition is not detected after detecting a stop condition
- When 1 is written to the ICCR1.IICRST bit with the ICCR1.ICE bit set to 0 (RIIC reset)

42.2.3 I²C-bus Mode Register 1 (ICMR1)

Address(es): RIIC0.ICMR1 0008 8302h, RIIC1.ICMR1 0008 8322h, RIIC2.ICMR1 0008 8342h



Bit	Symbol	Bit Name	Description	R/W
b2 to b0	BC[2:0]	Bit Counter	b2 b0 0 0 0: 9 bits 0 0 1: 2 bits 0 1 0: 3 bits 0 1 1: 4 bits 1 0 0: 5 bits 1 0 1: 6 bits 1 1 0: 7 bits 1 1 1: 8 bits	R/W*1
b3	BCWP	BC Write Protect	0: Enables a value to be written in the BC[2:0] bits. (This bit is read as 1.)	R/W*1
b6 to b4	CKS[2:0]	Internal Reference Clock Select	Select the internal reference clock (IICφ) source for the RIIC. b6 b4 0 0 0: PCLK/1 0 0 1: PCLK/2 0 1 0: PCLK/4 0 1 1: PCLK/8 1 0 0: PCLK/16 1 0 1: PCLK/32 1 1 0: PCLK/64 1 1 1: PCLK/128	R/W
b7	MTWP	MST/TRS Write Protect	0: Disables writing to the ICCR2.MST and TRS bits. 1: Enables writing to the ICCR2.MST and TRS bits.	R/W

Note 1. Rewrite the BC[2:0] bits and set the BCWP bit to 0 at the same time.

BC[2:0] Bits (Bit Counter)

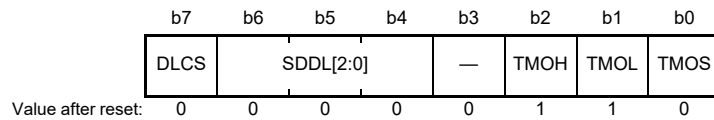
These bits function as a counter that indicates the number of bits remaining to be transferred at the detection of a rising edge on the SCLn line. Although these bits are writable and readable, it is not necessary to access these bits under normal conditions.

To write to these bits, specify the number of bits to be transferred plus one (data is transferred with an additional acknowledgment bit) between transferred bytes when the SCLn line is low.

The values of the BC[2:0] bits return to 000b at the end of a data transfer including the acknowledgment bit or when a start condition including a restart condition is detected.

42.2.4 I²C-bus Mode Register 2 (ICMR2)

Address(es): RIIC0.ICMR2 0008 8303h, RIIC1.ICMR2 0008 8323h, RIIC2.ICMR2 0008 8343h



Bit	Symbol	Bit Name	Description	R/W
b0	TMOS	Timeout Detection Time Select	0: Long mode is selected. 1: Short mode is selected.	R/W
b1	TMOL	Timeout L Count Control	0: Count-up is disabled while the SCLn line is low. 1: Count-up is enabled while the SCLn line is low.	R/W
b2	TMOH	Timeout H Count Control	0: Count-up is disabled while the SCLn line is high. 1: Count-up is enabled while the SCLn line is high.	R/W
b3	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b6 to b4	SDDL[2:0]	SDA Output Delay Counter	- When ICMR2.DLCS bit is 0 (IICφ) b6 b4 0 0 0: No output delay 0 0 1: 1 IICφ cycle 0 1 0: 2 IICφ cycles 0 1 1: 3 IICφ cycles 1 0 0: 4 IICφ cycles 1 0 1: 5 IICφ cycles 1 1 0: 6 IICφ cycles 1 1 1: 7 IICφ cycles - When ICMR2.DLCS bit is 1 (IICφ/2) b6 b4 0 0 0: No output delay 0 0 1: 1 or 2 IICφ cycles 0 1 0: 3 or 4 IICφ cycles 0 1 1: 5 or 6 IICφ cycles 1 0 0: 7 or 8 IICφ cycles 1 0 1: 9 or 10 IICφ cycles 1 1 0: 11 or 12 IICφ cycles 1 1 1: 13 or 14 IICφ cycles	R/W
b7	DLCS	SDA Output Delay Clock Source Select	0: The internal reference clock (IICφ) is selected as the clock source of the SDA output delay counter. 1: The internal reference clock divided by 2 (IICφ/2) is selected as the clock source of the SDA output delay counter.*1	R/W

Note 1. The DLCS bit setting of 1 (IICφ/2) only becomes valid when SCL pin is low. When SCL pin is high, the DLCS bit setting of 1 becomes invalid and the clock source becomes the internal reference clock (IICφ).

TMOS Bit (Timeout Detection Time Select)

This bit is used to select long mode or short mode for the timeout detection time when the timeout function is enabled (ICFER.TMOE bit is 1). When this bit is set to 0, long mode is selected. When this bit is set to 1, short mode is selected. In long mode, the timeout detection internal counter functions as a 16-bit counter. In short mode, the counter functions as a 14-bit counter. While the SCLn line is in the state that enables this counter as specified by bits TMOH and TMOL, the counter counts up in synchronization with the internal reference clock (IICφ) as a count source.

For details on the timeout function, refer to section 42.11.1, Timeout Function.

TMOL Bit (Timeout L Count Control)

This bit is used to enable or disable the internal counter of the timeout function to count up while the SCLn line is held low when the timeout function is enabled (ICFER.TMOE bit is 1).

TMOH Bit (Timeout H Count Control)

This bit is used to enable or disable the internal counter of the timeout function to count up while the SCLn line is held high when the timeout function is enabled (ICFER.TMOE bit is 1).

SDDL[2:0] Bits (SDA Output Delay Counter)

The SDA output can be delayed by the SDDL[2:0] setting. This counter works with the clock source selected by the DLCS bit. The setting of this function can be used for all types of SDA output, including the transmission of the acknowledgment bit.

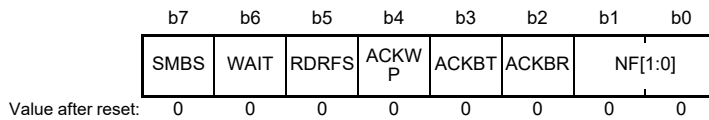
Set the SDA output delay time to meet the I²C-bus specification (within the data valid time/data valid acknowledge time*¹) or the SMBus specification (more than the data hold time (300 ns) and less than “clock low period – data setup time (250 ns)”). Note that, if a value outside the specification is set, communication with communication devices may malfunction or it may seemingly become a start condition or stop condition depending on the bus state.

For details on this function, refer to section 42.5, SDA Output Delay Function.

- Note 1. Data valid time/data valid acknowledge time
- 3,450 ns (up to 100 kbps: Standard-mode (Sm))
 - 900 ns (up to 400 kbps: Fast-mode (Fm))
 - 450 ns (up to 1 Mbps: Fast-mode Plus (Fm+))

42.2.5 I²C-bus Mode Register 3 (ICMR3)

Address(es): RIIC0.ICMR3 0008 8304h, RIIC1.ICMR3 0008 8324h, RIIC2.ICMR3 0008 8344h



Bit	Symbol	Bit Name	Description	R/W
b1, b0	NF[1:0]	Noise Filter Stage Select	b1 b0 0 0: Noise of up to one IIC ϕ cycle is filtered out (single-stage filter). 0 1: Noise of up to two IIC ϕ cycles is filtered out (2-stage filter). 1 0: Noise of up to three IIC ϕ cycles is filtered out (3-stage filter). 1 1: Noise of up to four IIC ϕ cycles is filtered out (4-stage filter).	R/W
b2	ACKBR	Received Acknowledge	0: 0 is received as the acknowledgment bit (ACK reception). 1: 1 is received as the acknowledgment bit (NACK reception).	R
b3	ACKBT	Transmit Acknowledge	0: 0 is to be sent as the acknowledgment bit (ACK transmission). 1: 1 is to be sent as the acknowledgment bit (NACK transmission).	R/W*1
b4	ACKWP	ACKBT Write Protect	0: Modification of the ACKBT bit is disabled. 1: Modification of the ACKBT bit is enabled.	R/W*1
b5	RDRFS	RDRF Flag Set Timing Select	0: The RDRF flag is set at the rising edge of the ninth SCL. (The SCLn line is not held low at the falling edge of the eighth clock pulse.) 1: The RDRF flag is set at the rising edge of the eighth SCL. (The SCLn line is held low at the falling edge of the eighth clock pulse.) Low-hold is released by writing a value to the ACKBT bit.	R/W*2
b6	WAIT	WAIT	0: No WAIT (The period between ninth clock pulse and first clock pulse is not held low.) 1: WAIT (The period between ninth clock pulse and first clock pulse is held low.) Low-hold is released by reading the ICDRR register.	R/W*2
b7	SMBS	SMBus/I ² C-bus Select	0: The I ² C-bus is selected. 1: The SMBus is selected.	R/W

Note 1. Write to the ACKBT bit only while the ACKWP bit is already 1. If it is attempted to write 1 to both the ACKWP and ACKBT bits at the same time, the ACKBT bit will not be set to 1.

Note 2. The WAIT and RDRFS bits are valid only in receive mode (invalid in transmit mode).

NF[1:0] Bits (Noise Filter Stage Select)

These bits are used to select the number of stages in the digital noise filter.

For details on the digital noise filter function, refer to section 42.6, Digital Noise Filters.

Note: Set the noise range to be filtered out by the noise filter within a range less than the SCLn line high period or low period. If the noise filter width is set to a value of [the shorter one of either SCL high width or SCL low width] – 1.5 × $t_{IIC\phi cyc}$ (cycle time of internal reference clock (IIC ϕ)) or a greater value, the serial clock is regarded as noise by the noise filter function of the RIIC, which may prevent the RIIC from operating normally.

ACKBR Bit (Received Acknowledge)

This bit is used to store the value of the acknowledgment bit received from the receiver in transmit mode.

[Setting condition]

- When 1 is received as the acknowledgment bit with the ICCR2.TRS bit set to 1

[Clearing conditions]

- When 0 is received as the acknowledgment bit with the ICCR2.TRS bit set to 1
- When 1 is written to the ICCR1.IICRST bit while the ICCR1.ICE bit is 0 (RIIC reset)

ACKBT Bit (Transmit Acknowledge)

This bit is used to set the bit to be sent at the acknowledgment timing in receive mode.

[Setting condition]

- When 1 is written to this bit with the ACKWP bit set to 1

[Clearing conditions]

- When 0 is written to this bit with the ACKWP bit set to 1
- When stop condition generation is detected (when a stop condition is detected with the ICCR2.SP bit set to 1)
- When 1 is written to the ICCR1.IICRST bit while the ICCR1.ICE bit is 0 (RIIC reset)

ACKWP Bit (ACKBT Write Protect)

This bit is used to control the modification of the ACKBT bit.

RDRFS Bit (RDRF Flag Set Timing Select)

This bit is used to select the RDRF flag set timing in receive mode and also to select whether to hold the SCLn line low at the falling edge of the eighth SCL.

When the RDRFS bit is 0, the SCLn line is not held low at the falling edge of the eighth SCL, and the RDRF flag is set to 1 at the rising edge of the ninth SCL.

When the RDRFS bit is 1, the RDRF flag is set to 1 at the rising edge of the eighth SCL and the SCLn line is held low at the falling edge of the eighth SCL. The low-hold of the SCLn line is released by writing a value to the ACKBT bit.

After data is received with this setting, the SCLn line is automatically held low before the acknowledgment bit is sent.

This enables processing to send ACK (ACKBT bit is 0) or NACK (ACKBT bit is 1) according to receive data.

WAIT Bit (WAIT)

This bit is used to control whether to hold the period between the ninth SCL and the first SCL low until the I²C-bus receive data register (ICDRR) is completely read each time single-byte data is received in receive mode.

When the WAIT bit is 0, the receive operation is continued without holding the period between the ninth and the first SCL low. When both the RDRFS and WAIT bits are 0, continuous receive operation is enabled with the double buffer.

When the WAIT bit is 1, the SCLn line is held low from the falling edge of the ninth SCL until the ICDRR register value is read each time single-byte data is received. This enables receive operation in byte units.

Note: When the value of the WAIT bit is to be read, be sure to read the ICDRR register beforehand.

SMBS Bit (SMBus/I²C-bus Select)

Setting this bit to 1 selects the SMBus and enables the IC SER.HOAE bit.

42.2.6 I²C-bus Function Enable Register (ICFER)

Address(es): RIIC0.ICFER 0008 8305h, RIIC1.ICFER 0008 8325h, RIIC2.ICFER 0008 8345h

	b7	b6	b5	b4	b3	b2	b1	b0
	FMPE	SCLE	NFE	NACKE	SALE	NALE	MALE	TMOE
Value after reset:	0	1	1	1	0	0	1	0

Bit	Symbol	Bit Name	Description	R/W
b0	TMOE	Timeout Function Enable	0: The timeout function is disabled. 1: The timeout function is enabled.	R/W
b1	MALE	Master Arbitration-Lost Detection Enable	0: Master arbitration-lost detection is disabled. (Disables the arbitration-lost detection function and does not clear the ICCR2.MST and TRS bits automatically when arbitration is lost.) 1: Master arbitration-lost detection is enabled. (Enables the arbitration-lost detection function and clears the ICCR2.MST and TRS bits automatically when arbitration is lost.)	R/W
b2	NALE	NACK Transmission Arbitration-Lost Detection Enable	0: NACK transmission arbitration-lost detection is disabled. 1: NACK transmission arbitration-lost detection is enabled.	R/W
b3	SALE	Slave Arbitration-Lost Detection Enable	0: Slave arbitration-lost detection is disabled. 1: Slave arbitration-lost detection is enabled.	R/W
b4	NACKE	NACK Reception Transfer Suspension Enable	0: Transfer operation is not suspended during NACK reception (transfer suspension disabled). 1: Transfer operation is suspended during NACK reception (transfer suspension enabled).	R/W
b5	NFE	Digital Noise Filter Enable	0: Digital noise filters are not used. 1: Digital noise filters are used.	R/W
b6	SCLE	SCL Synchronization Enable	0: SCL synchronization is disabled. 1: SCL synchronization is enabled.	R/W
b7	FMPE*1	Fast-Mode Plus Enable	0: No Fm+ slope control circuit is used for the SCLn pin and SDAn pin. 1: An Fm+ slope control circuit is used for the SCLn pin and SDAn pin.	R/W

Note 1. The Fast-mode Plus enable bit (FMPE) is only supported by RIIC0. In RIIC1 and RIIC2, bit 7 is reserved.

TMOE Bit (Timeout Function Enable)

This bit is used to enable or disable the timeout function.

For details on the timeout function, refer to section 42.11.1, Timeout Function.

MALE Bit (Master Arbitration-Lost Detection Enable)

This bit is used to specify whether to use the arbitration-lost detection function in master mode. Normally, set this bit to 1.

NALE Bit (NACK Transmission Arbitration-Lost Detection Enable)

This bit is used to specify whether to cause arbitration to be lost when ACK is detected during transmission of NACK in receive mode (such as when slaves with the same address exist on the bus or when two or more masters select the same slave device simultaneously with different number of receive bytes).

SALE Bit (Slave Arbitration-Lost Detection Enable)

This bit is used to specify whether to cause arbitration to be lost when a value different from the value being transmitted is detected on the bus in slave transmit mode (such as when slaves with the same address exist on the bus or when a mismatch with the transmit data occurs due to noise).

NACK Bit (NACK Reception Transfer Suspension Enable)

This bit is used to specify whether to continue or discontinue the data transfer when NACK is received in transmit mode. Normally, set this bit to 1.

When NACK is received with the NACK bit set to 1, the next transfer operation is suspended.

When the NACK bit is 0, the next transfer operation is continued regardless of the value of the received acknowledgment bit.

For details on the NACK reception transfer suspension function, refer to **section 42.8.2, NACK Reception Transfer Suspension Function**.

SCLE Bit (SCL Synchronization Enable)

This bit is used to specify whether the SCL output is to be synchronized with the SCL input. Normally, set this bit to 1. When the SCLE bit is set to 0 (SCL synchronization is disabled), the RIIC does not synchronize the SCL output with the SCL input. In this setting, the RIIC outputs the clock with the transfer rate set in registers ICBRH and ICBRL regardless of the SCLn line state. For this reason, if the load of the I²C-bus line is much larger than the specification value or if the SCL output overlaps in multiple masters, the short-cycle SCL that does not meet the specification may be output. When the SCL synchronization is not used, it also affects the generation of a start condition, restart condition, and stop condition, and the continuous output of additional SCL.

This bit must not be set to 0 except for checking the output of the set transfer rate.

FMPE Bit (Fast-Mode Plus Enable)

This bit is used to specify whether to use a slope control circuit for Fast-mode Plus (Fm+).

When this bit is set to 1, a slope control circuit conforming to the Fast-mode Plus (Fm+) slope control specification (tof) of the I²C-bus is selected. When this bit is set to 0, a slope control circuit conforming to the Standard-mode (Sm) and Fast-mode (Fm) slope control specification (tof) of the I²C-bus is selected.

Set this bit to 1 when using the transmission rate within a range up to 1 Mbps (Fast-mode Plus (Fm+)) of the I²C-bus specification. Set this bit to 0 when using the transmission rate at other rates (up to 100 kbps (Sm), up to 400 kbps (Fm)) or for SMBus (10 to 100 kbps).

42.2.7 I²C-bus Status Enable Register (ICSER)

Address(es): RIIC0.ICSER 0008 8306h, RIIC1.ICSER 0008 8326h, RIIC2.ICSER 0008 8346h

	b7	b6	b5	b4	b3	b2	b1	b0
	HOAE	—	DIDE	—	GCAE	SAR2E	SAR1E	SAR0E
Value after reset:	0	0	0	0	1	0	0	1

Bit	Symbol	Bit Name	Description	R/W
b0	SAR0E	Slave Address Register 0 Enable	0: Slave address in registers SARL0 and SARU0 is disabled. 1: Slave address in registers SARL0 and SARU0 is enabled.	R/W
b1	SAR1E	Slave Address Register 1 Enable	0: Slave address in registers SARL1 and SARU1 is disabled. 1: Slave address in registers SARL1 and SARU1 is enabled.	R/W
b2	SAR2E	Slave Address Register 2 Enable	0: Slave address in registers SARL2 and SARU2 is disabled. 1: Slave address in registers SARL2 and SARU2 is enabled.	R/W
b3	GCAE	General Call Address Enable	0: General call address detection is disabled. 1: General call address detection is enabled.	R/W
b4	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b5	DIDE	Device-ID Address Detection Enable	0: Device-ID address detection is disabled. 1: Device-ID address detection is enabled.	R/W
b6	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b7	HOAE	Host Address Enable	0: Host address detection is disabled. 1: Host address detection is enabled.	R/W

SARyE Bit (Slave Address Register y Enable) (y = 0 to 2)

This bit is used to enable or disable the slave address set in registers SARLy and SARUy.

When this bit is set to 1, the slave address set in registers SARLy and SARUy is enabled and is compared with the received slave address.

When this bit is set to 0, the slave address set in registers SARLy and SARUy is disabled and is ignored even if it matches the received slave address.

GCAE Bit (General Call Address Enable)

This bit is used to specify whether to ignore the general call address (0000 000b + 0 (write): All 0) when it is received.

When this bit is set to 1, if the received slave address matches the general call address, the RIIC recognizes the received slave address as the general call address independently of the slave addresses set in registers SARLy and SARUy (y = 0 to 2) and performs data receive operation.

When this bit is set to 0, the received slave address is ignored even if it matches the general call address.

DIDE Bit (Device-ID Address Detection Enable)

This bit is used to specify whether to recognize and execute the device-ID address when a device ID (1111 100b) is received in the first byte after a start condition or restart condition is detected.

When this bit is set to 1, if the received first byte matches the device ID, the RIIC recognizes that the device-ID address has been received. When the following R/W# bit is 0 (write), the RIIC recognizes the second and the following bytes as slave addresses and continues the receive operation.

When this bit is set to 0, the RIIC ignores the received first byte even if it matches the device ID address and recognizes the first byte as a normal slave address.

For details on the device-ID address detection, refer to section 42.7.3, Device-ID Address Detection.

HOAE Bit (Host Address Enable)

This bit is used to specify whether to ignore received host address (0001 000b) when the ICMR3.SMBS bit is 1.

When this bit is set to 1 while the ICMR3.SMBS bit is 1, if the received slave address matches the host address, the RIIC recognizes the received slave address as the host address independently of the slave addresses set in registers SARLy and SARUy (y = 0 to 2) and performs the receive operation.

When the ICMR3.SMBS bit or the HOAE bit is set to 0, the received slave address is ignored even if it matches the host address.

42.2.8 I²C-bus Interrupt Enable Register (ICIER)

Address(es): RIIC0.ICIER 0008 8307h, RIIC1.ICIER 0008 8327h, RIIC2.ICIER 0008 8347h

b7	b6	b5	b4	b3	b2	b1	b0
TIE	TEIE	RIE	NAKIE	SPIE	STIE	ALIE	TMOIE
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	TMOIE	Timeout Interrupt Request Enable	0: Timeout interrupt (TMOI) request is disabled. 1: Timeout interrupt (TMOI) request is enabled.	R/W
b1	ALIE	Arbitration-Lost Interrupt Request Enable	0: Arbitration-lost interrupt (ALI) request is disabled. 1: Arbitration-lost interrupt (ALI) request is enabled.	R/W
b2	STIE	Start Condition Detection Interrupt Request Enable	0: Start condition detection interrupt (STI) request is disabled. 1: Start condition detection interrupt (STI) request is enabled.	R/W
b3	SPIE	Stop Condition Detection Interrupt Request Enable	0: Stop condition detection interrupt (SPI) request is disabled. 1: Stop condition detection interrupt (SPI) request is enabled.	R/W
b4	NAKIE	NACK Reception Interrupt Request Enable	0: NACK reception interrupt (NAKI) request is disabled. 1: NACK reception interrupt (NAKI) request is enabled.	R/W
b5	RIE	Receive Data Full Interrupt Request Enable	0: Receive data full interrupt (RXI) request is disabled. 1: Receive data full interrupt (RXI) request is enabled.	R/W
b6	TEIE	Transmission End Interrupt Request Enable	0: Transmission end interrupt (TEI) request is disabled. 1: Transmission end interrupt (TEI) request is enabled.	R/W
b7	TIE	Transmit Data Empty Interrupt Request Enable	0: Transmit data empty interrupt (TXI) request is disabled. 1: Transmit data empty interrupt (TXI) request is enabled.	R/W

TMOIE Bit (Timeout Interrupt Request Enable)

This bit is used to enable or disable timeout interrupt (TMOI) requests when the ICSR2.TMOF flag is set to 1. A TMOI interrupt request is canceled by setting the TMOF flag or the TMOIE bit to 0.

ALIE Bit (Arbitration-Lost Interrupt Request Enable)

This bit is used to enable or disable arbitration-lost interrupt (ALI) requests when the ICSR2.AL flag is set to 1. An ALI interrupt request is canceled by setting the AL flag or the ALIE bit to 0.

STIE Bit (Start Condition Detection Interrupt Request Enable)

This bit is used to enable or disable start condition detection interrupt (STI) requests when the ICSR2.START flag is set to 1. An STI interrupt request is canceled by setting the START flag or the STIE bit to 0.

SPIE Bit (Stop Condition Detection Interrupt Request Enable)

This bit is used to enable or disable stop condition detection interrupt (SPI) requests when the ICSR2.STOP flag is set to 1. An SPI interrupt request is canceled by setting the STOP flag or the SPIE bit to 0.

NAKIE Bit (NACK Reception Interrupt Request Enable)

This bit is used to enable or disable NACK reception interrupt (NAKI) requests when the ICSR2.NACKF flag is set to 1. An NAKI interrupt request is canceled by setting the NACKF flag or the NAKIE bit to 0.

RIE Bit (Receive Data Full Interrupt Request Enable)

This bit is used to enable or disable receive data full interrupt (RXI) requests when the ICSR2.RDRF flag is set to 1.

TEIE Bit (Transmission End Interrupt Request Enable)

This bit is used to enable or disable transmission end interrupt (TEI) requests when the ICSR2.TEND flag is set to 1. An TEI interrupt request is canceled by setting the TEND flag or the TEIE bit to 0.

TIE Bit (Transmit Data Empty Interrupt Request Enable)

This bit is used to enable or disable transmit data empty interrupt (TXI) requests when the ICSR2.TDRE flag is set to 1.

42.2.9 I²C-bus Status Register 1 (ICSR1)

Address(es): RIIC0.ICSR1 0008 8308h, RIIC1.ICSR1 0008 8328h, RIIC2.ICSR1 0008 8348h

b7	b6	b5	b4	b3	b2	b1	b0
HOA	—	DID	—	GCA	AAS2	AAS1	AAS0
Value after reset: 0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	AAS0	Slave Address 0 Detection Flag	0: Slave address 0 is not detected. 1: Slave address 0 is detected.	R/(W) *1
b1	AAS1	Slave Address 1 Detection Flag	0: Slave address 1 is not detected. 1: Slave address 1 is detected.	R/(W) *1
b2	AAS2	Slave Address 2 Detection Flag	0: Slave address 2 is not detected. 1: Slave address 2 is detected.	R/(W) *1
b3	GCA	General Call Address Detection Flag	0: General call address is not detected. 1: General call address is detected.	R/(W) *1
b4	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b5	DID	Device-ID Address Detection Flag	0: Device-ID command is not detected. 1: Device-ID command is detected. • This bit is set to 1 when the first byte received immediately after a start condition is detected matches a value of (device ID (1111 100b) + 0 (write)).	R/(W) *1
b6	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b7	HOA	Host Address Detection Flag	0: Host address is not detected. 1: Host address is detected. • This bit is set to 1 when the received slave address matches the host address (0001 000b).	R/(W) *1

Note 1. Only 0 can be written to clear the flag.

AASy Flag (Slave Address y Detection Flag) (y = 0 to 2)

[Setting conditions]

For 7-bit address format: SARUy.FS bit = 0

- When the received slave address matches the SARLy.SVA[6:0] bits value with the ICSEr.SARyE bit set to 1 (slave address y detection enabled)

This flag is set to 1 at the rising edge of the ninth SCL in the first byte.

For 10-bit address format: SARUy.FS bit = 1

- When the received slave address matches a value of (11110b + SARUy.SVA[1:0] bits) and the following address matches the SARLy value with the ICSEr.SARyE bit set to 1 (slave address y detection enabled)

This flag is set to 1 at the rising edge of the ninth SCL in the second byte.

[Clearing conditions]

- When 0 is written to the AASy flag after reading the AASy flag to be 1
- When a stop condition is detected
- When 1 is written to the ICCR1.IICRST bit to apply an RIIC reset or an internal reset

For 7-bit address format: SARUy.FS bit = 0

- When the received slave address does not match the SARLy.SVA[6:0] bits value with the ICSEr.SARyE bit set to 1 (slave address y detection enabled)

This flag is set to 0 at the rising edge of the ninth SCL in the first byte.

For 10-bit address format: SARUy.FS bit = 1

- When the received slave address does not match a value of (11110b + SARUy.SVA[1:0] bits) with the ICSER.SARyE bit set to 1 (slave address y detection enabled)
This flag is set to 0 at the rising edge of the ninth SCL in the first byte.
- When the received slave address matches a value of (11110b + SARUy.SVA[1:0] bits) and the following address does not match the SARLy value with the ICSER.SARyE bit set to 1 (slave address y detection enabled)
This flag is set to 0 at the rising edge of the ninth SCL in the second byte.

GCA Flag (General Call Address Detection Flag)

[Setting condition]

- When the received slave address matches the general call address (0000 000b + 0 (write)) with the ICSER.GCAE bit set to 1 (general call address detection is enabled)
This flag is set to 1 at the rising edge of the ninth SCL in the first byte.

[Clearing conditions]

- When 0 is written to the GCA flag after reading GCA flag to be 1
- When a stop condition is detected
- When the received slave address does not match the general call address (0000 000b + 0 (write)) with the ICSER.GCAE bit set to 1 (general call address detection is enabled)
This flag is set to 0 at the rising edge of the ninth SCL in the first byte.
- When 1 is written to the ICCR1.IICRST bit to apply an RIIC reset or an internal reset

DID Flag (Device-ID Address Detection Flag)

[Setting condition]

- When the first byte received immediately after a start condition or restart condition is detected matches a value of (device ID (1111 100b) + 0 (write)) with the ICSER.DIDE bit set to 1 (device-ID address detection is enabled)
This flag is set to 1 at the rising edge of the ninth SCL in the first byte.

[Clearing conditions]

- When 0 is written to the DID flag after reading DID flag to be 1
- When a stop condition is detected
- When the first byte received immediately after a start condition or restart condition is detected does not match a value of (device ID (1111 100b)) with the ICSER.DIDE bit set to 1 (device-ID address detection is enabled)
This flag is set to 0 at the rising edge of the ninth SCL in the first byte.
- When the first byte received immediately after a start condition or restart condition is detected matches a value of (device ID (1111 100b) + 0 (write)) and the second byte does not match any of slave addresses 0 to 2 with the ICSER.DIDE bit set to 1 (device-ID address detection is enabled)
This flag is set to 0 at the rising edge of the ninth SCL in the second byte.
- When 1 is written to the ICCR1.IICRST bit to apply an RIIC reset or an internal reset

HOA Flag (Host Address Detection Flag)

[Setting condition]

- When the received slave address matches the host address (0001 000b) with the ICSER.HOAE bit set to 1 (host address detection is enabled)
This flag is set to 1 at the rising edge of the ninth SCL in the first byte.

[Clearing conditions]

- When 0 is written to the HOA flag after reading HOA flag to be 1
- When a stop condition is detected
- When the received slave address does not match the host address (0001 000b) with the ICSER.HOAE bit set to 1

(host address detection is enabled)

This flag is set to 0 at the rising edge of the ninth SCL in the first byte.

- When 1 is written to the ICCR1.IICRST bit to apply an RIIC reset or an internal reset

42.2.10 I²C-bus Status Register 2 (ICSR2)

Address(es): RIIC0.ICSR2 0008 8309h, RIIC1.ICSR2 0008 8329h, RIIC2.ICSR2 0008 8349h

	b7	b6	b5	b4	b3	b2	b1	b0
	TDRE	TEND	RDRF	NACKF	STOP	START	AL	TMOF
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	TMOF	Timeout Detection Flag	0: Timeout is not detected. 1: Timeout is detected.	R/(W) *1
b1	AL	Arbitration-Lost Flag	0: Arbitration is not lost. 1: Arbitration is lost.	R/(W) *1
b2	START	Start Condition Detection Flag	0: Start condition is not detected. 1: Start condition is detected.	R/(W) *1
b3	STOP	Stop Condition Detection Flag	0: Stop condition is not detected. 1: Stop condition is detected.	R/(W) *1
b4	NACKF	NACK Detection Flag	0: NACK is not detected. 1: NACK is detected.	R/(W) *1
b5	RDRF	Receive Data Full Flag	0: The ICDRR register contains no receive data. 1: The ICDRR register contains receive data.	R/(W) *1
b6	TEND	Transmission End Flag	0: Data is being transmitted. 1: Data has been transmitted.	R/(W) *1
b7	TDRE	Transmit Data Empty Flag	0: The ICDRT register contains transmit data. 1: The ICDRT register contains no transmit data.	R

Note 1. Only 0 can be written to clear the flag.

TMOF Flag (Timeout Detection Flag)

This flag is set to 1 when the RIIC recognizes timeout after the SCLn line state remains unchanged for a certain period.

[Setting condition]

- When the SCLn line state remains unchanged for the period specified by bits ICMR2.TMOH, TMOL, and TMOS while the ICFER.TMOE bit is 1 (the timeout function is enabled) in master mode or in slave mode and the received slave address matches.

[Clearing conditions]

- When 0 is written to the TMOF bit after reading TMOF = 1
- When 1 is written to the ICCR1.IICRST bit to apply an RIIC reset or an internal reset

AL Flag (Arbitration-Lost Flag)

This flag shows that bus mastership has been lost (loss in arbitration) due to a bus conflict or some other reason when a start condition is generated or an address and data are transmitted. The RIIC monitors the level on the SDAn line during transmission and, if the level on the line does not match the value of the bit being output, sets the value of the AL flag to 1 to indicate that the bus is occupied by another device.

The RIIC can also detect loss of arbitration during NACK transmission in receive mode or during data transmission in slave mode.

[Setting conditions]

When master arbitration-lost detection is enabled: ICFER.MALE = 1

- When the internal SDA output state does not match the SDAn line level at the rising edge of SCL except for the ACK period during data (including slave address) transmission in master transmit mode (when the SDAn line is driven low while the internal SDA output is high (the SDAn pin is in the high-impedance state))
- When a start condition is detected while the ICCR2.ST bit is 1 (requests to generate a start condition) or the internal SDA output state does not match the SDAn line level
- When the ICCR2.ST bit is set to 1 (requests to generate a start condition) with the ICCR2.BBSY flag set to 1.

When NACK arbitration-lost detection is enabled: ICFER.NALE = 1

- When the internal SDA output state does not match the SDAn line level at the rising edge of SCL in the ACK period during NACK transmission in receive mode

When slave arbitration-lost detection is enabled: ICFER.SALE = 1

- When the internal SDA output state does not match the SDAn line level at the rising edge of SCL except for the ACK period during data transmission in slave transmit mode

[Clearing conditions]

- When 0 is written to the AL flag after reading AL = 1
- When 1 is written to the ICCR1.IICRST bit to apply an RIIC reset or an internal reset

Table 42.4 Relationship between Arbitration-Lost Generation Sources and Arbitration-Lost Enable Functions

ICFER			ICSR2	Error	Arbitration-Lost Generation Source
MALE	NALE	SALE	AL		
1	x	x	1	Start condition generation error	When internal SDA output state does not match SDAn line level when a start condition is detected while the ICCR2.ST bit is 1
					When ICCR2.ST bit is set to 1 with ICCR2.BBSY flag set to 1
			1	Transmit data mismatch	When transmit data (including slave address) does not match the bus state in master transmit mode
x	1	x	1	NACK transmission mismatch	When ACK is detected during transmission of NACK in master receive mode or slave receive mode
x	x	1	1	Transmit data mismatch	When transmit data does not match the bus state in slave transmit mode

x: Don't care

START Flag (Start Condition Detection Flag)

[Setting condition]

- When a start condition (or a restart condition) is detected

[Clearing conditions]

- When 0 is written to the START bit after reading START = 1
- When a stop condition is detected
- When 1 is written to the ICCR1.IICRST bit to apply an RIIC reset or an internal reset

STOP Flag (Stop Condition Detection Flag)

[Setting condition]

- When a stop condition is detected

[Clearing conditions]

- When 0 is written to the STOP bit after reading STOP = 1
- When 1 is written to the ICCR1.IICRST bit to apply an RIIC reset or an internal reset

NACKF Flag (NACK Detection Flag)

[Setting condition]

- When ACK is not received (NACK is received) from the receiver in transmit mode with the ICFER.NACKE bit set to 1 (transfer suspension enabled)

[Clearing conditions]

- When 0 is written to the NACKF bit after reading NACKF = 1
- When 1 is written to the ICCR1.IICRST bit to apply an RIIC reset or an internal reset

Note: When the NACKF flag is set to 1, the RIIC suspends data transmission/reception. Writing to the ICDRT register in transmit mode or reading from the ICDRR register in receive mode with the NACKF flag set to 1 does not enable data transmit/receive operation. To restart data transmission/reception, set the NACKF flag to 0.

RDRF Flag (Receive Data Full Flag)

[Setting conditions]

- When receive data has been transferred from the ICDRS register to the ICDRR register
This flag is set to 1 at the rising edge of the eighth or ninth SCL (selected by the ICMR3.RDRFS bit)
- When the received slave address matches after a start condition (or a restart condition) is detected with the ICCR2.TRS bit set to 0

[Clearing conditions]

- When 0 is written to the RDRF bit after reading RDRF = 1
- When data is read from the ICDRR register
- When 1 is written to the ICCR1.IICRST bit to apply an RIIC reset or an internal reset

TEND Flag (Transmission End Flag)

[Setting condition]

- At the rising edge of the ninth SCL while the TDRE flag is 1

[Clearing conditions]

- When 0 is written to the TEND bit after reading TEND = 1
- When data is written to the ICDRT register
- When a stop condition is detected
- When 1 is written to the ICCR1.IICRST bit to apply an RIIC reset or an internal reset

TDRE Flag (Transmit Data Empty Flag)

[Setting conditions]

- When data has been transferred from the ICDRT register to the ICDRS register and the ICDRT register becomes empty
- When the ICCR2.TRS bit is set to 1
- When the received slave address matches while the TRS bit is 1

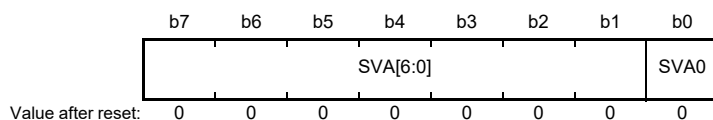
[Clearing conditions]

- When data is written to the ICDRT register
- When the ICCR2.TRS bit is set to 0
- When 1 is written to the ICCR1.IICRST bit to apply an RIIC reset or an internal reset

Note: The NACKF flag becoming 1 while the ICFER.NACKE bit is 1 suspends data transmission and reception by the RIIC. Even if the next data for transmission has already been written to the ICDRT register (the TDRE flag is 0), the data in the ICDRT register is retained but not transferred to the ICDRS register. At this point, the TDRE flag does not become 1.

42.2.11 Slave Address Register Ly (SARLy) (y = 0 to 2)

Address(es): RIIC0.SARL0 0008 830Ah, RIIC1.SARL0 0008 832Ah, RIIC2.SARL0 0008 834Ah,
RIIC0.SARL1 0008 830Ch, RIIC1.SARL1 0008 832Ch, RIIC2.SARL1 0008 834Ch,
RIIC0.SARL2 0008 830Eh, RIIC1.SARL2 0008 832Eh, RIIC2.SARL2 0008 834Eh



Bit	Symbol	Bit Name	Description	R/W
b0	SVA0	10-Bit Address LSB	Set a slave address	R/W
b7 to b1	SVA[6:0]	7-Bit Address/10-Bit Address Lower Bits	Set a slave address	R/W

SVA0 Bit (10-Bit Address LSB)

When the 10-bit address format is selected (SARUy.FS bit is 1), this bit functions as the LSB of a 10-bit address and forms the lower 8 bits of a 10-bit address in combination with the SVA[6:0] bits.

When the ICSEr.SARyE bit is set to 1 (SARLy and SARUy enabled) and the SARUy.FS bit is 1, this bit is valid. While the SARUy.FS bit or SARyE bit is 0, the setting of this bit is ignored.

SVA[6:0] Bits (7-Bit Address/10-Bit Address Lower Bits)

When the 7-bit address format is selected (SARUy.FS bit is 0), these bits function as a 7-bit address. When the 10-bit address format is selected (SARUy.FS bit is 1), these bits function as the lower 8 bits of a 10-bit address in combination with the SVA0 bit.

While the ICSEr.SARyE bit is 0, the setting of these bits is ignored.

42.2.12 Slave Address Register Uy (SARUy) (y = 0 to 2)

Address(es): RIIC0.SARU0 0008 830Bh, RIIC1.SARU0 0008 832Bh, RIIC2.SARU0 0008 834Bh,
RIIC0.SARU1 0008 830Dh, RIIC1.SARU1 0008 832Dh, RIIC2.SARU1 0008 834Dh,
RIIC0.SARU2 0008 830Fh, RIIC1.SARU2 0008 832Fh, RIIC2.SARU2 0008 834Fh

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	SVA[1:0]		FS
Value after reset:	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	FS	7-Bit/10-Bit Address Format Select	0: The 7-bit address format is selected. 1: The 10-bit address format is selected.	R/W
b2, b1	SVA[1:0]	10-Bit Address Upper Bits	Set a slave address	R/W
b7 to b3	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

FS Bit (7-Bit/10-Bit Address Format Select)

This bit is used to select 7-bit address or 10-bit address for slave address y (in registers SARLy and SARUy).

When the ICSEr.SARyE bit is set to 1 (registers SARLy and SARUy enabled) and the SARUy.FS bit is 0, the 7-bit address format is selected for slave address y, the SARLy.SVA[6:0] bits setting is valid, and the settings of the SVA[1:0] bits and the SARLy.SVA0 bit are ignored.

When the ICSEr.SARyE bit is set to 1 (registers SARLy and SARUy enabled) and the SARUy.FS bit is 1, the 10-bit address format is selected for slave address y and the settings of the SVA[1:0] bits and SARLy are valid.

While the ICSEr.SARyE bit is 0 (registers SARLy and SARUy disabled), the setting of the SARUy.FS bit is invalid.

SVA[1:0] Bits (10-Bit Address Upper Bits)

When the 10-bit address format is selected (FS = 1), these bits function as the upper 2 bits of a 10-bit address.

When the ICSEr.SARyE bit is set to 1 (SARLy and SARUy enabled) and the SARUy.FS bit is 1, these bits are valid.

While the SARUy.FS bit or SARyE bit is 0, the setting of these bits is ignored.

42.2.13 I²C-bus Bit Rate Low-Level Register (ICBRL)

Address(es): RIIC0.ICBRL 0008 8310h, RIIC1.ICBRL 0008 8330h, RIIC2.ICBRL 0008 8350h



Bit	Symbol	Bit Name	Description	R/W
b4 to b0	BRL[4:0]	Bit Rate Low Period	Low period of SCL	R/W
b7 to b5	—	Reserved	These bits are read as 1. The write value should be 1.	R/W

ICBRL is a 5-bit register to set the low period of SCL.

It also works to generate the data setup time for automatic SCL low-hold operation (refer to section 42.8, Automatic Low-Hold Function for SCL); when the RIIC is used only in slave mode, this register needs to be set to a value longer than the data setup time*1.

ICBRL counts the low period with the internal reference clock (IIC ϕ) specified by the ICMR1.CKS[2:0] bits.

If the digital noise filter is enabled (the ICFER.NFE bit is 1), set the ICBRL register to a value at least one greater than the number of stages in the noise filter. Regarding the number of stages in the noise filter, see the description of the ICMR3.NF[1:0] bits.

Note 1. Data setup time (t_{SU}: DAT)

250 ns (up to 100 kbps: Standard-mode (Sm))

100 ns (up to 400 kbps: Fast-mode (Fm))

50 ns (up to 1 Mbps: Fast-mode plus (Fm+))

42.2.14 I²C-bus Bit Rate High-Level Register (ICBRH)

Address(es): RIIC0.ICBRH 0008 8311h, RIIC1.ICBRH 0008 8331h, RIIC2.ICBRH 0008 8351h



Bit	Symbol	Bit Name	Description	R/W
b4 to b0	BRH[4:0]	Bit Rate High Period	High period of SCL	R/W
b7 to b5	—	Reserved	These bits are read as 1. The write value should be 1.	R/W

ICBRH is a 5-bit register to set the high period of SCL. ICBRH is valid in master mode. If the RIIC is used only in slave mode, this register need not to set the high period.

ICBRH counts the high period with the internal reference clock ($IIC\phi$) specified by the ICMR1.CKS[2:0] bits.

If the digital noise filter is enabled (the ICFER.NFE bit is 1), set the ICBRH register to a value at least one greater than the number of stages in the noise filter. Regarding the number of stages in the noise filter, see the description of the ICMR3.NF[1:0] bits.

The I²C transfer rate and the SCL duty are calculated using the following expression.

Transfer rate = $1 / \{[(ICBRH + 1) + (ICBRL + 1)] / IIC\phi + SCLn \text{ line rise time } [tr] + SCLn \text{ line fall time } [tf]\}$

Duty cycle = $\{SCLn \text{ line rise time } [tr]^2 + (ICBRH + 1) / IIC\phi\} / \{SCLn \text{ line fall time } [tf]^2 + (ICBRL + 1) / IIC\phi\}$

Note 1. $IIC\phi = PCLK \times \text{Division ratio}$

Note 2. The SCLn line rise time [tr] and SCLn line fall time [tf] depend on the total bus line capacitance [Cb] and the pull-up resistor [Rp]. For details, refer to the I²C-bus specification from NXP Semiconductors.

Table 42.5 lists examples of ICBRH/ICBRL settings.

Table 42.5 Examples of ICBRH/ICBRL Settings for Transfer Rate

Transfer Rate (kbps)	Operating Frequency PCLK (MHz)								
	8			10			12.5		
	CKS[2:0]	ICBRH	ICBRL	CKS[2:0]	ICBRH	ICBRL	CKS[2:0]	ICBRH	ICBRL
10	100b	22 (F6h)	25 (F9h)	101b	13 (EDh)	15 (EFh)	101b	16 (F0h)	20 (F4h)
50	010b	16 (F0h)	19 (F3h)	010b	21 (F5h)	24 (F8h)	011b	12 (ECh)	15 (EFh)
100	001b	15 (EFh)	18 (F2h)	001b	19 (F3h)	23 (F7h)	001b	24 (F8h)	29 (FDh)
400	000b	4 (E4h)	10 (EAh)	000b	5 (E5h)	12 (ECh)	000b	7 (E7h)	16 (F0h)
1000	000b	2 (E2h)	3 (E3h)	000b	2 (E2h)	4 (E4h)	000b	3 (E3h)	6 (E6h)

Transfer Rate (kbps)	Operating Frequency PCLK (MHz)								
	16			20			25		
	CKS[2:0]	ICBRH	ICBRL	CKS[2:0]	ICBRH	ICBRL	CKS[2:0]	ICBRH	ICBRL
10	101b	22 (F6h)	25 (F9h)	110b	13 (EDh)	15 (EFh)	110b	16 (F0h)	20 (F4h)
50	011b	16 (F0h)	19 (F3h)	011b	21 (F5h)	24 (F8h)	100b	12 (ECh)	15 (EFh)
100	010b	15 (EFh)	18 (F2h)	010b	19 (F3h)	23 (F7h)	010b	24 (F8h)	29 (FDh)
400	000b	9 (E9h)	20 (F4h)	000b	11 (EBh)	25 (F9h)	001b	7 (E7h)	16 (F0h)
1000	000b	4 (E4h)	7 (E7h)	000b	5 (E5h)	9 (E9h)	000b	6 (E6h)	12 (ECh)

Transfer Rate (kbps)	Operating Frequency PCLK (MHz)								
	30			32			33		
	CKS[2:0]	ICBRH	ICBRL	CKS[2:0]	ICBRH	ICBRL	CKS[2:0]	ICBRH	ICBRL
10	110b	20 (F4h)	24 (F8h)	110b	22 (F6h)	25 (F9h)	110b	22 (F6h)	26 (FAh)
50	100b	15 (EFh)	18 (F2h)	100b	16 (F0h)	19 (F3h)	100b	17 (F1h)	20 (F4h)
100	011b	14 (EEh)	17 (F1h)	011b	15 (EFh)	18 (F2h)	011b	16 (F0h)	19 (F3h)
400	001b	8 (E8h)	19 (F3h)	001b	9 (E9h)	20 (F4h)	001b	9 (E9h)	21 (F5h)
1000	000b	7 (E7h)	14 (EEh)	000b	8 (E8h)	16 (F0h)	000b	8 (E8h)	16 (F0h)

Transfer Rate (kbps)	Operating Frequency PCLK (MHz)								
	40			50			60		
	CKS[2:0]	ICBRH	ICBRL	CKS[2:0]	ICBRH	ICBRL	CKS[2:0]	ICBRH	ICBRL
10	111b	13 (EDh)	15 (EFh)	111b	16 (F0h)	20 (F4h)	111b	20 (F4h)	24 (F8h)
50	100b	21 (F5h)	24 (F8h)	100b	26 (FAh)	31 (FFh)	101b	15 (EFh)	18 (F2h)
100	011b	19 (F3h)	23 (F7h)	011b	24 (F8h)	29 (FDh)	100b	14 (EEh)	17 (F1h)
400	001b	11 (EBh)	25 (F9h)	010b	7 (E7h)	16 (F0h)	010b	8 (E8h)	19 (F3h)
1000	000b	10 (EAh)	20 (F4h)	000b	12 (ECh)	24 (F8h)	000b	15 (EFh)	29 (FDh)

Note: ICBRH/ICBRL settings in these tables are calculated using the following values:
 SCLn line rise time (tr): 100 kbps or less (Sm): 1000 ns, 400 kbps or less (Fm): 300 ns, 1 Mbps or less (Fm+): 120 ns
 SCLn line fall time (tf): 400 kbps or less (Sm/Fm): 300 ns, 1 Mbps or less (Fm+): 120 ns
 For the specified values of rise time (tr) and fall time (tf) of the SCLn signal, refer to the I²C-bus specification from NXP Semiconductors.

42.2.15 I²C-bus Transmit Data Register (ICDRT)

Address(es): RIIC0.ICDRT 0008 8312h, RIIC1.ICDRT 0008 8332h, RIIC2.ICDRT 0008 8352h



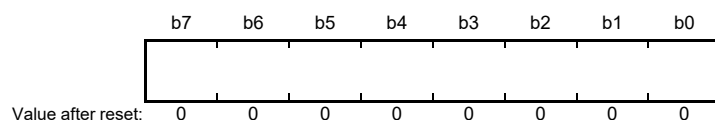
When the ICDRT register detects a space in the I²C-bus shift register (ICDRS), it transfers the transmit data that has been written to the ICDRT register to the ICDRS register and starts transmitting data in transmit mode.

The double-buffer structure of the ICDRT register and the ICDRS register allows continuous transmit operation if the next transmit data has been written to the ICDRT register while the ICDRS register data is being transmitted.

The ICDRT register can always be read and written. Write transmit data to the ICDRT register once when a transmit data empty interrupt (TXI) request is generated.

42.2.16 I²C-bus Receive Data Register (ICDRR)

Address(es): RIIC0.ICDRR 0008 8313h, RIIC1.ICDRR 0008 8333h, RIIC2.ICDRR 0008 8353h



When 1 byte of data has been received, the received data is transferred from the I²C-bus shift register (ICDRS) to the ICDRR register to enable the next data to be received.

The double-buffer structure of the ICDRS register and the ICDRR register allows continuous receive operation if the received data has been read from the ICDRR register while the ICDRS register is receiving data.

The ICDRR register cannot be written. Read data from the ICDRR register once when a receive data full interrupt (RXI) request is generated.

If the ICDRR register receives the next receive data before the current data is read from the ICDRR register (while the ICSR2.RDRF flag is 1), the RIIC automatically holds the SCL line low one cycle before the RDRF flag is set to 1 next.

42.2.17 I²C-bus Shift Register (ICDRS)

The ICDRS register is an 8-bit shift register to transmit and receive data.

During transmission, transmit data is transferred from the ICDRT register to the ICDRS register and is sent from the SDAn pin. During reception, data is transferred from the ICDRS register to the ICDRR register after 1 byte of data has been received.

The ICDRS register cannot be accessed directly.

42.3 Operation

42.3.1 Communication Data Format

The I²C-bus format consists of 8-bit data and 1-bit acknowledgment. The first byte following a start condition or restart condition is an address byte used to specify a slave device with which the master device communicates. The specified slave is valid until a new slave is specified or a stop condition is generated.

Figure 42.2 shows the I²C-bus format, and Figure 42.3 shows the I²C-bus timing.

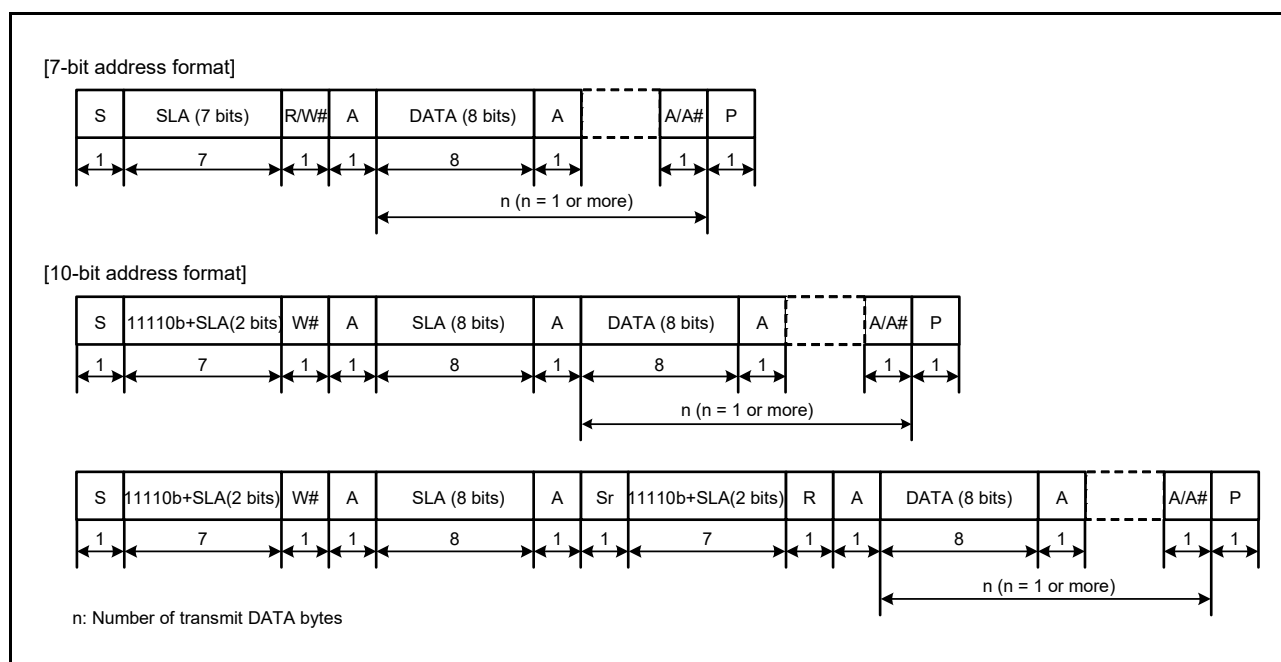


Figure 42.2 I²C-bus Format

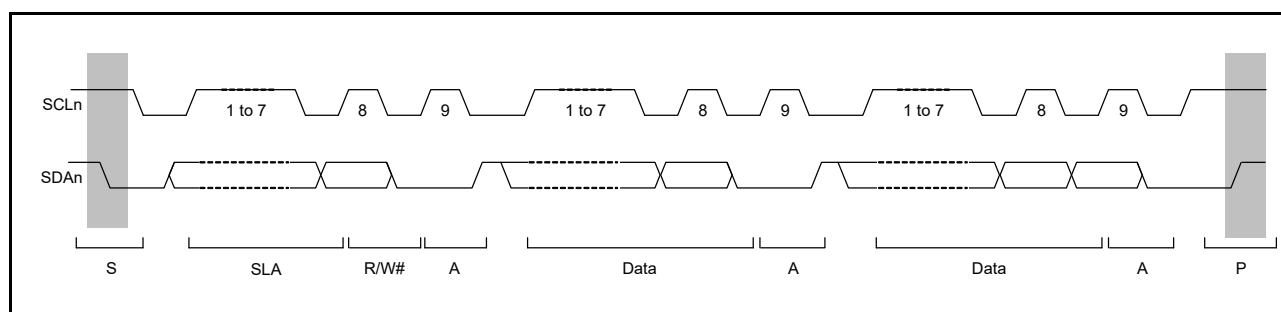


Figure 42.3 I²C-bus Timing (SLA = 7 Bits)

- S: Start condition. The master device drives the SDAn line low from high while the SCLn line is high.
- SLA: Slave address, by which the master device selects a slave device.
- R/W#: Indicates the direction of data transfer: from the slave device to the master device when R/W is 1, or from the master device to the slave device when R/W is 0.
- A: Acknowledge. The receiver drives the SDAn line low. (In master transmit mode, the slave device returns acknowledge. In master receive mode, the master device returns acknowledge.)
- A#: Not Acknowledge. The receiver drives the SDAn line high.
- Sr: Restart condition. The master device drives the SDAn line low from high after the setup time has elapsed with the SCLn line high.
- DATA: Transmitted or received data
- P: Stop condition. The master device drives the SDAn line high from low while the SCLn line is high.

42.3.2 Initial Settings

Before starting data transmission and reception, initialize the RIIC according to the procedure in Figure 42.4.

Set the ICCR1.ICE bit to 1 (internal reset) after setting the ICCR1.IICRST bit to 1 (RIIC reset) with the ICCR1.ICE bit set to 0 (SCLn and SDAn pins in inactive state). This initializes the various flags and internal state of the ICSR1 register. After that, set registers SARLy, SARUy, ICSEr, ICMR1, ICBRH, and ICBRL (y = 0 to 2), and set the other registers as necessary (for initial settings of the RIIC, see Figure 42.4). When the necessary register settings have been completed, set the ICCR1.IICRST bit to 0 (releases the RIIC reset). This step is not necessary if initialization of the RIIC has already been completed.

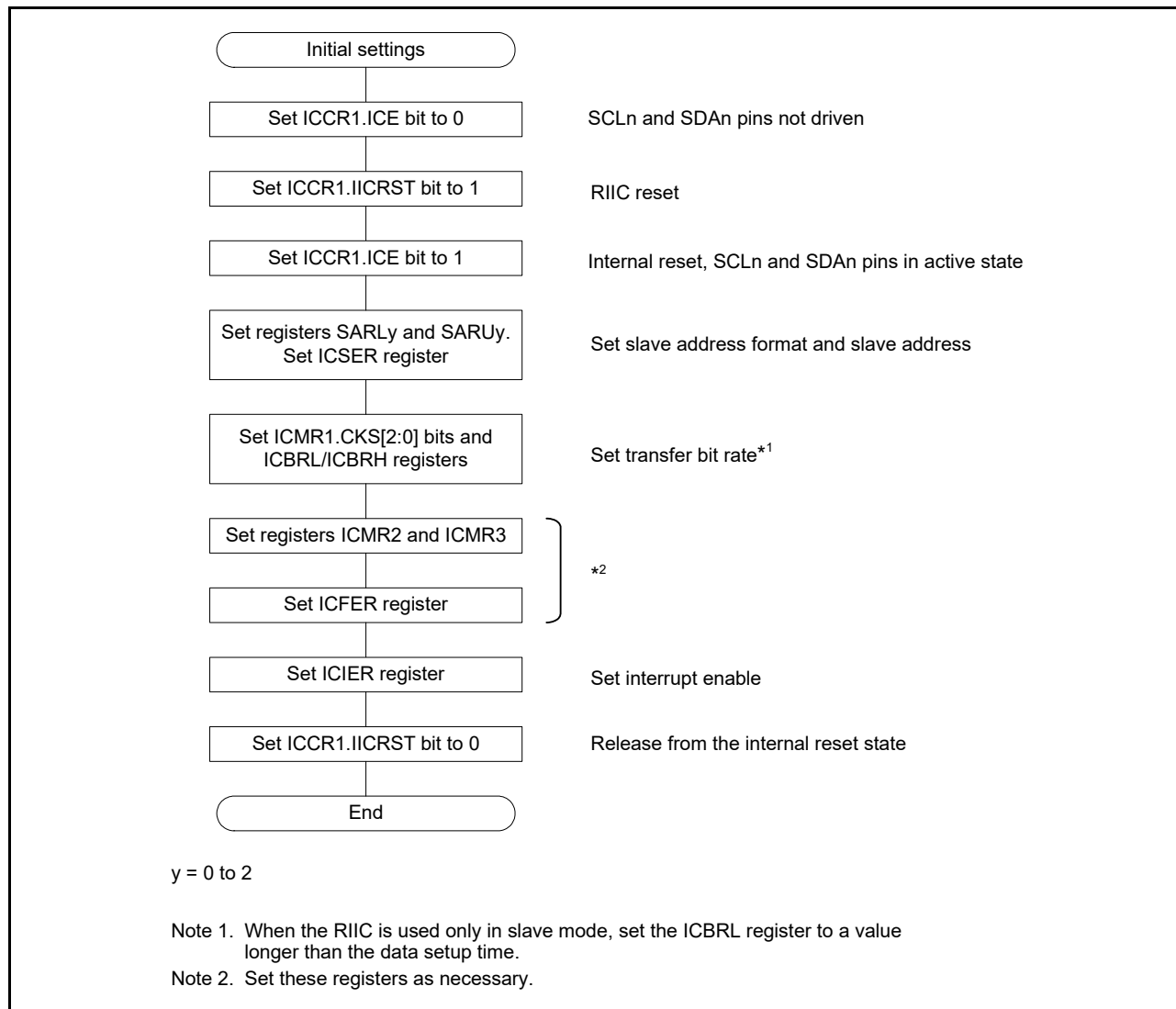


Figure 42.4 Example of RIIC Initialization Flowchart

42.3.3 Master Transmit Operation

In master transmit operation, the RIIC generates clock signals and sends data as the master device, and the slave device returns acknowledgments. Figure 42.5 shows an example of usage of master transmission and Figure 42.6 to Figure 42.8 show the timing of operations in master transmission.

The following describes the procedure and operations for master transmission.

- (1) Initial settings. For details, refer to **section 42.3.2, Initial Settings**.
- (2) Read the ICCR2.BBSY flag to check that the bus is open, and then set the ICCR2.ST bit to 1 (requests to generate a start condition). Upon receiving the request, the RIIC generates a start condition. At the same time, the BBSY flag and the ICSR2.START flag are automatically set to 1 and the ST bit is automatically set to 0. At this time, if the start condition is detected and the internal levels for the SDA output state and the levels on the SDA line have matched while the ST bit is 1, the RIIC recognizes that generating of the start condition as requested by the ST bit has been successfully completed, and bits MST and TRS in the ICCR2 register are automatically set to 1, placing the RIIC in master transmit mode. The ICSR2.TDRE flag is also automatically set to 1 in response to setting of the TRS bit to 1.
- (3) Check that the ICSR2.TDRE flag is 1, and then write the value for transmission (the slave address and the R/W# bit) to the ICDRT register. Once the data for transmission are written to the ICDRT register, the TDRE flag is automatically set to 0, the data are transferred from the ICDRT register to the ICDRS register, and the TDRE flag is again set to 1. After the byte containing the slave address and R/W# bit has been transmitted, the value of the TRS bit is automatically updated to select master transmit or master receive mode in accord with the value of the transmitted R/W# bit. If the value of the R/W# bit was 0, the RIIC continues in master transmit mode. Because the ICSR2.NACKF flag being 1 at this time indicates that no slave device recognized the address or there was an error in communications, write 1 to the ICCR2.SP bit to generate a stop condition. For data transmission with an address in the 10-bit format, start by writing 1111 0b, the 2 higher-order bits of the slave address, and W to the ICDRT register as the first address transmission. Then, as the second address transmission, write the 8 lower-order bits of the slave address to the ICDRT register.
- (4) After confirming that the ICSR2.TDRE flag is 1, write the data for transmission to the ICDRT register. The RIIC automatically holds the SCLn line low until the data for transmission are ready or a stop condition is generated.
- (5) After all bytes of data for transmission have been written to the ICDRT register, wait until the ICSR2.NACKF or ICSR2.TEND flag becomes 1, and then set the ICCR2.SP bit to 1 (requests to generate a stop condition). Upon receiving a stop condition generation request, the RIIC generates the stop condition.
- (6) Upon detecting the stop condition, the RIIC automatically sets bits MST and TRS in the ICCR2 register to 00b and enters slave receive mode. Furthermore, it automatically sets the TDRE and TEND flags to 0, and sets the ICSR2.STOP flag to 1.
- (7) After checking that the ICSR2.STOP flag is 1, set the ICSR2.NACKF and STOP flags to 0 for the next transfer operation.

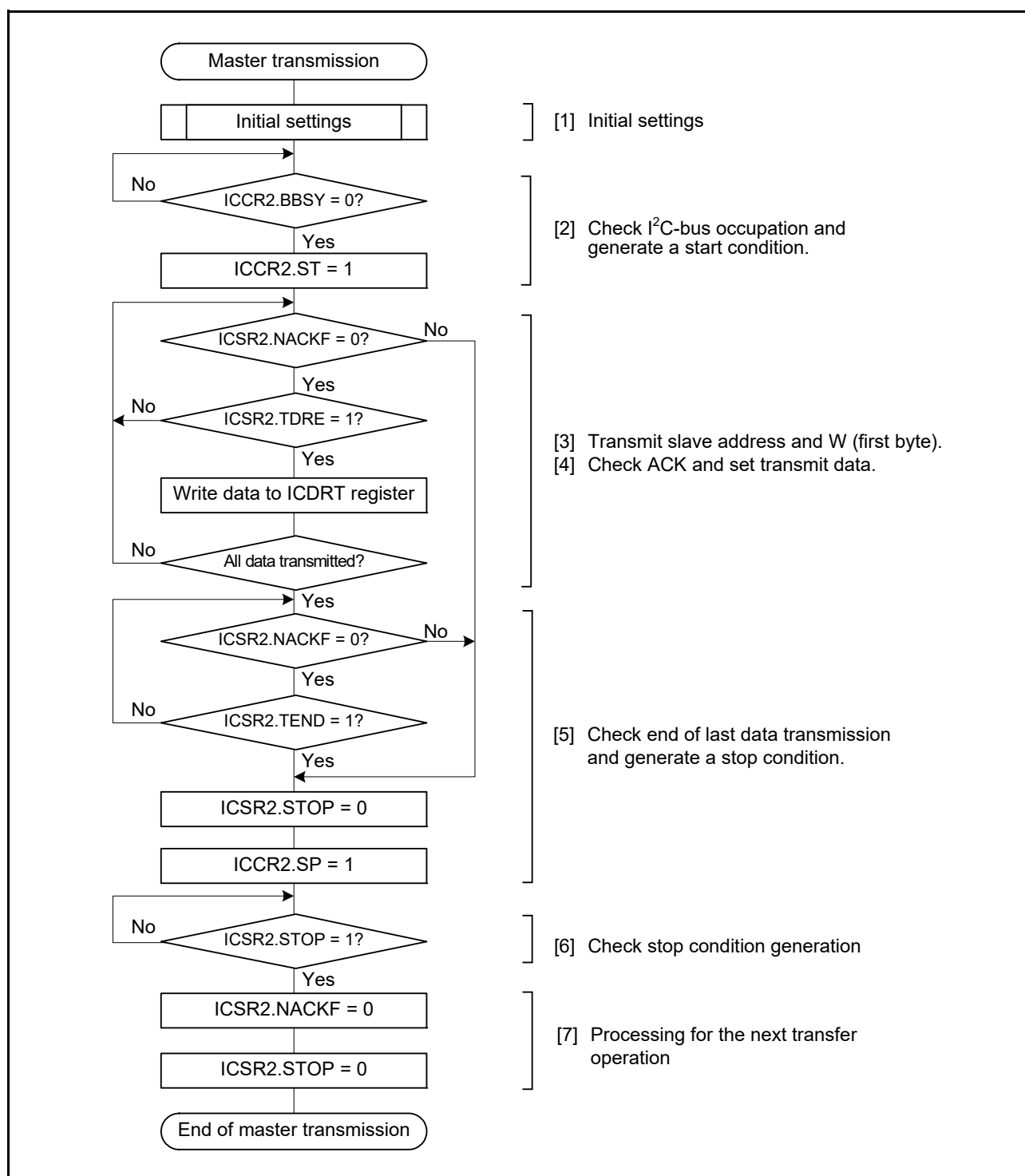


Figure 42.5 Example of Master Transmission Flowchart

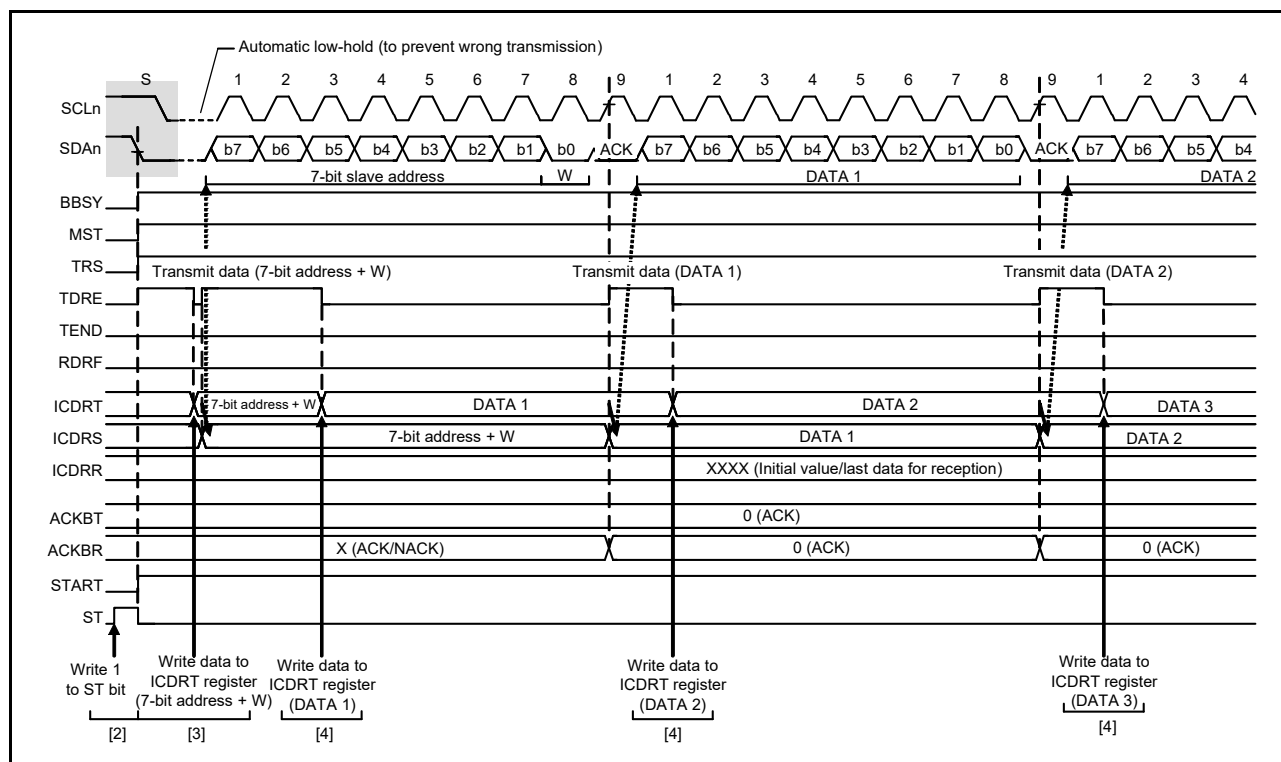


Figure 42.6 Master Transmit Operation Timing (1) (7-Bit Address Format)

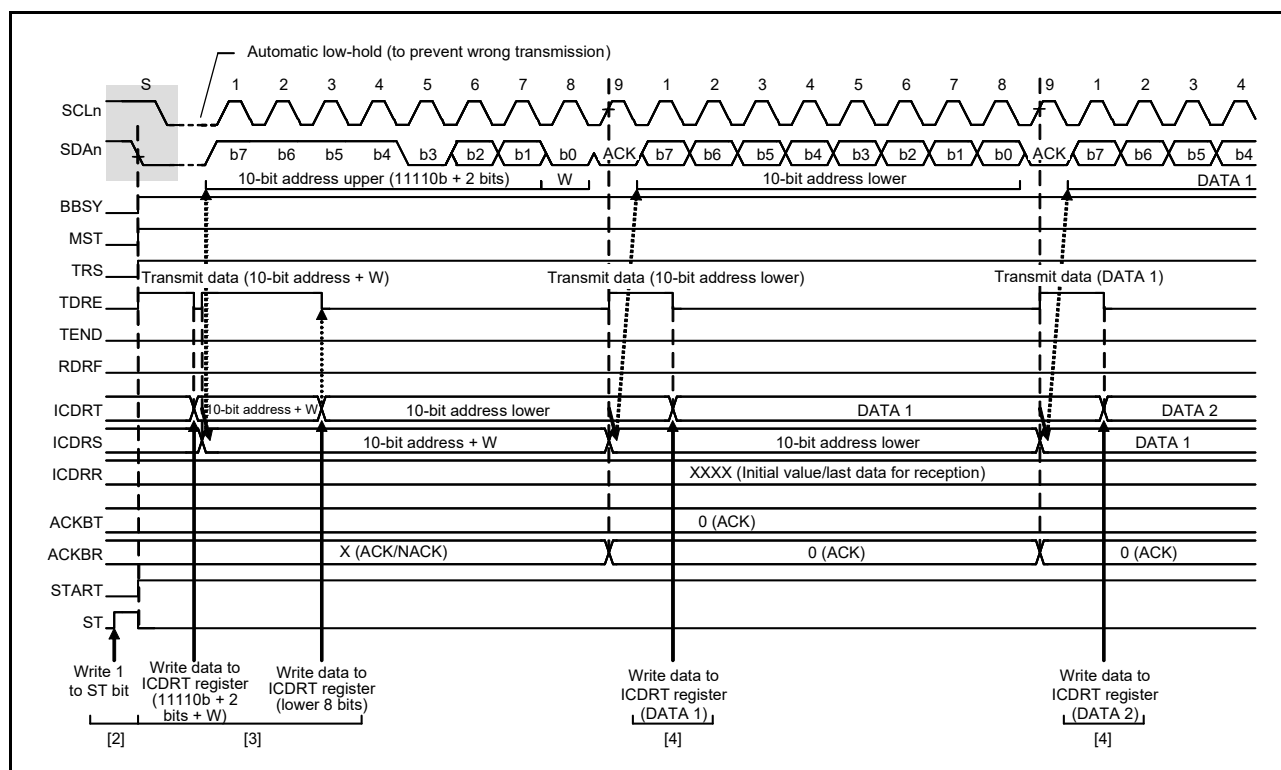


Figure 42.7 Master Transmit Operation Timing (2) (10-Bit Address Format)

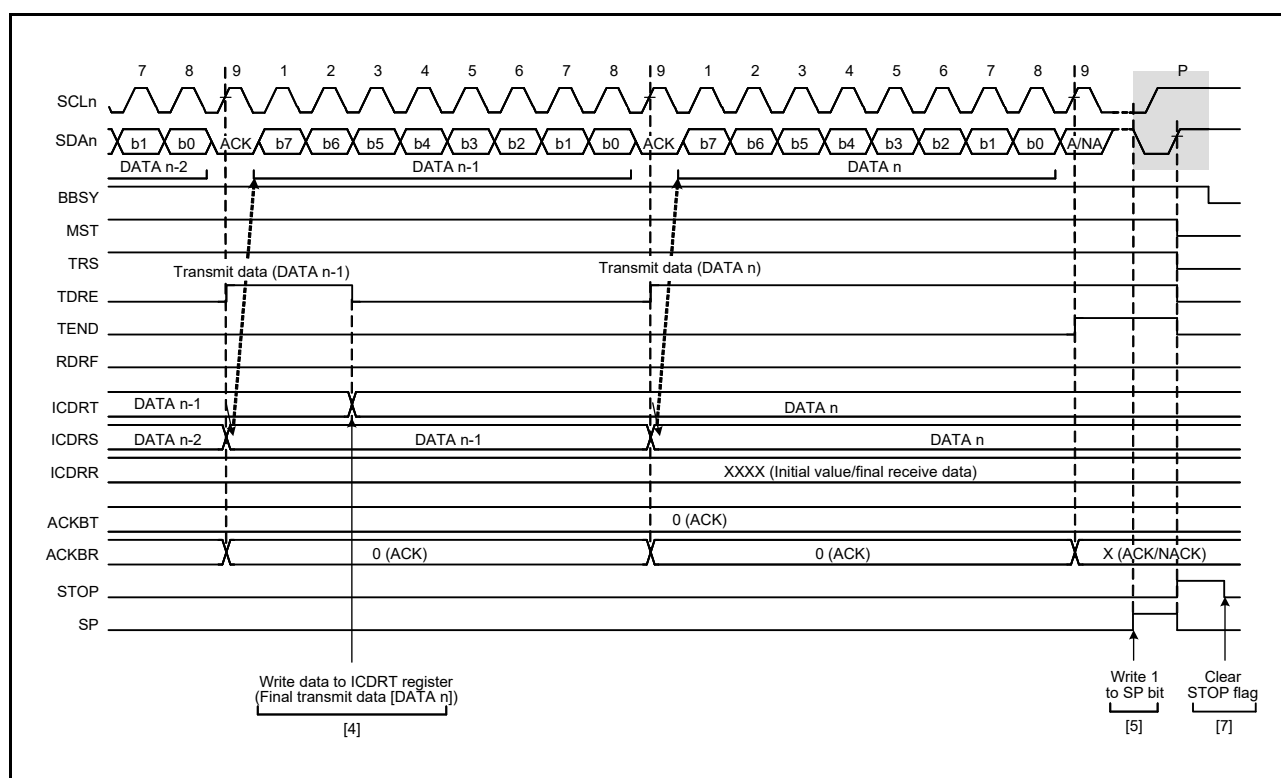


Figure 42.8 Master Transmit Operation Timing (3)

42.3.4 Master Receive Operation

In master receive operation, the RIIC as a master device generates clock signals, receives data from the slave device, and returns acknowledgments. Because the RIIC must start by sending a slave address to the corresponding slave device, this part of the procedure is performed in master transmit mode, but the subsequent steps are in master receive mode.

Figure 42.9 and Figure 42.10 show examples of usage of master reception (7-bit address format) and Figure 42.11 to Figure 42.13 show the timing of operations in master reception.

The following describes the procedure and operations for master reception.

- (1) Initial settings. For details, refer to section 42.3.2, Initial Settings.
- (2) Read the ICCR2.BBSY flag to check that the bus is open, and then set the ICCR2.ST bit to 1 (requests to generate a start condition). Upon receiving the request, the RIIC generates a start condition. When the RIIC detects the start condition, the BBSY flag and the ICSR2.START flag are automatically set to 1 and the ST bit is automatically set to 0. At this time, if the start condition is detected and the levels for the SDA output and the levels on the SDA_n line have matched while the ST bit is 1, the RIIC recognizes that generating of the start condition as requested by the ST bit has been successfully completed, and bits MST and TRS in the ICCR2 register are automatically set to 1, placing the RIIC in master transmit mode. The ICSR2.TDRE flag is also automatically set to 1 in response to setting of the TRS bit to 1.
- (3) Check that the ICSR2.TDRE flag is 1, and then write the value for transmission (the first byte indicates the slave address and value of the R/W# bit) to the ICDRT register. Once the data for transmission are written to the ICDRT register, the TDRE flag is automatically set to 0, the data are transferred from the ICDRT register to the ICDRS register, and the TDRE flag is again set to 1. Once the byte containing the slave address and R/W# bit has been transmitted, the value of the ICCR2.TRS bit is automatically updated to select transmit or receive mode in accord with the value of the transmitted R/W# bit. If the value of the R/W# bit was 1, the TRS bit is set to 0 on the rising edge of the ninth SCL, placing the RIIC in master receive mode. At this time, the TDRE flag is set to 0 and the ICSR2.RDRF flag is automatically set to 1.

Because the ICSR2.NACKF flag being 1 at this time indicates that no slave device recognized the address or there was an error in communications, write 1 to the ICCR2.SP bit to generate a stop condition.

For master reception from a device with a 10-bit address, start by using master transmission to send the 10-bit address, and then generate a restart condition. After that, transmitting 1111 0b, the two higher-order bits of the slave address, and the R bit places the RIIC in master receive mode.

- (4) Dummy read the ICDRR register after confirming that the ICSR2.RDRF flag is 1; this makes the RIIC start output of the SCL and start data reception.
- (5) After 1 byte of data has been received, the ICSR2.RDRF flag is set to 1 on the rising edge of the eighth or ninth SCL (the clock signal) as selected by the ICMR3.RDRFS bit. Reading the ICDRR register at this time will produce the received data, and the RDRF flag is automatically set to 0 at the same time. Furthermore, the value of the acknowledgment bit received during the ninth SCL is returned as the value set in the ICMR3.ACKBT bit. Furthermore, if the next byte to be received is the next to last byte, set the ICMR3.WAIT bit to 1 (for wait insertion) before reading the ICDRR register (containing the second byte from last). As well as enabling NACK output even in the case of delays in processing to set the ICMR3.ACKBT bit to 1 (NACK) in step (6), due to other interrupts, etc., this fixes the SCLn line to low on the falling edge of the ninth clock pulse in reception of the last byte, so the state is such that generating a stop condition is possible.
- (6) When the ICMR3.RDRFS bit is 0 and the slave device must be notified that it is to end transfer for data reception after transfer of the next (final) byte, set the ICMR3.ACKBT bit to 1 (NACK).
- (7) After reading the byte before last from the ICDRR register, if the value of the ICSR2.RDRF flag is confirmed to be 1, write 1 to the ICCR2.SP bit (requests to generate a stop condition) and then read the last byte from the ICDRR register. When the ICDRR register is read, the RIIC is released from the wait state and generates the stop condition after low-level output in the ninth clock pulse is completed or the SCLn line is released from the low-hold state.
- (8) Upon detecting the stop condition, the RIIC automatically sets bits MST and TRS in the ICCR2 register to 00b and enters slave receive mode. Furthermore, detection of the stop condition leads to setting of the ICSR2.STOP flag to 1.
- (9) After checking that the ICSR2.STOP flag is 1, set the ICSR2.NACKF and STOP flags to 0 for the next transfer operation.

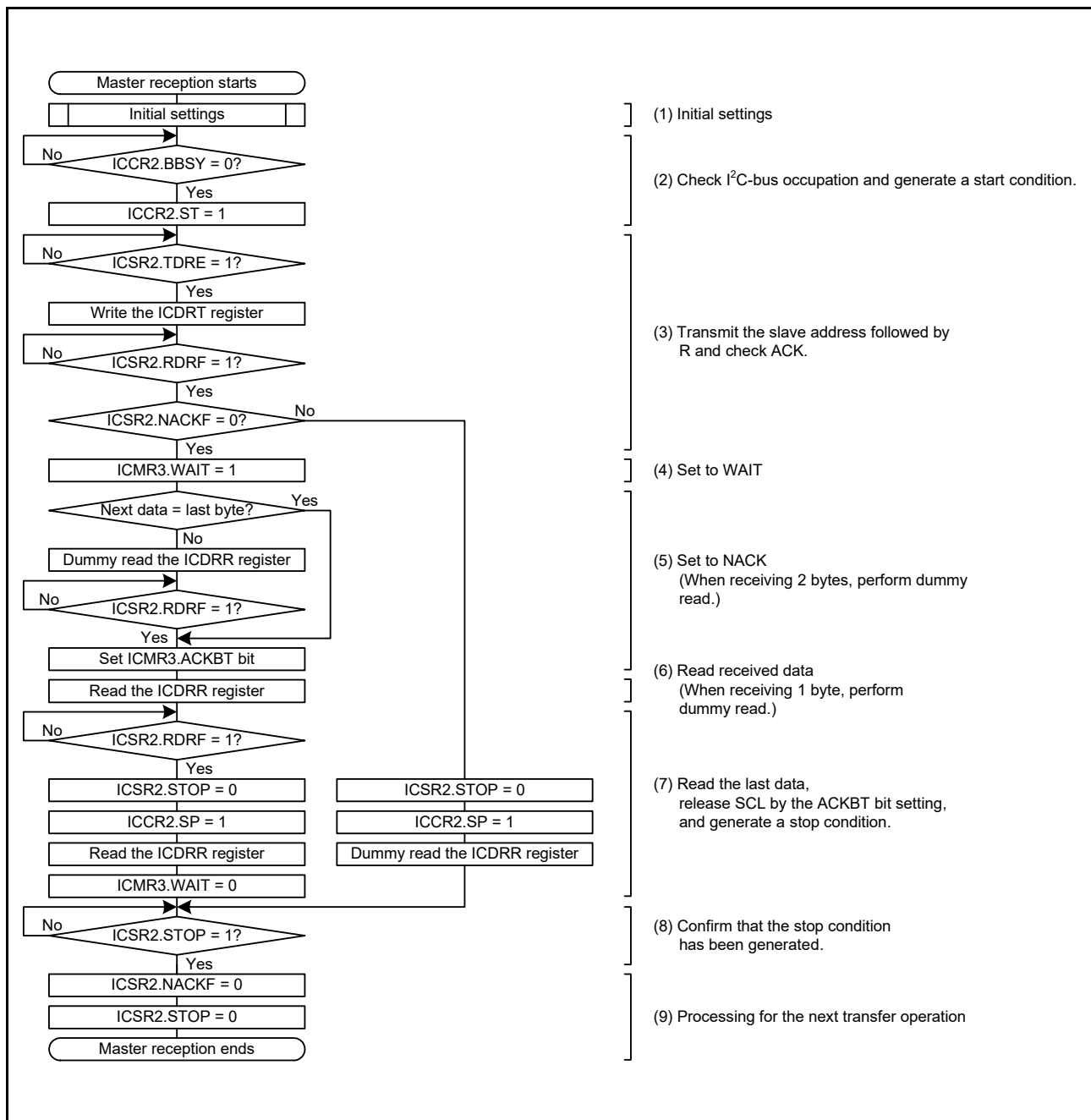
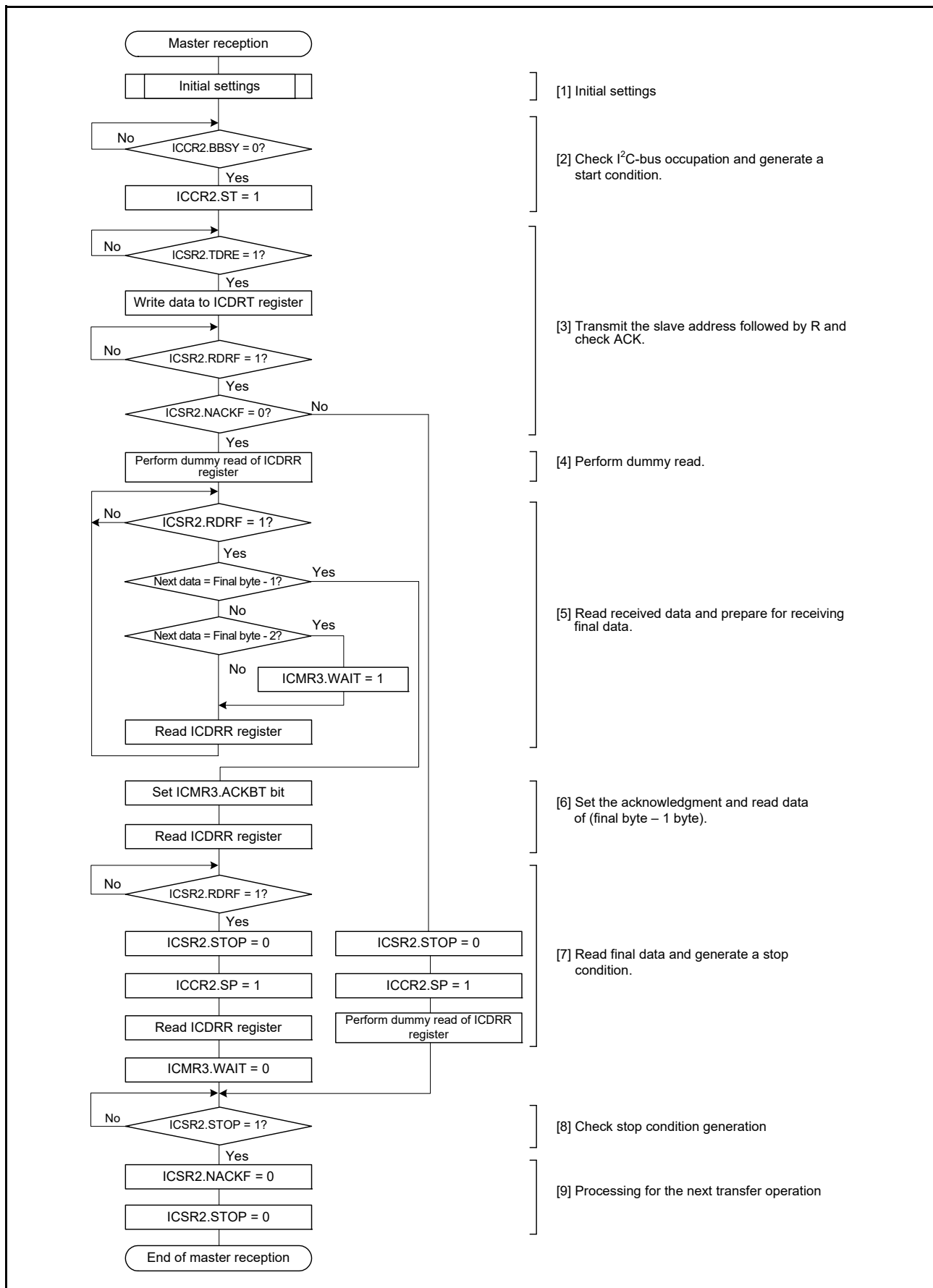


Figure 42.9 Example of Master Reception (7-Bit Address Format, 1 or 2 bytes)



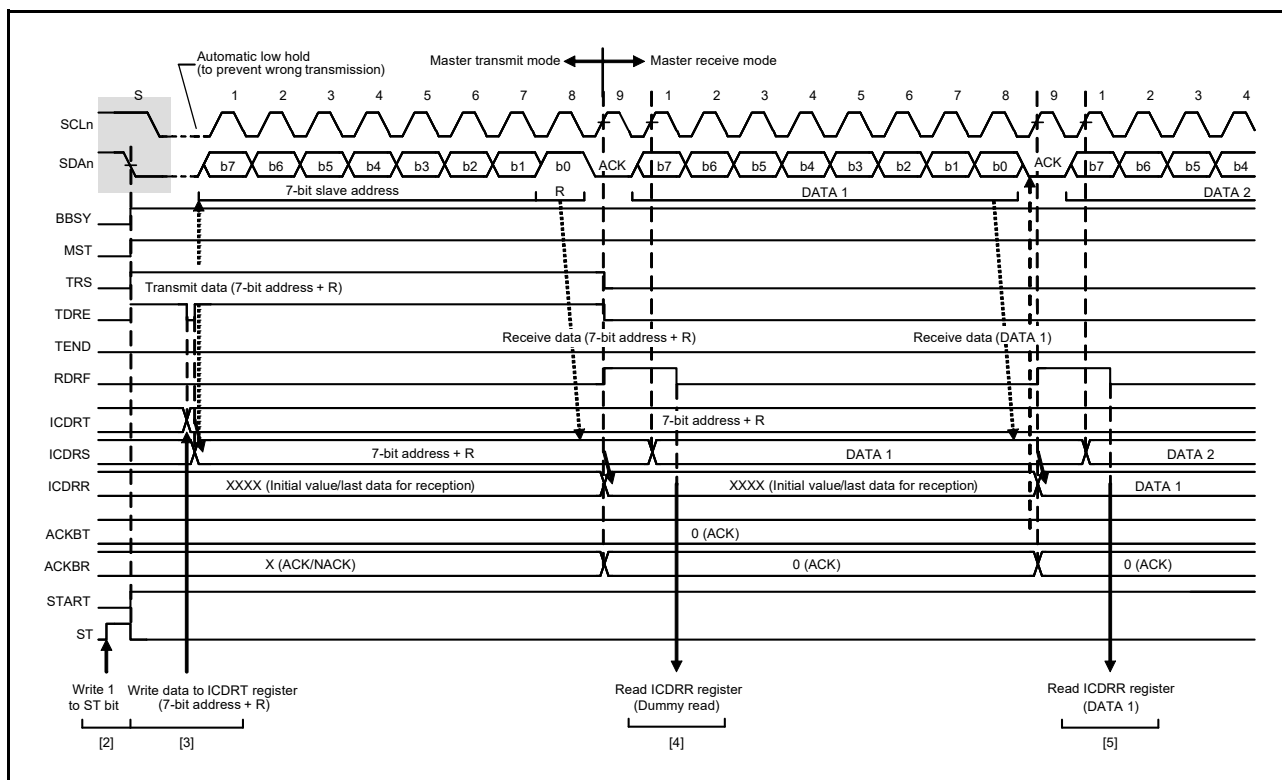


Figure 42.11 Master Receive Operation Timing (1) (7-Bit Address Format, When RDRFS bit is 0)

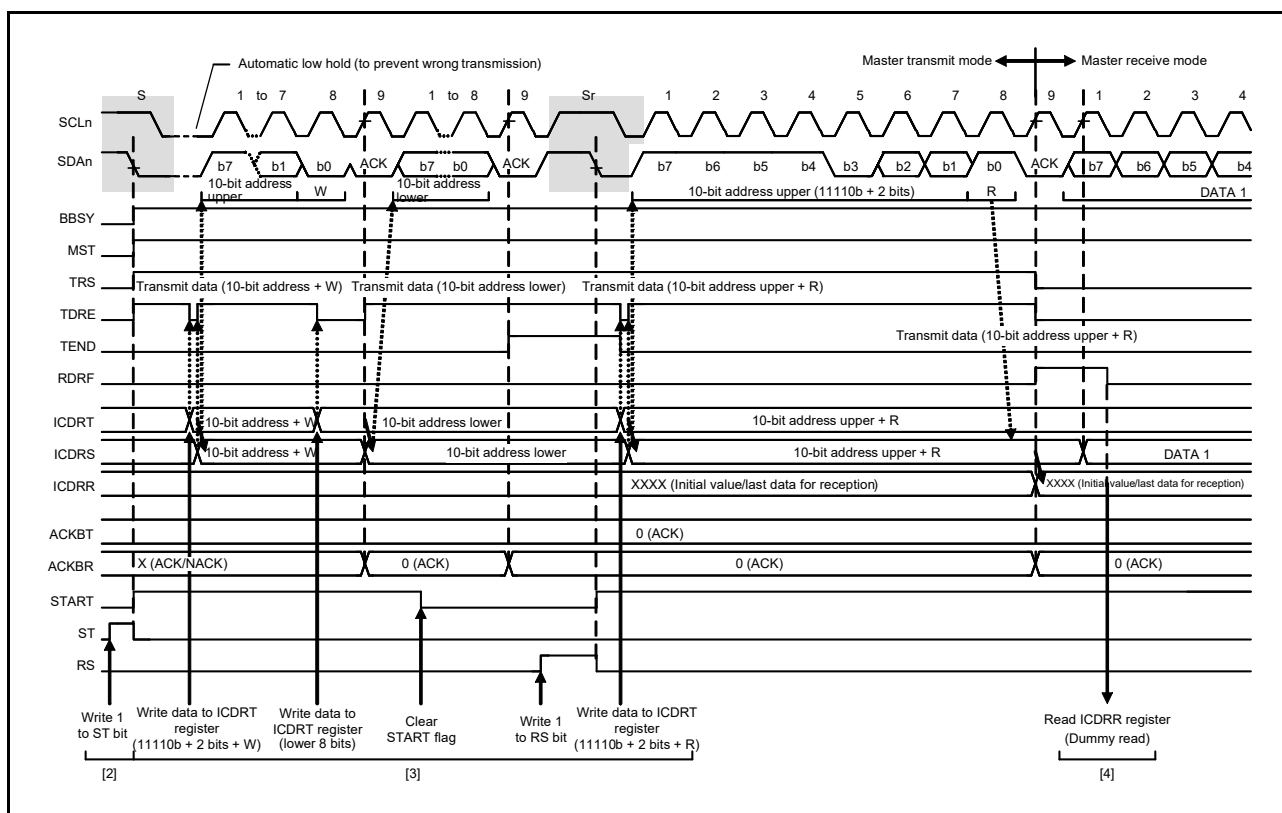


Figure 42.12 Master Receive Operation Timing (2) (10-Bit Address Format, When RDRFS bit is 0)

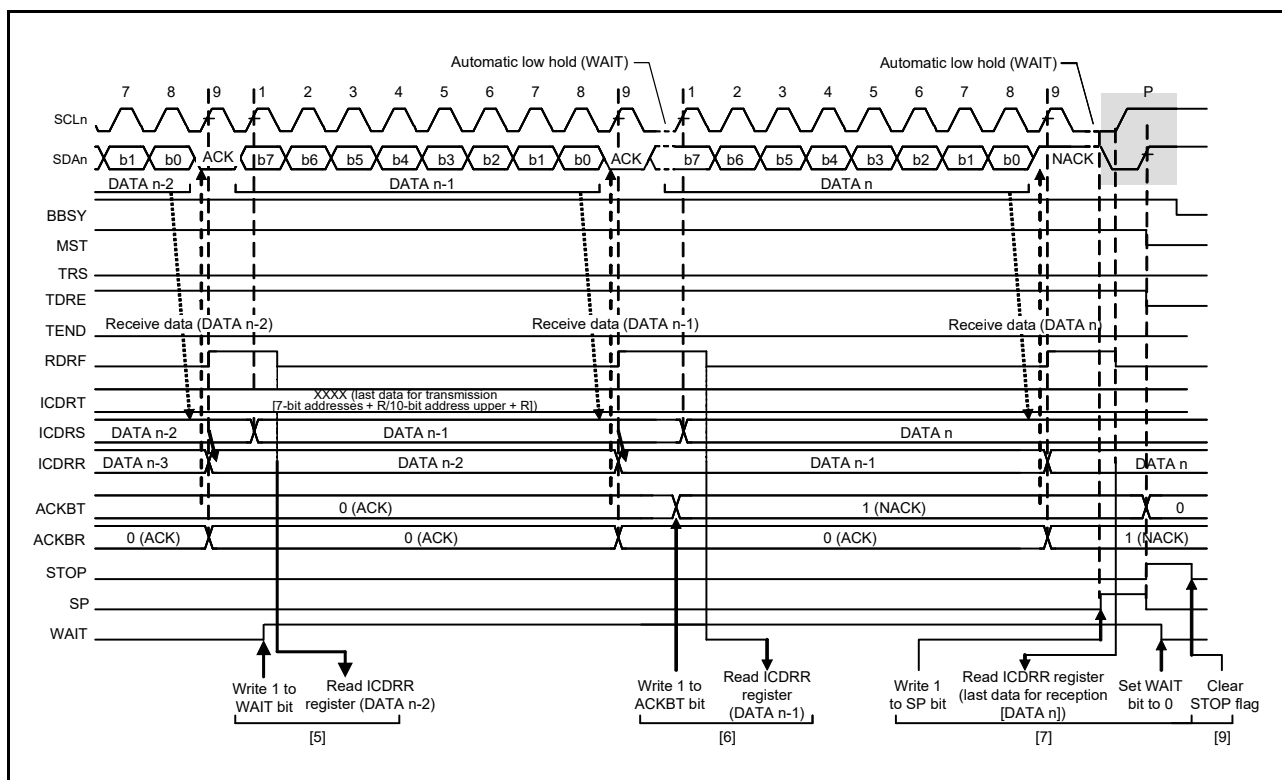


Figure 42.13 Master Receive Operation Timing (3) (When RDRFS bit is 0)

42.3.5 Slave Transmit Operation

In slave transmit operation, the master device outputs the SCL, the RIIC transmits data as a slave device, and the master device returns acknowledgments.

Figure 42.14 shows an example of usage of slave transmission and Figure 42.15 and Figure 42.16 show the timing of operations in slave transmission.

The following describes the procedure and operations for slave transmission.

- (1) Initial settings. For details, refer to **section 42.3.2, Initial Settings**.
After initial settings, the RIIC will stay in the standby state until it receives a slave address that it matches.
- (2) After receiving a matching slave address, the RIIC sets one of the corresponding bits ICSR1.HOA, GCA, and AASy (y = 0 to 2) to 1 on the rising edge of the ninth SCL (the clock signal) and outputs the value set in the ICMR3.ACKBT bit to the acknowledgment bit on the ninth SCL. If the value of the R/W# bit that was also received at this time is 1, the RIIC automatically places itself in slave transmit mode by setting both the ICCR2.TRS bit and the ICSR2.TDRE flag to 1.
- (3) After the ICSR2.TDRE flag is confirmed to be 1, write the data for transmission to the ICDRT register. At this time, if the RIIC does not receive ACK from the master device (receives an NACK signal) while the ICFER.NACKF bit is 1, the RIIC suspends transfer of the next data.
- (4) Wait until the ICSR2.TEND flag is set to 1 while the ICSR2.TDRE flag is 1, after the ICSR2.NACKF flag is set to 1 or the last byte for transmission is written to the ICDRT register. When the ICSR2.NACKF flag or the TEND flag is 1, the RIIC drives the SCLn line low on the falling edge of ninth SCL.
- (5) When the ICSR2.NACKF flag or the ICSR2.TEND flag is 1, dummy read the ICDRR register to complete the processing. This releases the SCLn line.
- (6) Upon detecting the stop condition, the RIIC automatically sets bits ICSR1.HOA, GCA, and AASy (y = 0 to 2), flags ICSR2.TDRE and TEND, and the ICCR2.TRS bit to 0, and enters slave receive mode.
- (7) After checking that the ICSR2.STOP flag is 1, set the ICSR2.NACKF and STOP flags to 0 for the next transfer operation.

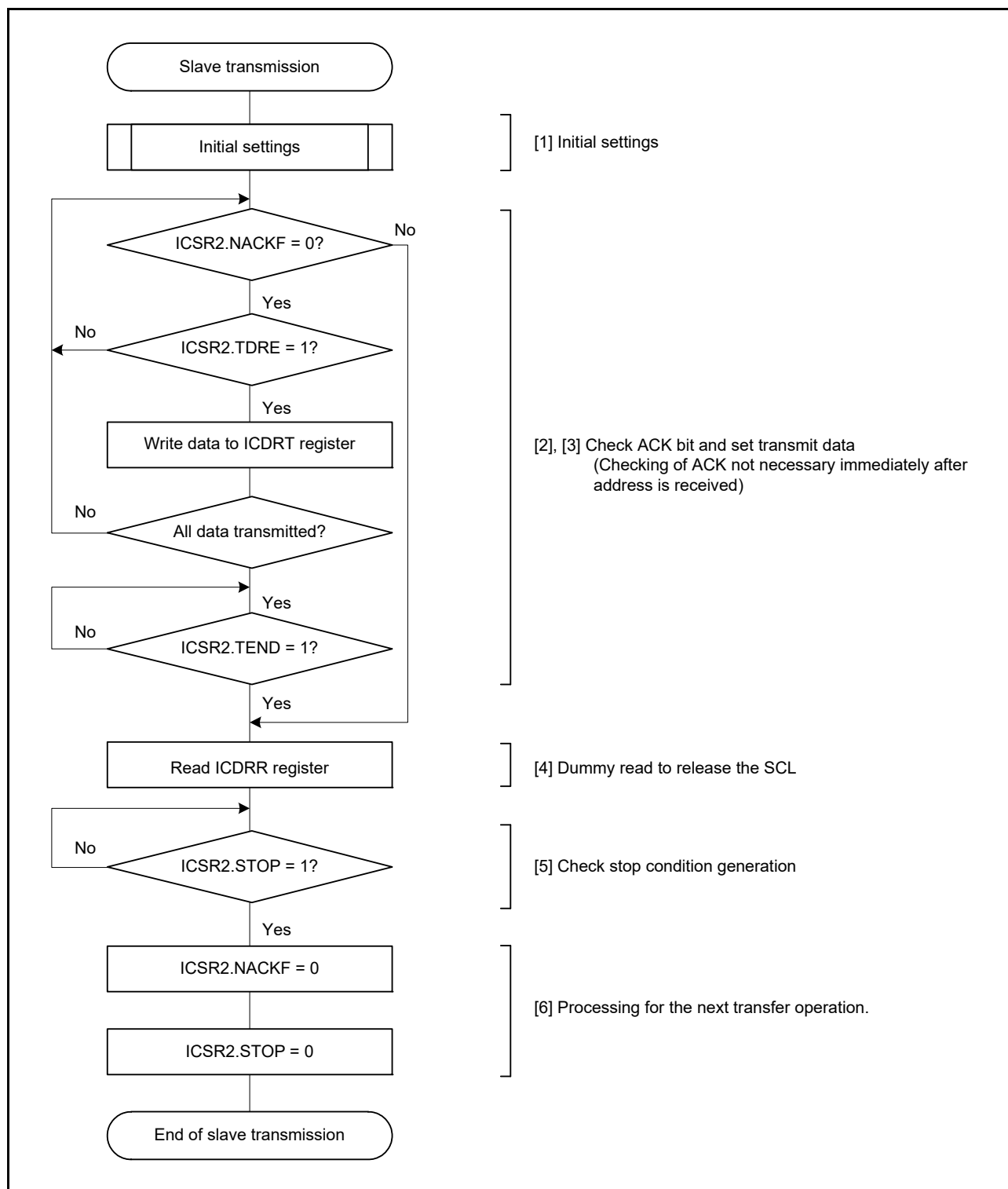


Figure 42.14 Example of Slave Transmission Flowchart

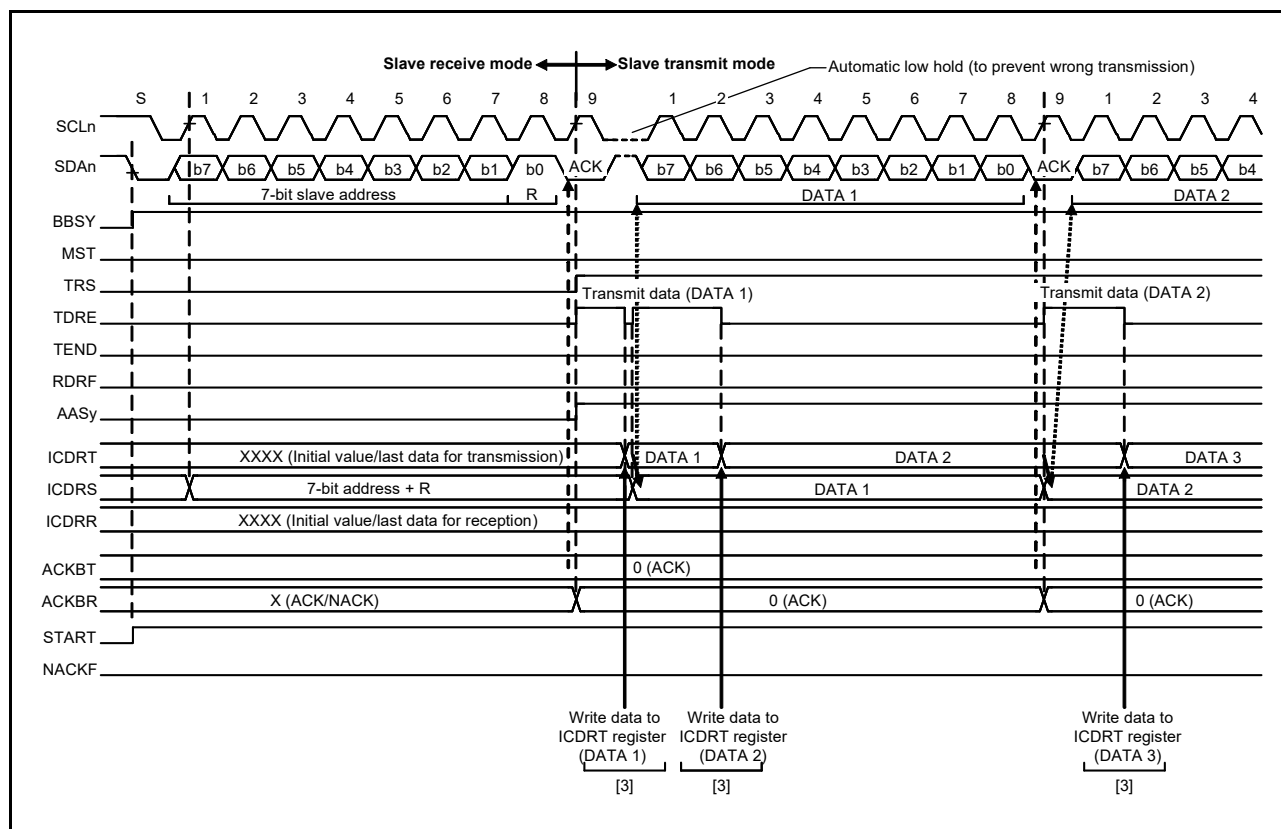


Figure 42.15 Slave Transmit Operation Timing (1) (7-Bit Address Format)

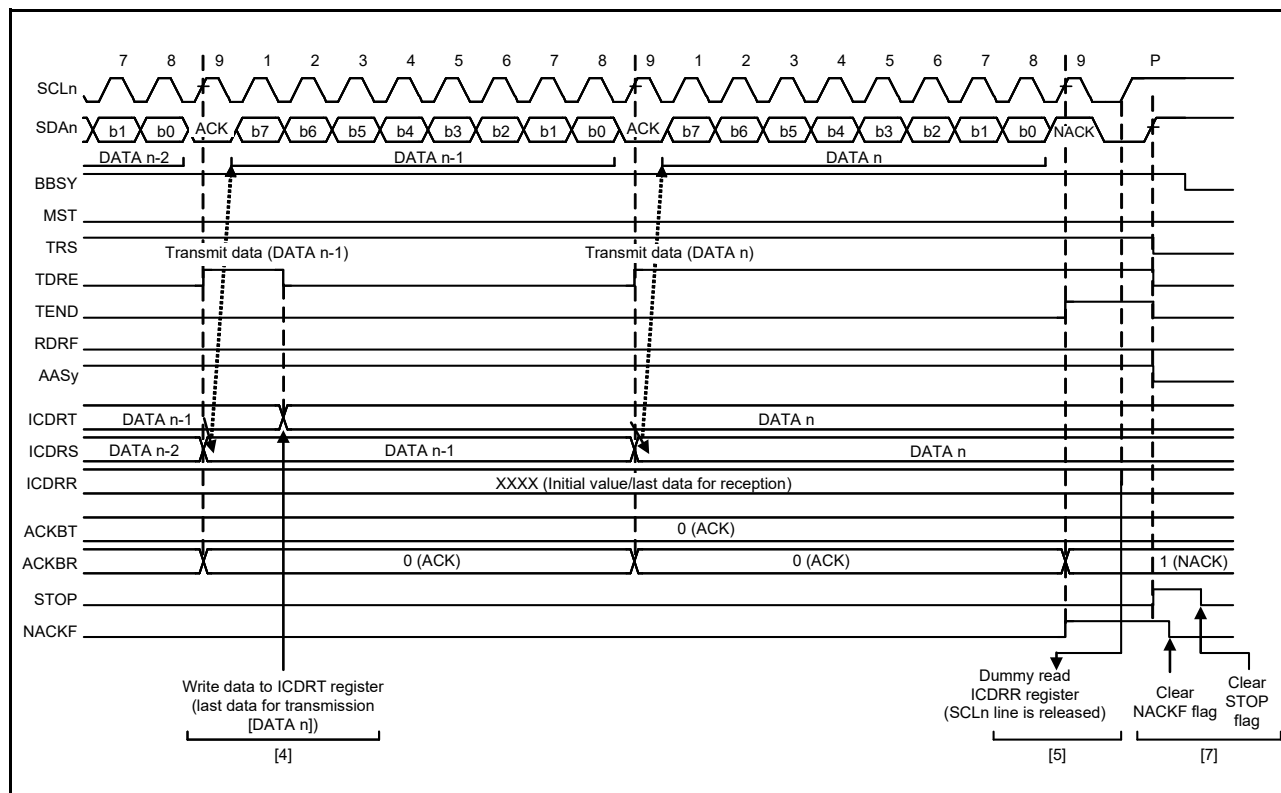


Figure 42.16 Slave Transmit Operation Timing (2)

42.3.6 Slave Receive Operation

In slave receive operation, the master device outputs the SCL and transmit data, and the RIIC returns acknowledgments as a slave device.

Figure 42.17 shows an example of usage of slave reception and Figure 42.18 and Figure 42.19 show the timing of operations in slave reception.

The following describes the procedure and operations for slave reception.

- (1) Initial settings. For details, refer to section 42.3.2, Initial Settings.
After initial settings, the RIIC will stay in the standby state until it receives a slave address that it matches.
- (2) After receiving a matching slave address, the RIIC sets one of the corresponding bits ICSR1.HOA, GCA, and AASy (y = 0 to 2) to 1 on the rising edge of the ninth SCL (the clock signal) and outputs the value set in the ICMR3.ACKBT bit to the acknowledgment bit on the ninth SCL. If the value of the R/W# bit that was also received at this time is 0, the RIIC continues to place itself in slave receive mode and sets the ICSR2.RDRF flag to 1.
- (3) After the ICSR2.STOP flag is confirmed to be 0 and the ICSR2.RDRF flag to be 1, dummy read the ICDRR register (the dummy value consists of the slave address and R/W# bit when the 7-bit address format is selected, or the lower 8 bits when the 10-bit address format is selected).
- (4) When the ICDRR register is read, the RIIC automatically sets the ICSR2.RDRF flag to 0. If reading of the ICDRR register is delayed and a next byte is received while the RDRF flag is still set to 1, the RIIC holds the SCLn line low from one SCL cycle before the timing with which RDRF should be set. In this case, reading the ICDRR register releases the SCLn line from being held low.
When the ICSR2.STOP flag is 1 and the ICSR2.RDRF flag is also 1 or when all the data is completely received, read the ICDRR register.
- (5) Upon detecting the stop condition, the RIIC automatically clears bits ICSR1.HOA, GCA, and AASy (y = 0 to 2) to 0.
- (6) After checking that the ICSR2.STOP flag is 1, set the ICSR2.STOP flag to 0 for the next transfer operation.

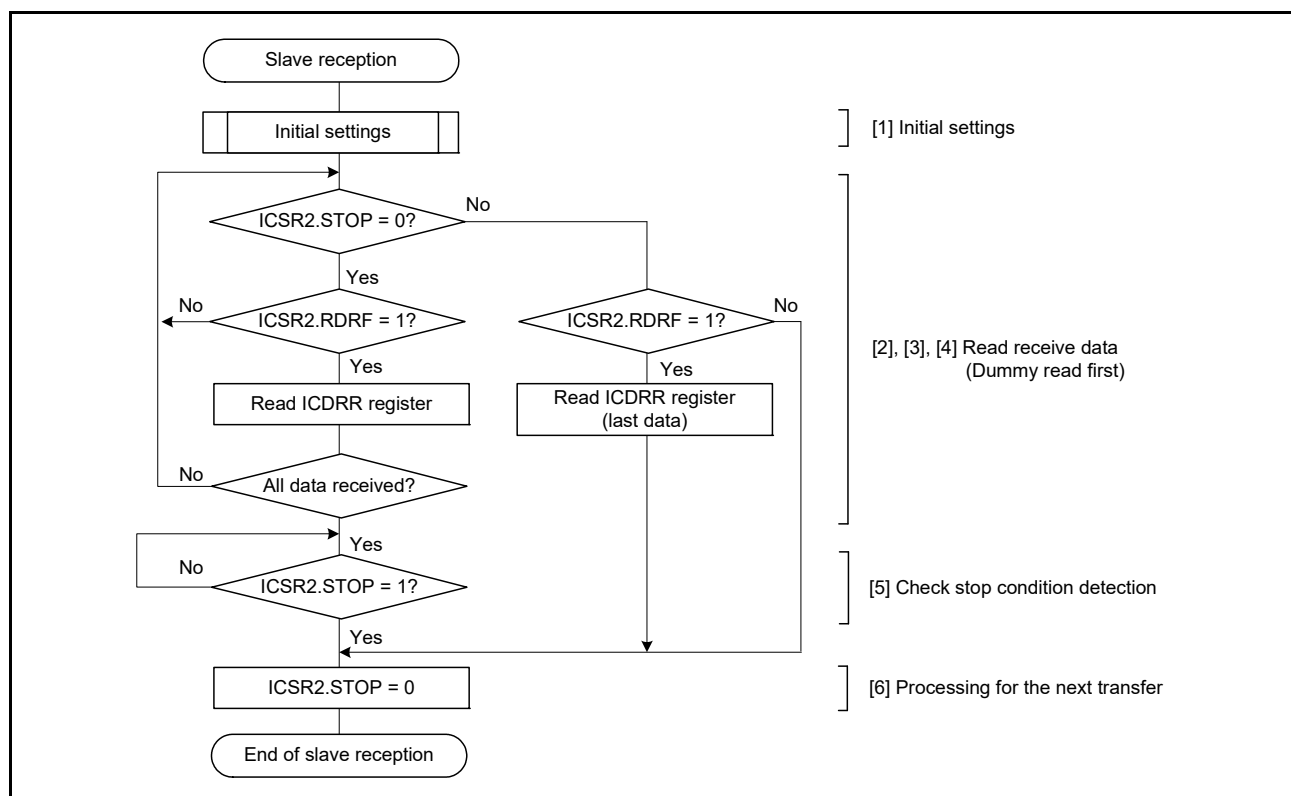


Figure 42.17 Example of Slave Reception Flowchart

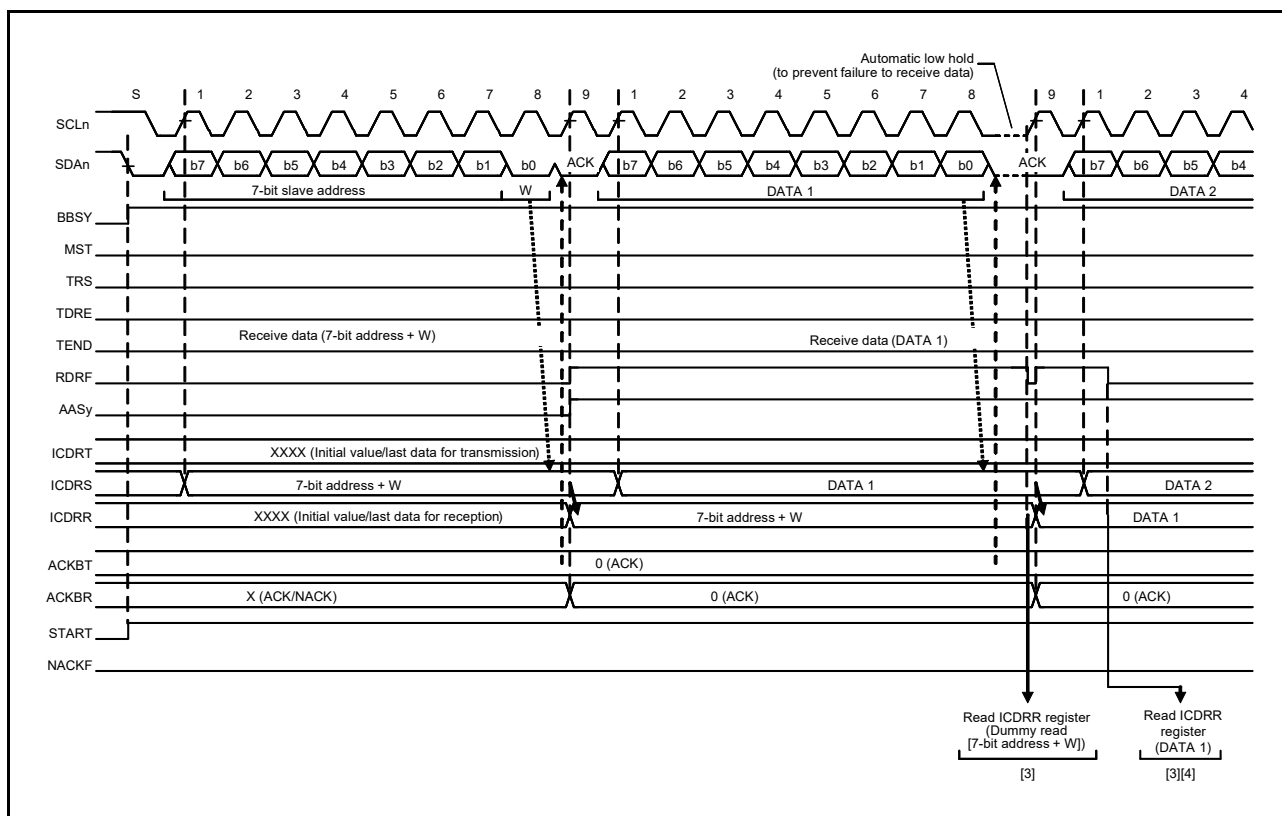


Figure 42.18 Slave Receive Operation Timing (1) (7-Bit Address Format, when RDRFS bit is 0)

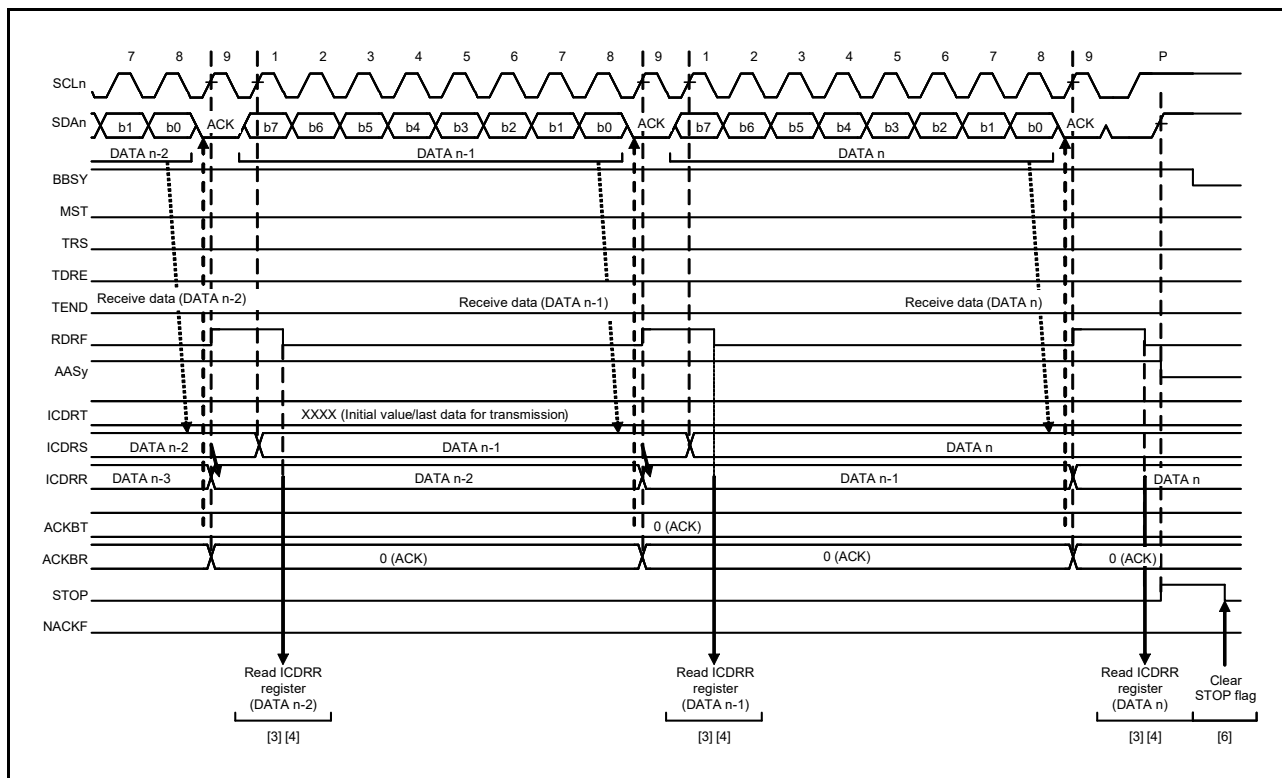


Figure 42.19 Slave Receive Operation Timing (2) (when RDRFS bit is 0)

42.4 SCL Synchronization Circuit

In generation of the SCL, the RIIC starts counting out the value for width at high level specified in the ICBRH register when it detects a rising edge on the SCLn line and drives the SCLn line low once counting of the width at high level is complete. When the RIIC detects the falling edge of the SCLn line, it starts counting out the width at low period specified in the ICBRL register, and then stops driving the SCLn line (releases the line) once counting of the width at low level is complete. The SCL is thus generated.

If multiple master devices are connected to the I²C-bus, a collision of SCL signals may arise due to contention with another master device. In such cases, the master devices have to synchronize their SCL signals. Since synchronization of the SCL signals must be handled bit by bit, the RIIC is equipped with a facility (the SCL synchronization circuit) for obtaining bit-by-bit synchronization of the SCL signals by monitoring the SCLn line while in master mode.

When the RIIC has detected a rising edge on the SCLn line and thus started counting out the width at high level specified in the ICBRH register, and the level on the SCLn line falls because an SCL signal is being generated by another master device, the RIIC stops counting when it detects the falling edge, drives the level on the SCLn line low, and starts counting out the width at low level specified in the ICBRL register. When the RIIC finishes counting out the width at low level, it stops driving the SCLn line low (i.e. releases the line). At this time, if the width at low level of the SCL signal from the other master device is longer than the width at low level set in the RIIC, the width at low level of the SCL signal will be extended. Once the width at low level for the other master device has ended, the SCL signal rises because the SCLn line has been released. When the RIIC finishes outputting the low period of the SCL, the SCLn line is released and the SCL rises. That is, in cases of contention of SCL signals from more than one master, the width at high level of the SCL signal is synchronized with that of the clock having the narrower width, and the width at low level of the SCL signal is synchronized with that of the clock having the broader width. However, such synchronization of the SCL signal is only enabled when the ICFER.SCLE bit is set to 1.

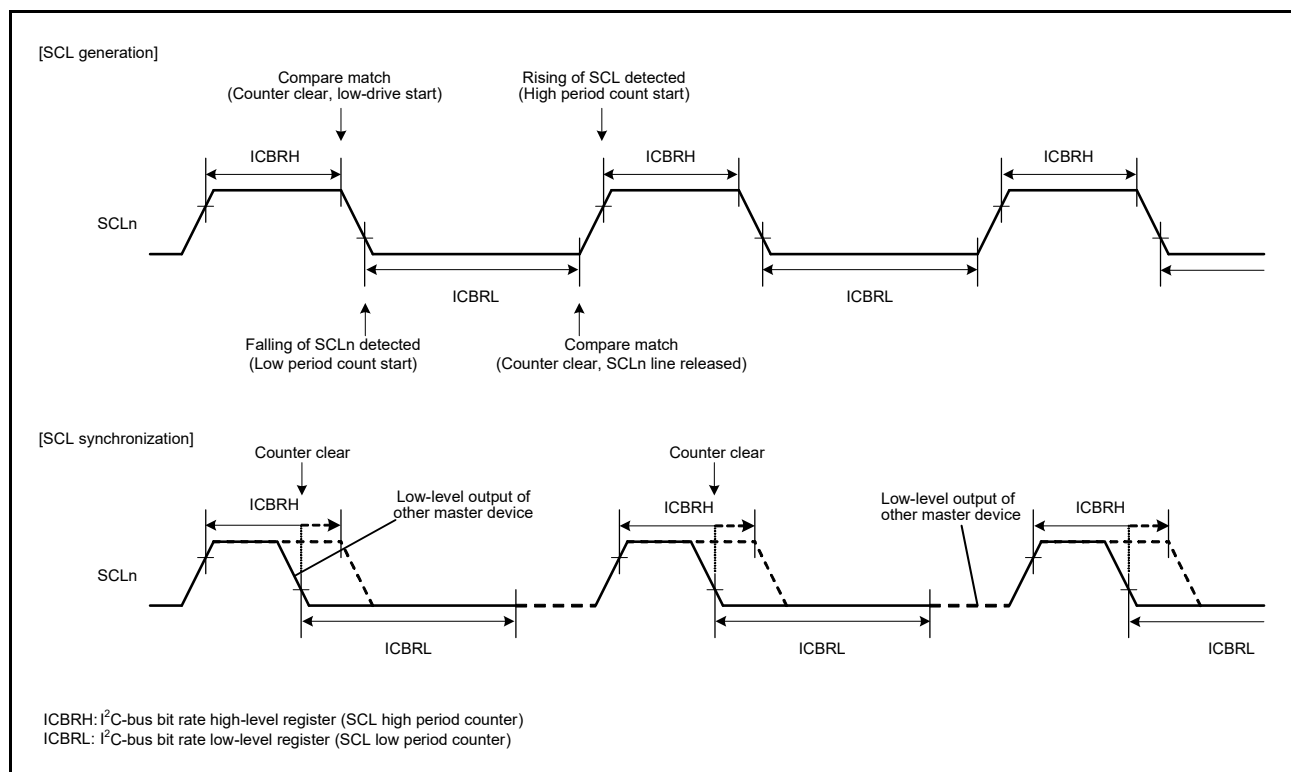


Figure 42.20 Generation and Synchronization of the SCL Signal from the RIIC

42.6 Digital Noise Filters

The states of the SCLn and SDAn pins are conveyed to the internal circuitry through digital noise filters. Figure 42.22 is a block diagram of the digital noise filter.

The on-chip digital noise filter of each RIIC consists of four flip-flop circuit stages connected in series and a match detection circuit.

The number of effective stages in the digital noise filter is selected by the ICMR3.NF[1:0] bits. The selected number of effective stages determines the noise-filtering capability as a period from one to four cycles of IIC ϕ .

The input signal to the SCLn pin (or SDAn pin) is sampled on falling edges of the IIC ϕ signal. When the input signal level matches the output level for the number of effective flip-flop circuit stages selected by the ICMR3.NF[1:0] bits, the signal level is conveyed to the subsequent stages. If the signal levels do not match, the previous value is retained.

If the ratio between the frequency of the internal operating clock (PCLK) and the transfer rate is relatively small (e.g. data transfer at 400 kbps with PCLK = 4 MHz), the characteristics of the digital noise filter may lead to the elimination of needed signals as noise.

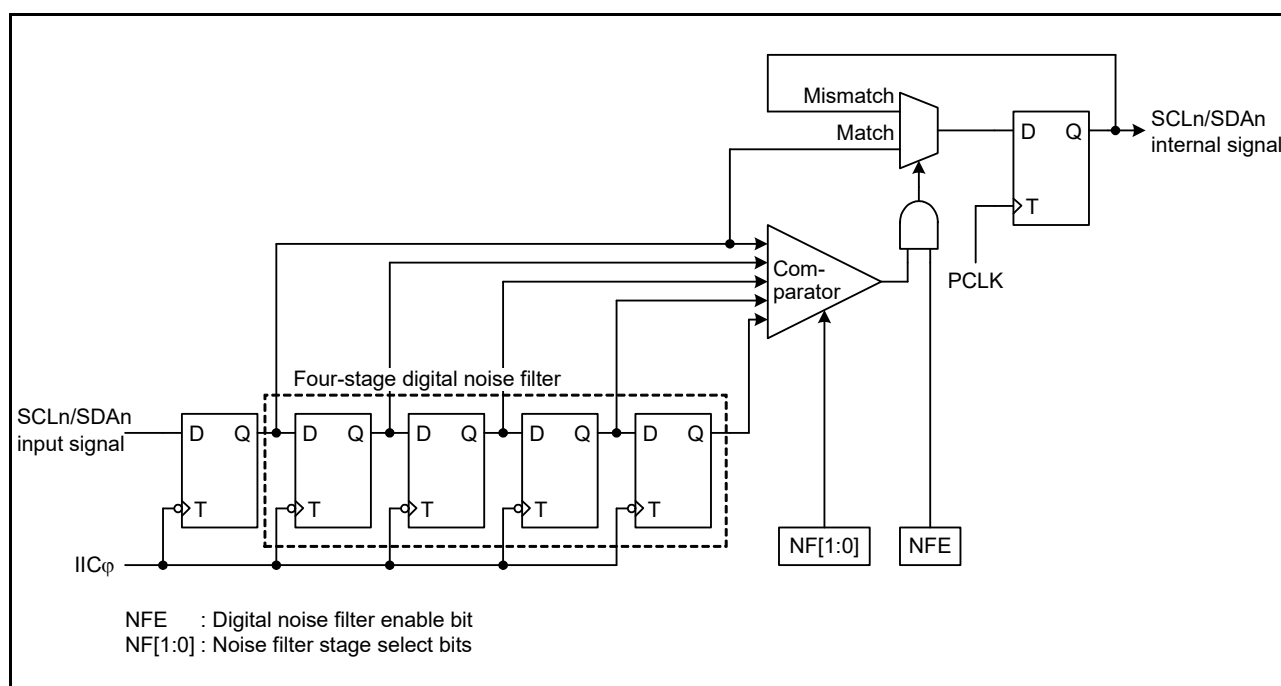


Figure 42.22 Block Diagram of the Digital Noise Filter

42.7 Address Match Detection

The RIIC can set three unique slave addresses in addition to the general call address and host address, and also can set 7-bit or 10-bit slave addresses.

42.7.1 Slave-Address Match Detection

The RIIC can set three unique slave addresses, and has a slave address detection function for each unique slave address. When the ICSER.SARyE bit ($y = 0$ to 2) is set to 1, the slave addresses set in registers SARUy and SARLy ($y = 0$ to 2) can be detected.

When the RIIC detects a match with its set slave address, the corresponding ICSR1.AASy flag ($y = 0$ to 2) is set to 1 on the rising edge of the ninth SCL, and the ICSR2.RDRF flag or the ICSR2.TDRE flag is set to 1 according to the level of the R/W# bit. This causes a receive data full interrupt (RXI) or transmit data empty interrupt (TXI) to be generated. The AASy flag is used to identify which slave address has been specified.

Figure 42.23 to Figure 42.25 show the AASy flag set timing in three cases.

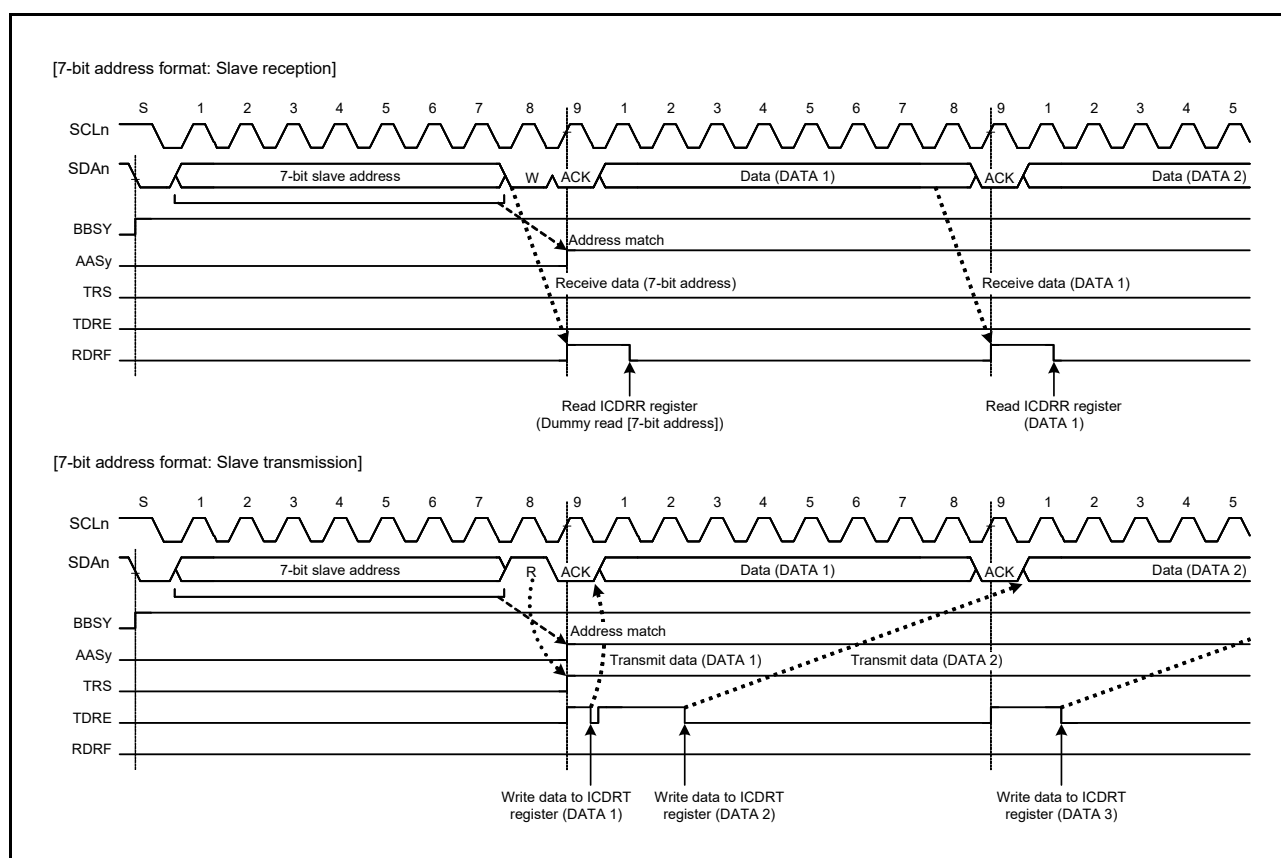


Figure 42.23 AASy Flag Set Timing with 7-Bit Address Format Selected

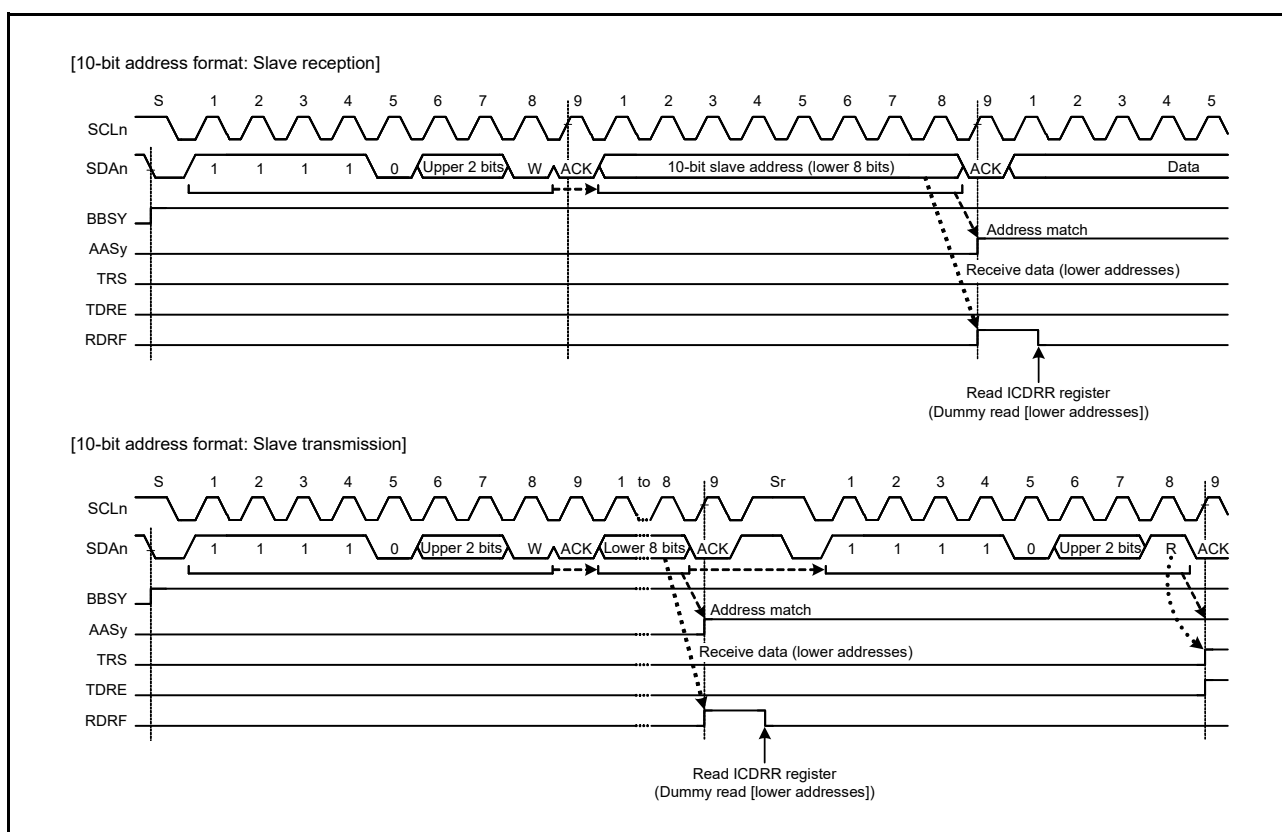


Figure 42.24 AASy Flag Set Timing with 10-Bit Address Format Selected

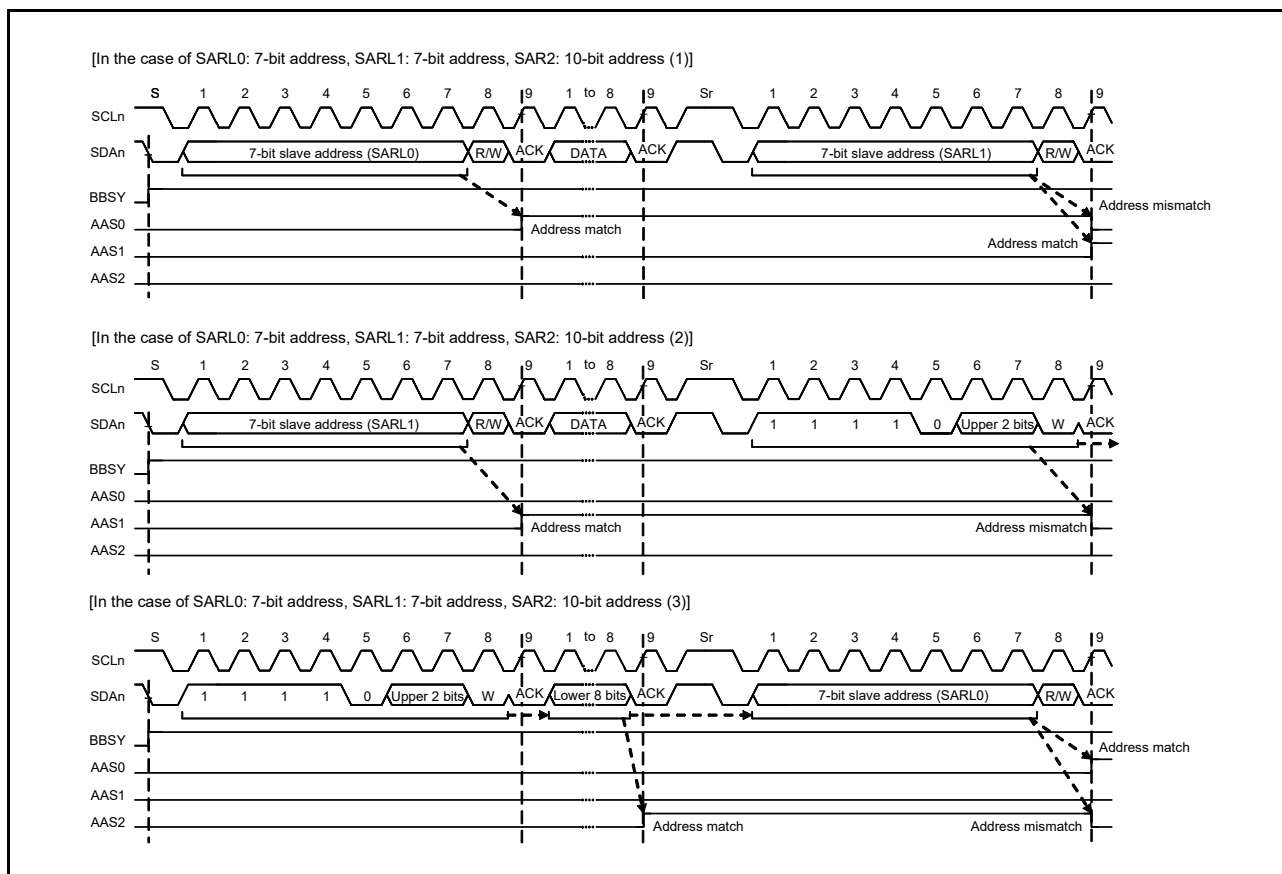


Figure 42.25 AASy Flag Set/Clear Timing with 7-Bit/10-Bit Address Formats Mixed

42.7.2 Detection of the General Call Address

The RIIC also has a facility for detecting the general call address (0000 000b + 0 (write)). This is enabled by setting the ICSER.GCAE bit to 1.

If the address following a start or restart condition is 0000 000b + 1 (read) (start byte), the RIIC recognizes this as the address of a slave device with an “all-zero” address but not as the general call address.

When the RIIC detects the general call address, both the ICSR1.GCA flag and the ICSR2.RDRF flag are set to 1 on the rising edge of the ninth SCL. This leads to the generation of a receive data full interrupt (RXI). The value of the GCA flag can be confirmed to recognize that the general call address has been transmitted.

Operation after detection of the general call address is the same as normal slave receive operation.

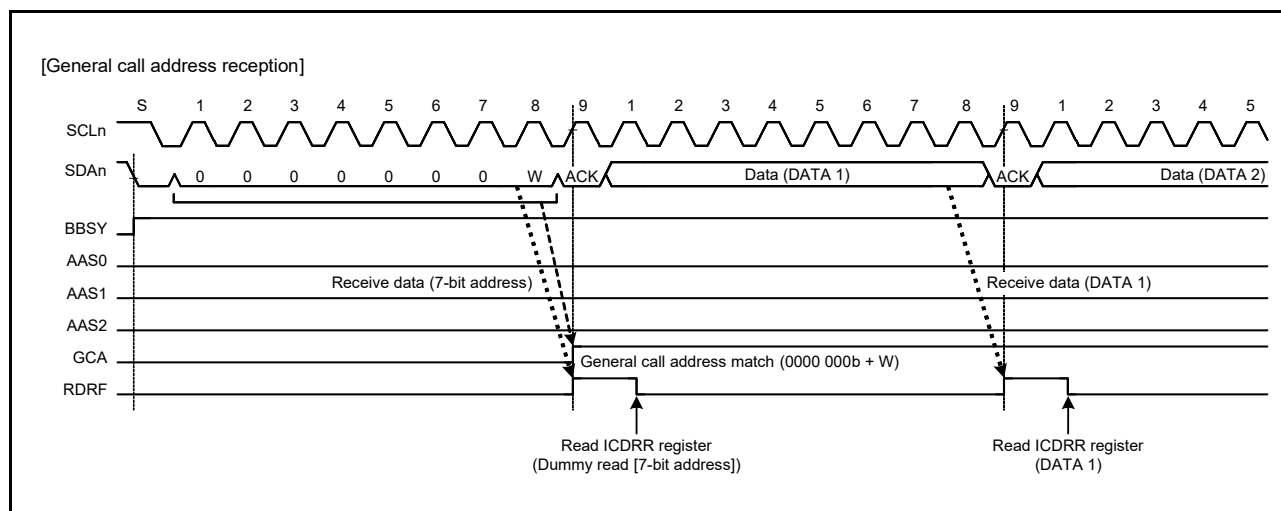


Figure 42.26 Timing of GCA Flag Setting during Reception of General Call Address

42.7.3 Device-ID Address Detection

The RIIC module has a function to detect device-ID addresses complying with the I²C-bus specification. When the RIIC receives 1111 100b as the first seven bits of the first byte following a start condition or a restart condition while the ICSER.DIDE bit is 1, the RIIC recognizes the address as a device-ID address, sets the ICSR1.DID flag to 1 on the rising edge of the ninth SCL when the following R/W# bit is 0, and then compares the second and following bytes with its own slave address. If the received address matches the value in the slave address register, the RIIC sets the corresponding ICSR1.AASy flag (y = 0 to 2) to 1.

After that, when the first byte received after a start or restart condition is generated matches the device ID address (1111 100b) again and the following R/W# bit is 1, the RIIC does not compare the second and subsequent bytes and sets the ICSR2.TDRE flag to 1.

In the device-ID address detection function, the RIIC sets the DID flag to 0 if a match with the RIIC's own slave address is not obtained or a match with the device ID address is not obtained after a match with the RIIC's own slave address and the detection of a restart condition. If the first byte after detection of a start or restart condition matches the device ID address (1111 100b) and the R/W# bit is 0, the RIIC sets the DID flag to 1 and compares the second and subsequent bytes with the RIIC's slave address. If the R/W# bit is 1, the DID flag holds the previous value and the RIIC does not compare the second and subsequent bytes. Therefore, the reception of a device-ID address can be checked by reading the DID flag after confirming that TDRE flag is 1.

Furthermore, prepare the device-ID fields (3 bytes: 12 bits indicating the manufacturer + 9 bits identifying the part + 3 bits indicating the revision) that must be sent to the host after reception of a continuous device-ID field as normal data for transmission. For details of the information that must be included in device-ID fields, contact NXP Semiconductors.

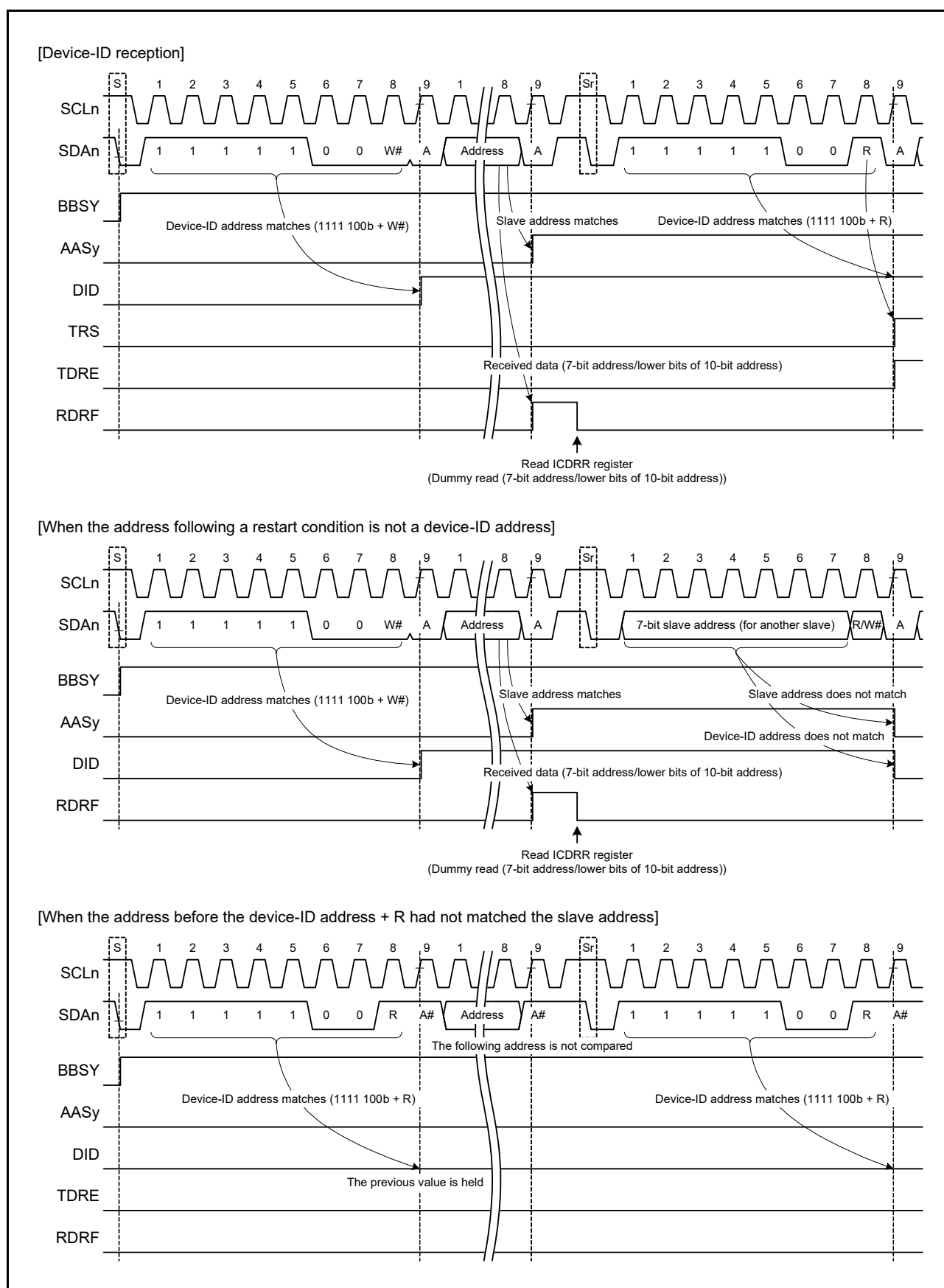


Figure 42.27 Set/Clear Timing of the AASy and DID Flags during Reception of Device-ID Address

42.7.4 Host Address Detection

The RIIC has a function to detect the host address while the SMBus is operating. When the ICSER.HOAE bit is set to 1 while the ICMR3.SMBS bit is 1, the RIIC can detect the host address (0001 000b) in slave receive mode (bits MST and TRS in the ICCR2 register are 00b).

When the RIIC detects the host address, the ICSR1.HOA flag is set to 1 at the rising edge of the ninth SCL, and at the same time, the ICSR2.RDRF flag is set to 1 when the R/W# bit is 0 (Wr bit). This causes a receive data full interrupt (RXI) to be generated. The HOA flag is used to recognize that the host address was sent from the smart battery or other devices.

If the bit following the host address (0001 000b) is an Rd bit (R/W# bit is 1), the RIIC can also detect the host address. After the host address is detected, the RIIC operates in the same manner as normal slave operation.

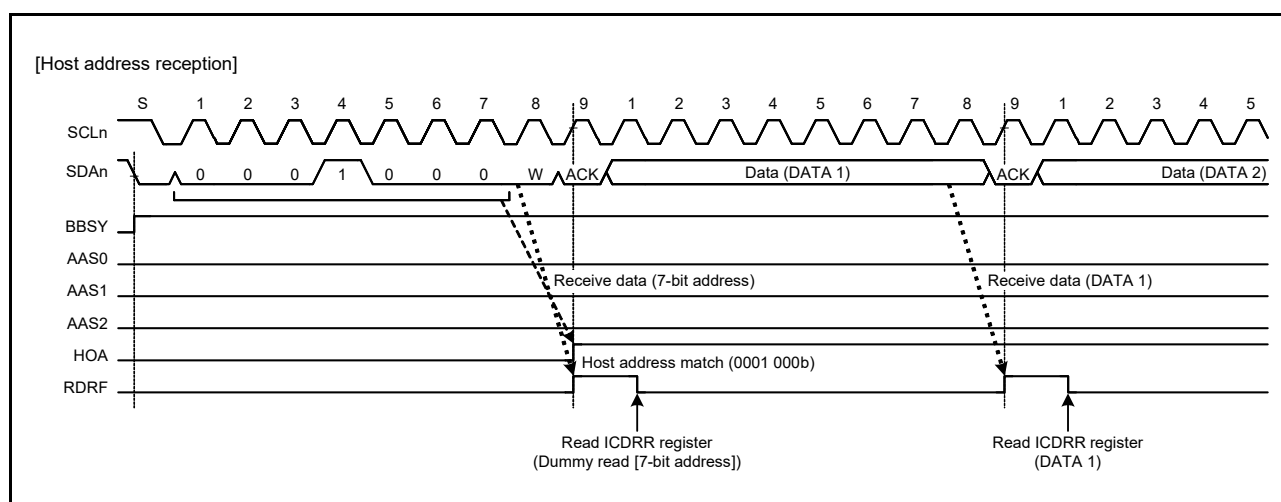


Figure 42.28 HOA Flag Set Timing during Reception of Host Address

42.8 Automatic Low-Hold Function for SCL

42.8.1 Function to Prevent Wrong Transmission of Transmit Data

If the shift register (ICDRS) is empty when data have not been written to the I²C-bus transmit data register (ICDRT) with the RIIC in transmission mode (ICCR2.TRS bit is 1), the SCLn line is automatically held low over the intervals shown below. This low period is extended until data for transmission have been written, which prevents the unintended transmission of erroneous data.

Master transmit mode

- Low period after a start condition or restart condition is generated
- Low period between the ninth clock pulse of one transfer and the first clock pulse of the next

Slave transmit mode

- Low period between the ninth clock pulse of one transfer and the first clock pulse of the next

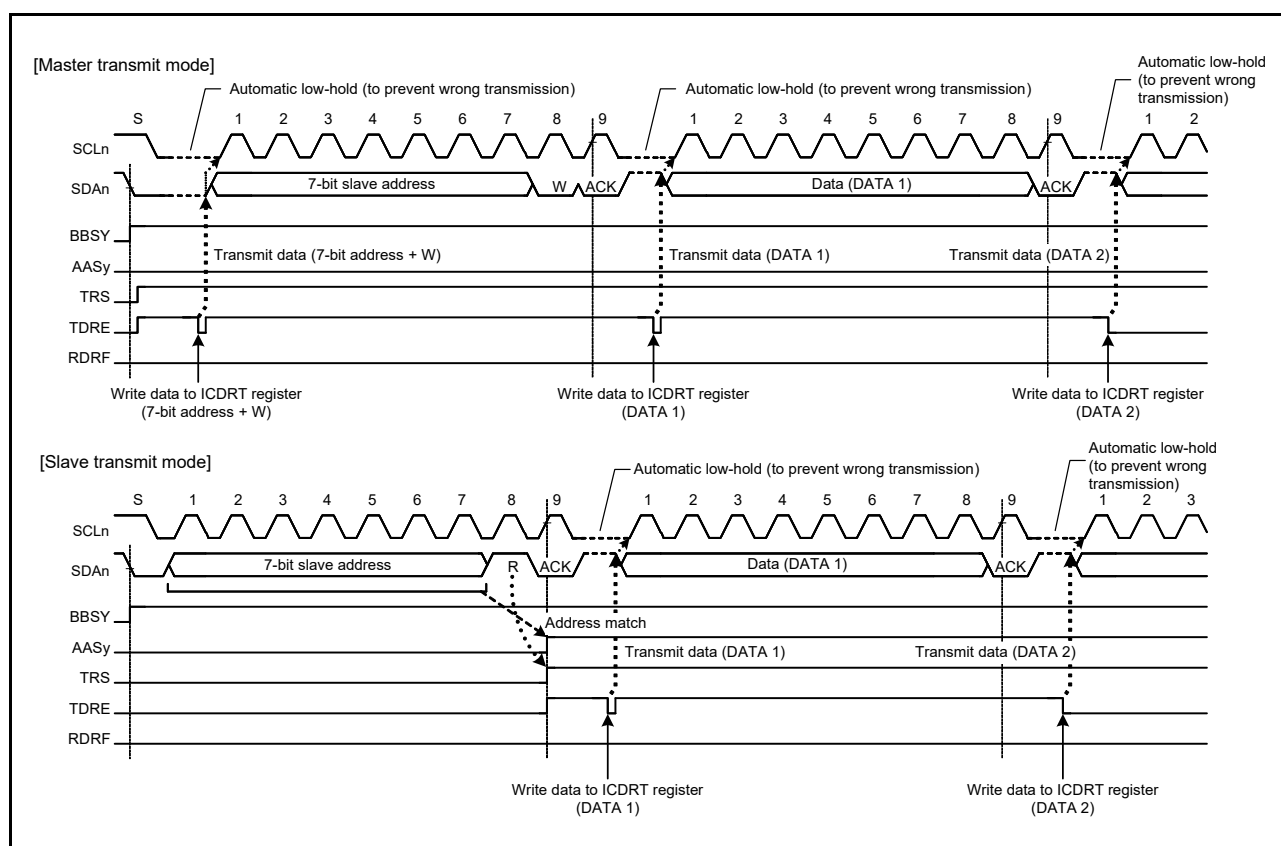


Figure 42.29 Automatic Low-Hold Operation in Transmit Mode

42.8.2 NACK Reception Transfer Suspension Function

The RIIC has a function to suspend transfer operation when NACK is received in transmit mode (ICCR2.TRS bit is 1). This function is enabled when the ICFER.NACKE bit is set to 1 (transfer suspension enabled). If the next transmit data has already been written (ICSR2.TDRE flag is 0) when NACK is received, next data transmission at the falling edge of the ninth SCL is automatically suspended. This prevents the SDA_n line output level from being held low when the MSB of the next transmit data is 0.

If the data transmission is suspended (ICSR2.NACKF flag is 1) by this function, the following data transmission and data reception are not started. To resume data transfer, set the NACKF flag to 0. In master transmit mode, restart data transfer by setting the NACKF flag to 0 after generating a restart condition, or restart data transfer from a start condition after generating a stop condition then setting the NACKF flag to 0.

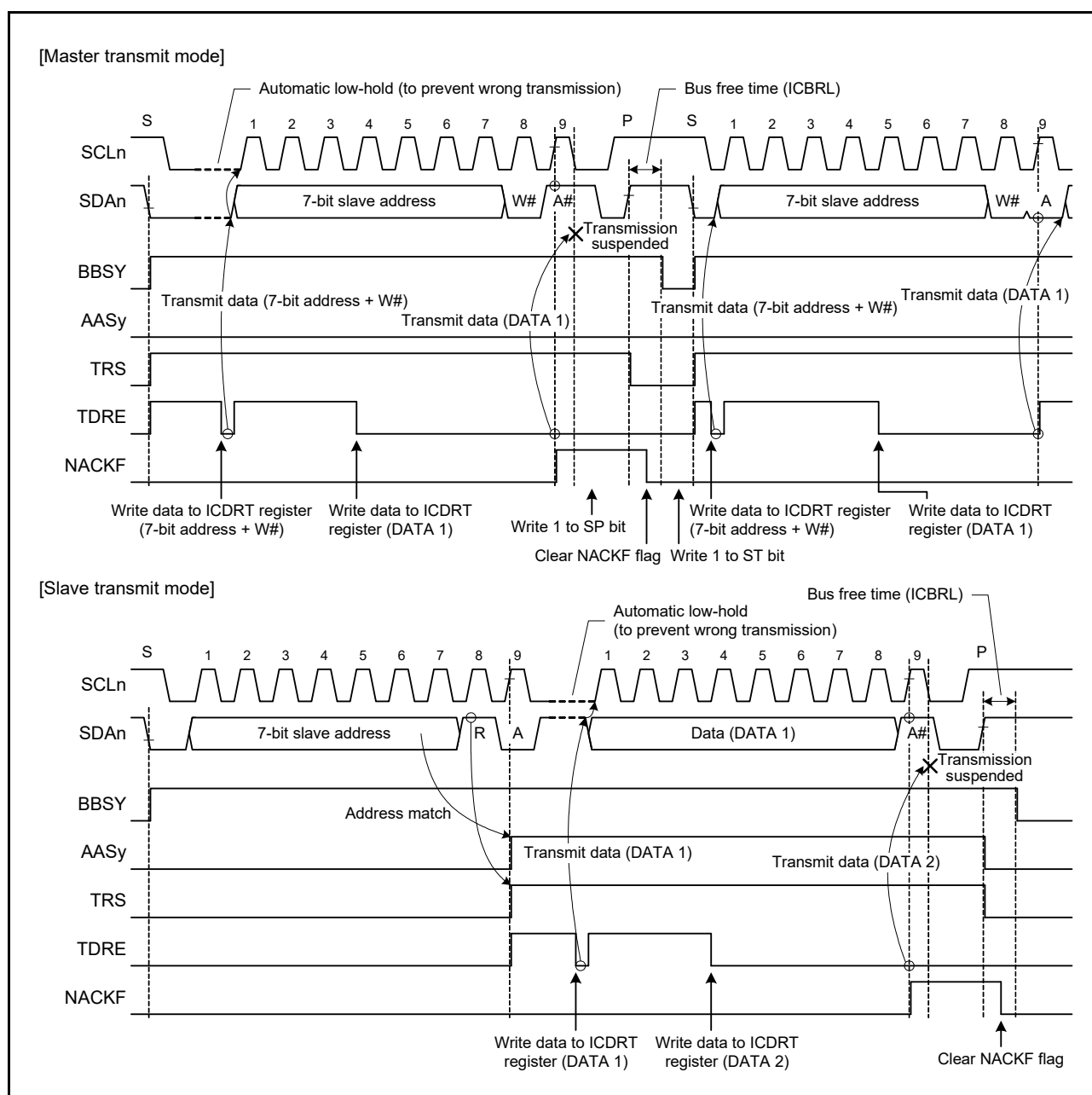


Figure 42.30 Suspension of Data Transmission When NACK is Received (NACKE = 1)

42.8.3 Function to Prevent Failure to Receive Data

If response processing is delayed when receive data (ICDRR) read is delayed for a period of one transfer byte or more with receive data full (ICSR2.RDRF flag is 1) in receive mode (ICCR2.TRS bit is 0), the RIIC holds the SCLn line low automatically immediately before the next data is received to prevent failure to receive data.

This function to prevent failure to receive data using the automatic low-hold function is also enabled even if the read processing of the final receive data is delayed and, in the meantime, the RIIC's own slave address is designated after a stop condition is generated. This function does not disturb other communication because the RIIC does not hold the SCLn line low when a mismatch with its own slave address occurs after a stop condition is generated.

Sections in which the SCLn line is held low can be selected with a combination of the WAIT and RDRFS bits in the ICMR3 register.

(1) 1-Byte Receive Operation and Automatic Low-Hold Function Using the WAIT Bit

When the ICMR3.WAIT bit is set to 1, the RIIC performs 1-byte receive operation using the WAIT bit function.

Furthermore, when the ICMR3.RDRFS bit is 0, the RIIC automatically sends the ICMR3.ACKBT bit value for the acknowledgment bit in the period from the falling edge of the eighth SCL to the falling edge of the ninth SCL, and automatically holds the SCLn line low at the falling edge of the ninth SCL using the WAIT bit function. This low-hold is released by reading data from the ICDRR register, which enables byte-wise receive operation.

The WAIT bit function is enabled for receive bytes after a match with the RIIC's own slave address (including the general call address and host address) is obtained in master receive mode or slave receive mode.

(2) 1-Byte Receive Operation (ACK/NACK Transmission Control) and Automatic Low-Hold Function Using the RDRFS Bit

When the ICMR3.RDRFS bit is set to 1, the RIIC performs 1-byte receive operation using the RDRFS bit function.

When the RDRFS bit is set to 1, the ICSR2.RDRF flag (receive data full) is set to 1 at the rising edge of the eighth SCL, and the SCLn line is automatically held low at the falling edge of the eighth SCL. This low-hold is released by writing a value to the ICMR3.ACKBT bit, but cannot be released by reading data from the ICDRR register, which enables receive operation by the ACK/NACK transmission control according to the data received in byte units.

The RDRFS bit function is enabled for receive bytes after a match with the RIIC's own slave address (including the general call address and host address) is obtained in master receive mode or slave receive mode.

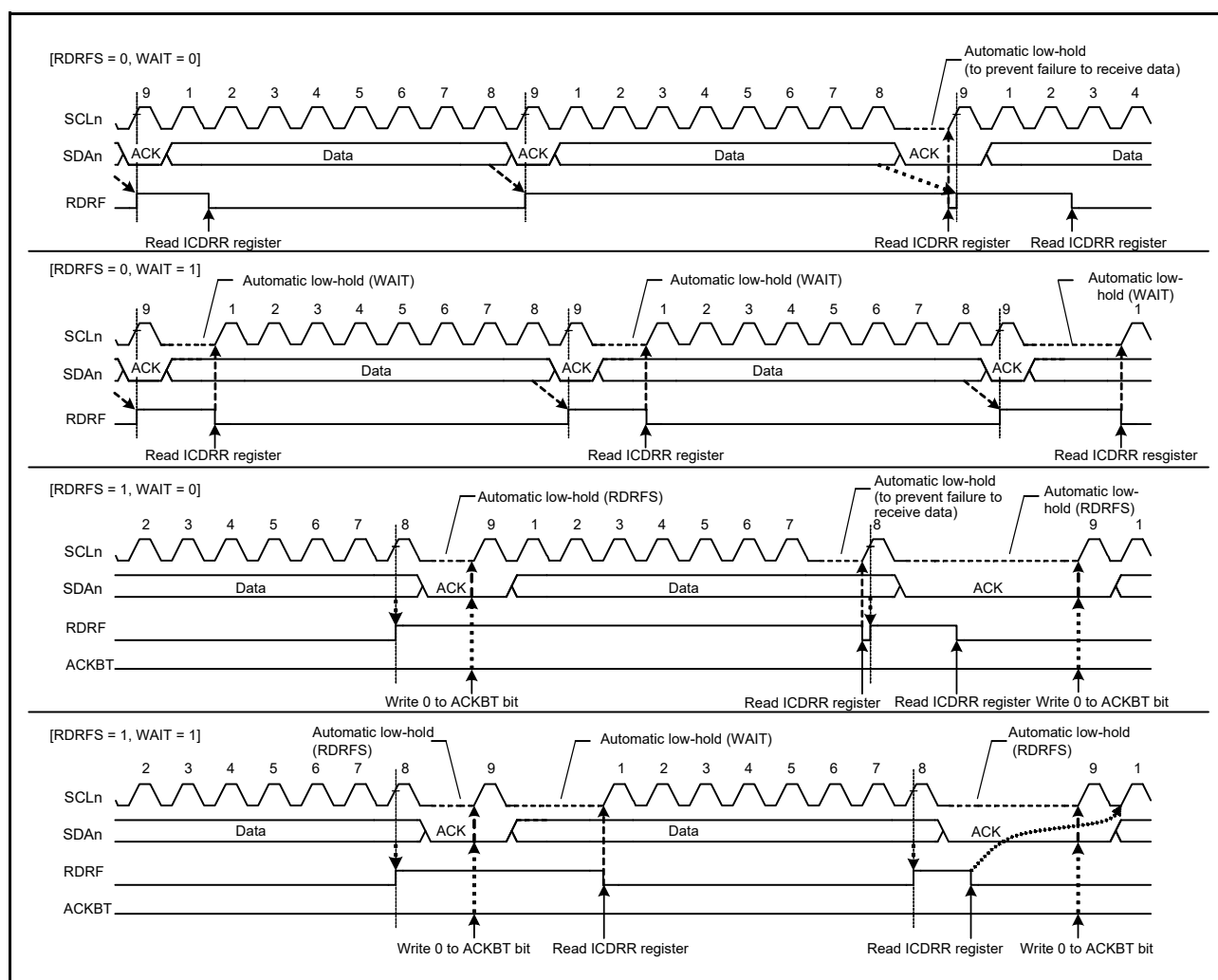


Figure 42.31 Automatic Low-Hold Operation in Receive Mode (Using RDRFS and WAIT Bits)

42.9 Arbitration-Lost Detection Functions

In addition to the normal arbitration-lost detection function defined by the I²C-bus specification, the RIIC has functions to prevent double-generation of a start condition, to detect arbitration-lost during transmission of NACK, and to detect arbitration-lost in slave transmit mode.

42.9.1 Master Arbitration-Lost Detection (MALE Bit)

The RIIC drives the SDA_n line low to generate a start condition. However, if the SDA_n line has already been driven low by another master device generating a start condition, the RIIC causes arbitration to be lost, so priority is given to transfer by the other master device. Similarly, if the ICCR2.ST bit is set to 1 while the ICCR2.BBSY flag is 1 (bus busy state), arbitration is lost, so priority is given to transfer by the other master device. No start condition is generated in this case. When a start condition is generated successfully, if the data for transmission including the address bits (i.e. the internal SDA output level) and the level on the SDA_n line do not match (the high output as the internal SDA output; i.e. the SDA_n pin is in the high-impedance state) and the low is detected on the SDA_n line, the RIIC loses the arbitration. After a master arbitration-lost is generated, the RIIC immediately enters slave receive mode. If a slave address (including the general call address) matches its own address at this time, the RIIC continues in slave operation.

A master arbitration-lost is detected when the following conditions are met while the ICFER.MALE bit is 1 (master arbitration-lost detection enabled).

Conditions for detecting master arbitration-lost

- Non-matching of the internal level for output on SDA and the level on the SDA_n line after a start condition was generated by setting the ICCR2.ST bit to 1 while the ICCR2.BBSY flag was set to 0 (erroneous generation of a start condition)
- Setting of the ICCR2.ST bit to 1 while the BBSY flag is set to 1 (start condition double-generation error)
- When the transmit data excluding acknowledgment bit (internal SDA output level) does not match the level on the SDA_n line in master transmit mode (bits MST and TRS in the ICCR2 register = 11b)

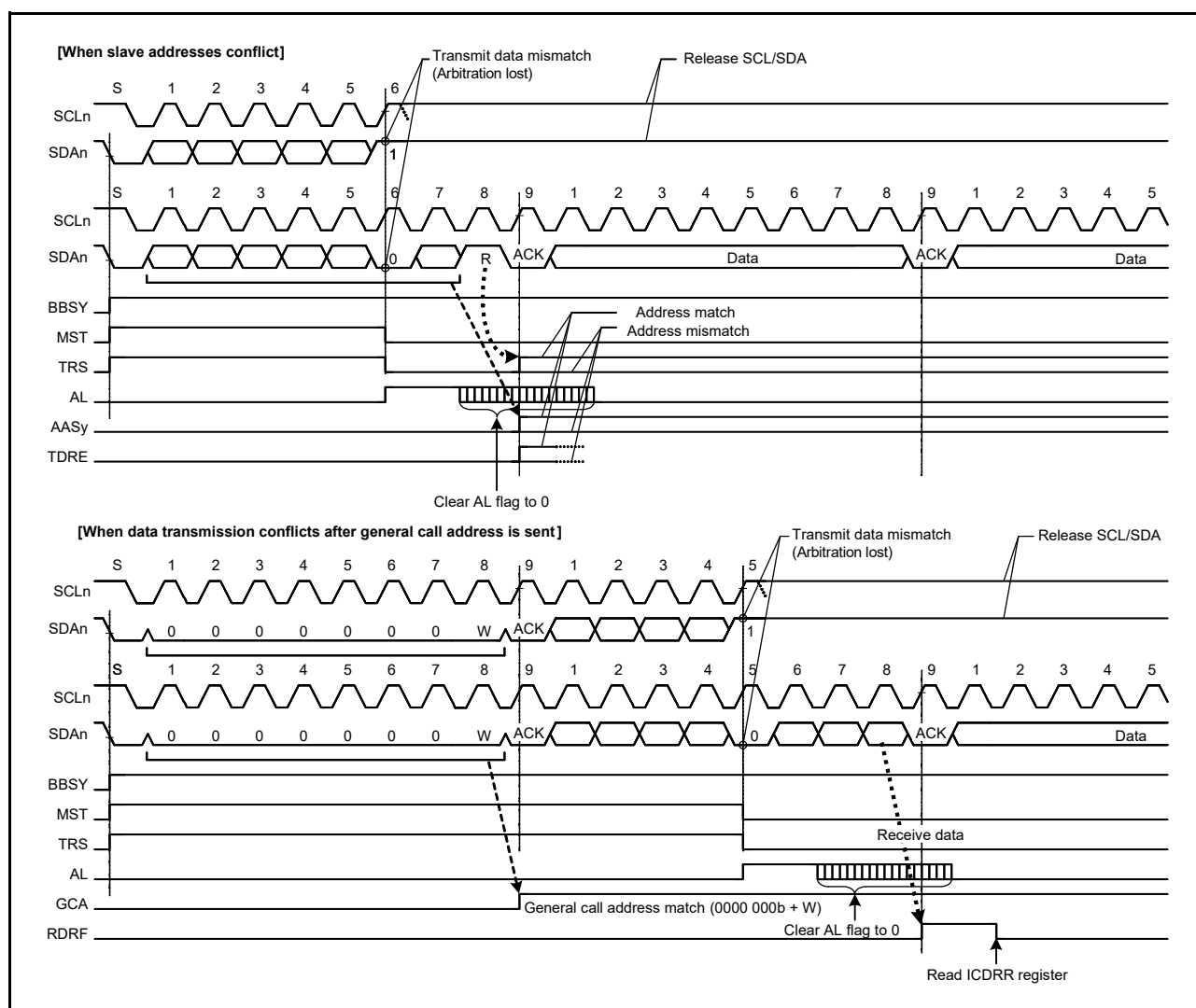


Figure 42.32 Examples of Master Arbitration-Lost Detection (MALE = 1)

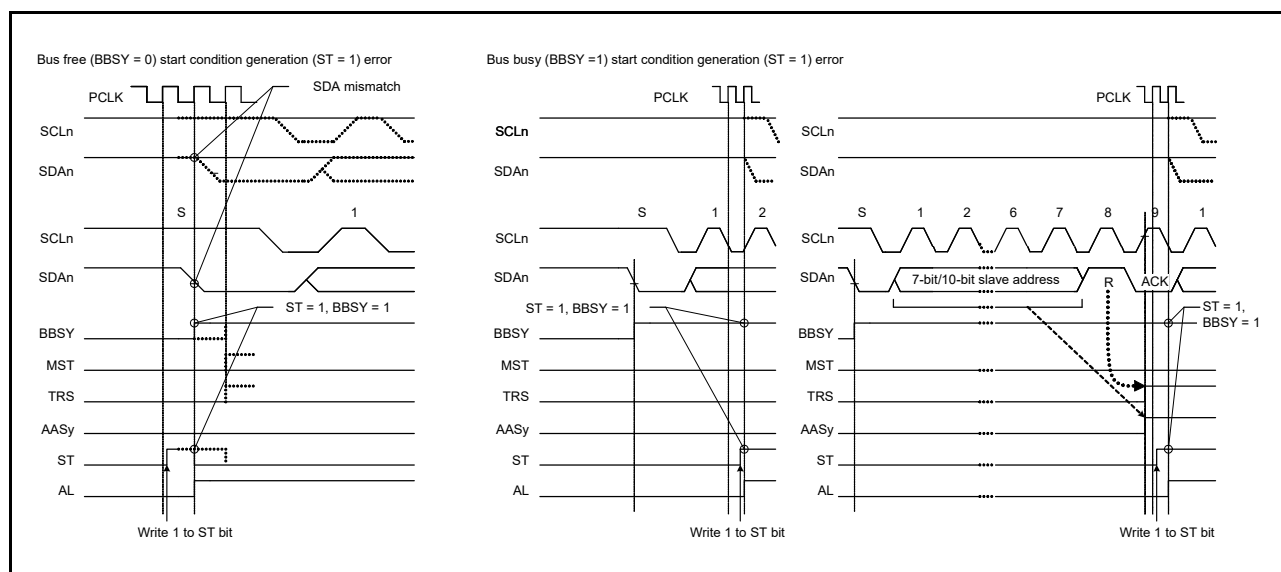


Figure 42.33 Arbitration-Lost When a Start Condition is Generated (MALE = 1)

42.9.2 NACK Transmission Arbitration-Lost Detection Function (NALE Bit)

The RIIC has a function to cause arbitration to be lost if the internal SDA output level does not match the level on the SDA_n line (the high output as the internal SDA output; i.e. the SDA_n pin is in the high-impedance state) and the low is detected on the SDA_n line during transmission of NACK in receive mode. Arbitration is lost due to a conflict of NACK transmission and ACK transmission when two or more master devices receive data from the same slave device simultaneously in a multi-master system. Such conflict occurs when multiple master devices send/receive the same information through a single slave device. Figure 42.34 shows an example of NACK transmission arbitration-lost detection.

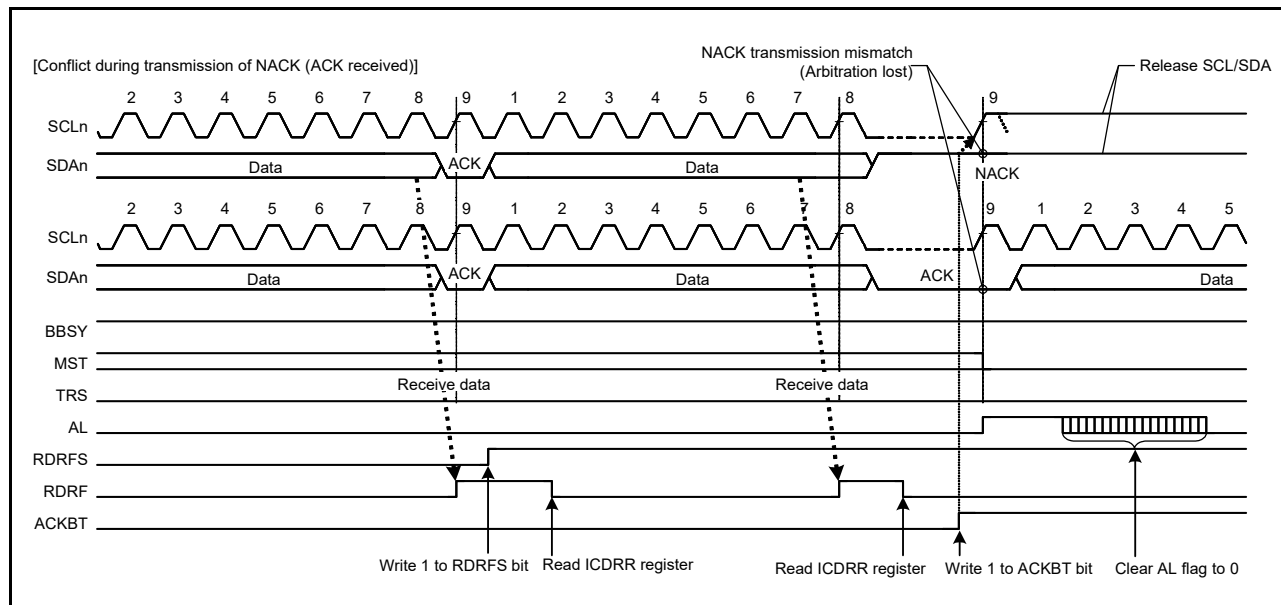


Figure 42.34 Example of NACK transmission arbitration-lost Detection (NALE = 1)

The following explains arbitration-lost detection using an example where two master devices (master A and master B) and a single slave device are connected through the bus. In this example, master A receives two bytes of data from the slave device, and master B receives four bytes of data from the slave device.

If master A and master B access the slave device simultaneously, because the slave address is identical, arbitration is not lost in both master A and master B during access to the slave device. Therefore, both master A and master B recognize that they have obtained the bus mastership and operate as such. Here, master A sends NACK when it has received second byte of data from the slave device. Meanwhile, master B sends ACK because it has not received necessary four bytes of data. At this time, the NACK transmission from master A and the ACK transmission from master B conflict. In general, if a conflict like this occurs, master A cannot detect ACK transmitted by master B and generates a stop condition. Therefore, the generation of the stop condition conflicts with the SCL output of master B, which disturbs communication.

When the RIIC receives ACK during transmission of NACK, it detects a defeat in conflict with other master devices and causes arbitration to be lost.

If a NACK transmission arbitration-lost occurs, the RIIC immediately cancels the slave address matched state and enters slave receive mode. This prevents a stop condition from being generated, preventing a communication failure on the bus. Also, in the ARP command processing of SMBus, it is possible to omit surplus processing (FFh transmission processing) after NACK transmission when the UDID (Unique Device Identifier) of “Assign Address” does not match and after the NACK transmission of the “Get UDID (General)” after the “Assign Address” is confirmed.

The RIIC detects NACK transmission arbitration-lost when the following condition is met with the ICFER.NALE bit set to 1 (NACK transmission arbitration-lost is enabled).

Condition for detecting NACK transmission arbitration-lost

- When the internal SDA output level does not match the SDA_n line (ACK is received) during transmission of NACK (ICMR3.ACKBT bit = 1)

42.9.3 Slave Arbitration-Lost Detection (SALE Bit)

The RIIC has a function to cause arbitration to be lost if the data for transmission (i.e. the internal SDA output level) and the level on the SDA_n line do not match (the high output as the internal SDA output; i.e. the SDA_n pin is in the high-impedance state and the low is detected on the SDA_n line in slave transmit mode). The slave arbitration-lost detection function is mainly used when transmitting a UDID (Unique Device Identifier) of SMBus.

When the slave arbitration-lost occurs, the RIIC is immediately released from the slave address matched state and enters slave receive mode. This function can detect a data conflict during UDID transmission of SMBus and omit surplus processing (FFh transmission processing) after the data conflict.

The RIIC detects slave arbitration-lost when the following condition is met with the ICFER.SALE bit set to 1 (slave arbitration-lost detection enabled).

Condition for detecting slave arbitration-lost

- When transmit data excluding acknowledgment bit (internal SDA output level) does not match the SDA_n line in slave transmit mode (bits MST and TRS in the ICCR2 register are 01b)

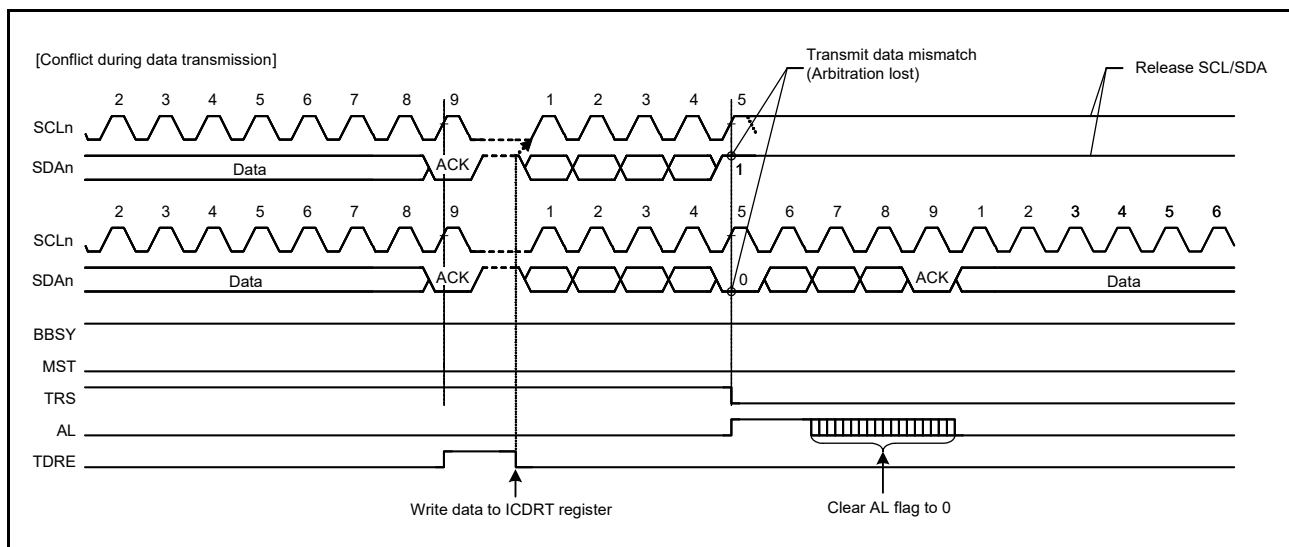


Figure 42.35 Example of Slave Arbitration-Lost Detection (SALE = 1)

42.10 Start Condition/Restart Condition/Stop Condition Generating Function

42.10.1 Generating a Start Condition

The RIIC generates a start condition when the ICCR2.ST bit is set to 1.

When the ST bit is set to 1, a start condition generation request is made and the RIIC generates a start condition when the ICCR2.BBSY flag is 0 (bus free state). When a start condition is generated normally, the RIIC automatically shifts to the master transmit mode.

A start condition is generated in the following sequence.

Start condition generation

- (1) Drive the SDAn line low (high to low).
- (2) Secure the time set in the ICBRH register and the start condition hold time.
- (3) Drive the SCLn line low (high to low).
- (4) Detect the low level on the SCLn line and secure the low period of the signal on the SCLn line set in the ICBRL register.

42.10.2 Generating a Restart Condition

The RIIC generates a restart condition when the ICCR2.RS bit is set to 1.

When the RS bit is set to 1, a restart condition generation request is made and the RIIC generates a restart condition when the ICCR2.BBSY flag is 1 (bus busy state) and the ICCR2.MST bit is 1 (master mode).

A restart condition is generated in the following sequence.

Restart condition generation

- (1) Release the SDAn line.
- (2) Secure the low period of the signal on the SCLn line set in the ICBRL register.
- (3) Release the SCLn line (low to high).
- (4) Detect the high level on the SCLn line and secure the time set in the ICBRL register and the restart condition setup time.
- (5) Drive the SDAn line low (high to low).
- (6) Secure the time set in the ICBRH register and the restart condition hold time.
- (7) Drive the SCLn line low (high to low).
- (8) Detect the low level on the SCLn line and secure the low period of the signal on the SCLn line set in the ICBRL register.

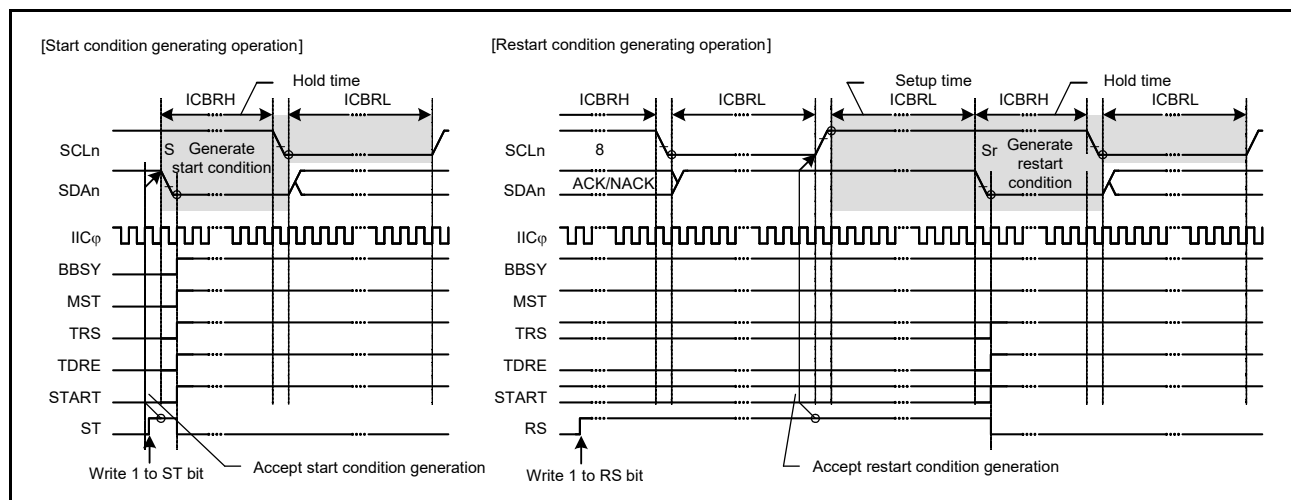


Figure 42.36 Start Condition/Restart Condition Generation Timing (ST and RS Bits)

42.10.3 Generating a Stop Condition

The RIIC generates a stop condition when the ICCR2.SP bit is set to 1.

When the SP bit is set to 1, a stop condition generation request is made and the RIIC generates a stop condition when the ICCR2.BBSY flag is 1 (bus busy state) and the ICCR2.MST bit is 1 (master mode).

A stop condition is generated in the following sequence.

Stop condition generation

- (1) Drive the SDA_n line low (high to low).
- (2) Secure the low period of the signal on the SCL_n line set in the ICBRL register.
- (3) Release the SCL_n line (low to high).
- (4) Detect the high level on the SCL_n line and secure the time set in the ICBRH register and the stop condition setup time.
- (5) Release the SDA_n line (low to high).
- (6) Secure the time set in the ICBRL register and the bus free time.
- (7) Set the BBSY flag to 0 (to release the bus mastership).

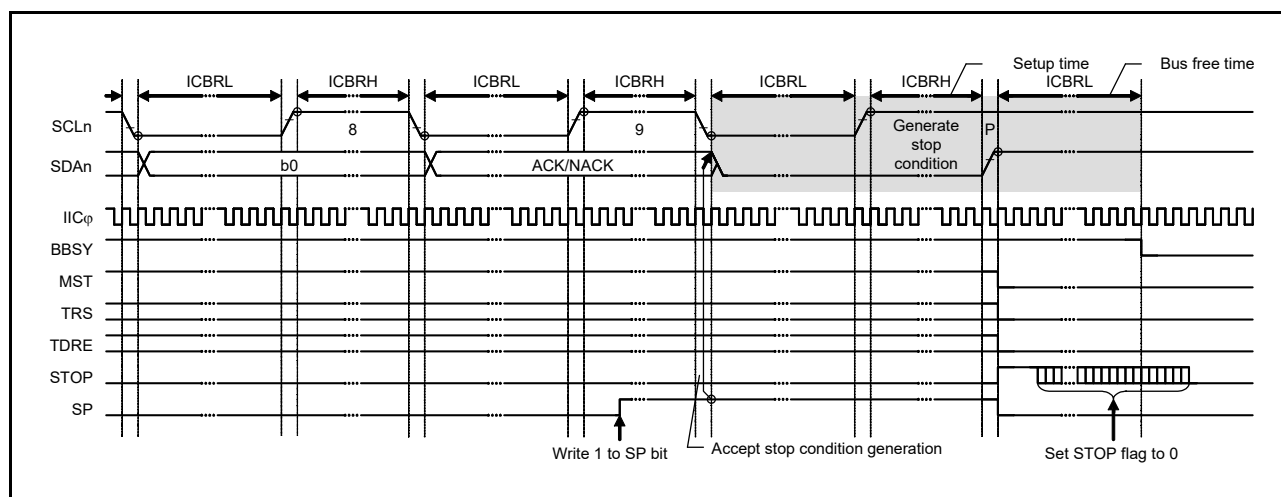


Figure 42.37 Stop Condition Generation Timing (SP Bit)

42.11 Bus Hanging

If the clock signals from the master and slave devices go out of synchronization due to noise or other factors, the I²C-bus might hang with a fixed level on the SCLn line and/or SDAn line.

As measures against the bus hanging, the RIIC has a timeout function to detect hanging by monitoring the SCLn line, a function for the output of an additional SCL to release the bus from a hung state due to clock signals being out of synchronization, the RIIC reset function, and internal reset function.

By checking bits SCLO, SDAO, SCLI, and SDAI in the ICCR1 register, it is possible to see whether the RIIC or its partner in communications is placing the low level on the SCLn or SDAn lines.

42.11.1 Timeout Function

The RIIC includes a timeout function for detecting when the SCLn line has been stuck longer than the predetermined time. The RIIC can detect an abnormal bus state by monitoring that the SCLn line is stuck low or high for a predetermined time.

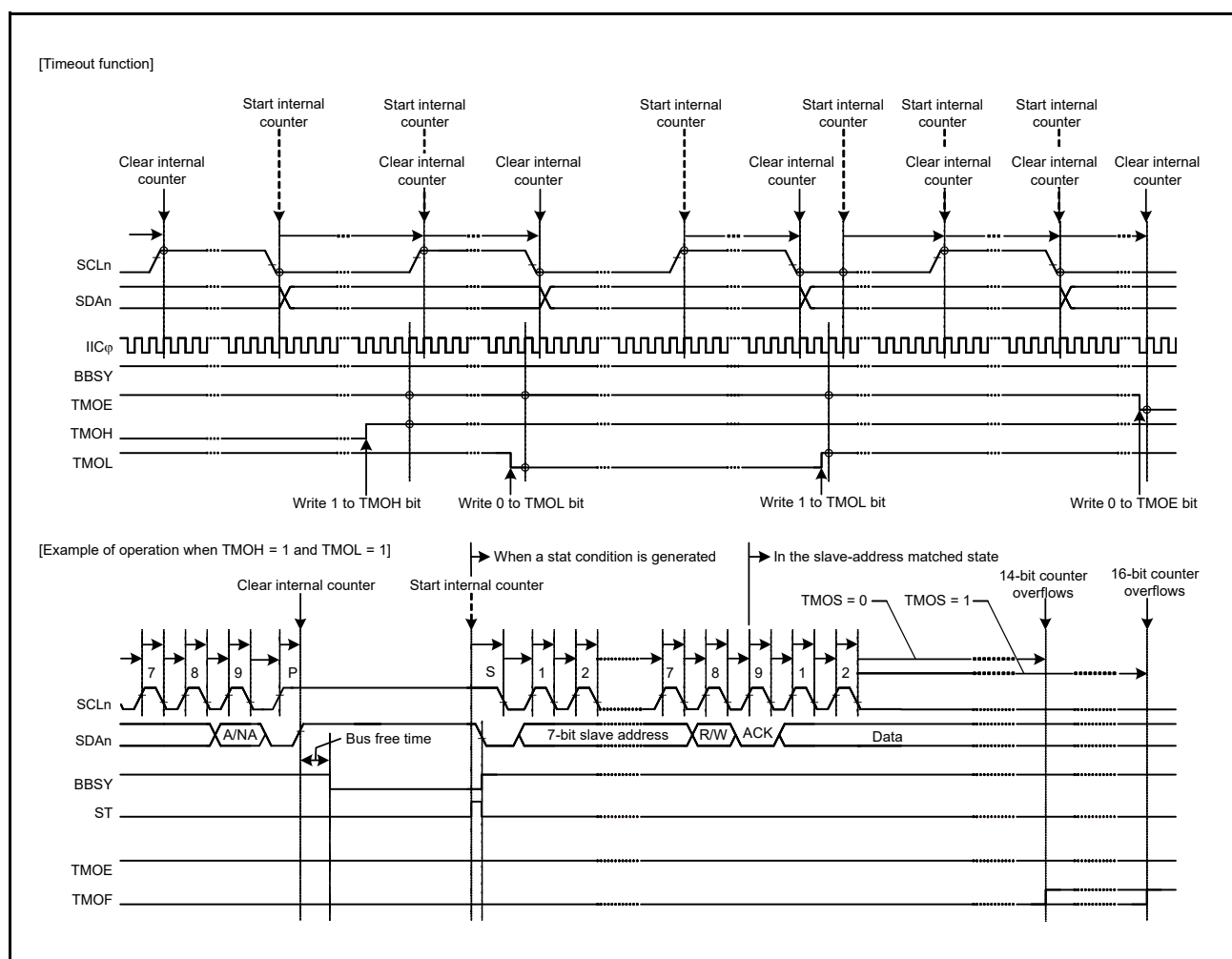
The timeout function monitors the SCLn line state and counts the low period or high period using the internal counter. The timeout function resets the internal counter each time the SCLn line changes (rising or falling), but continues to count unless the SCLn line changes. If the internal counter overflows due to no SCLn line change, the RIIC can detect the timeout and report the bus hung state.

This timeout function is enabled when the ICFER.TMOE bit is 1. It detects a hung state that the SCLn line is stuck low or high during the following conditions:

- The bus is busy (ICCR2.BBSY flag is 1) in master mode (ICCR2.MST bit is 1).
- The RIIC's own slave address is detected (ICSR1 register is not 00h) and the bus is busy (ICCR2.BBSY flag is 1) in slave mode (ICCR2.MST bit is 0).
- The bus is free (ICCR2.BBSY flag is 0) while generation of a start condition is requested (ICCR2.ST bit is 1).

The internal counter of the timeout function works using the internal reference clock (IIC ϕ) set by the ICMR1.CKS[2:0] bits as a count source. It functions as a 16-bit counter when long mode is selected (ICMR2.TMOS bit is 0) or a 14-bit counter when short mode is selected (TMOS bit is 1).

The SCLn line level (low/high or both levels) during which this counter is activated can be selected by the setting of bits TMOH and TMOL in the ICMR2 register. If both bits TMOL and TMOH are set to 0, the internal counter does not work.



42.11.2 Additional SCL Output Function

In master mode, the RIIC module has a facility for the output of additional SCL to release the SDA_n line from being held low by the slave device due to the master being out of synchronization with the slave device.

This function is mainly used in master mode to release the SDA_n line from the state of being stuck low by including additional SCL output from the RIIC with single cycles of the SCL as the unit if the RIIC cannot generate a stop condition because the slave device is holding the SDA_n line low. Do not use this function in normal situations. Using it when communications are proceeding correctly will lead to malfunctions.

When the ICCR1.CLO bit is set to 1, an additional clock pulse at the frequency set by the ICMR1.CKS[2:0] bits and the ICBRH and ICBRL registers is output from the SCL_n pin. After output of this clock pulse, the CLO bit automatically becomes 0. The SCL_n pin is held low when the ICCR2.BBSY flag is 1 and held high when the BBSY flag is 0.

Consecutive additional clock pulses can be output by writing 1 to the CLO bit after confirming the CLO bit to be 0.

When the RIIC module is in master mode and the slave device is holding the SDA_n line low because synchronization with the slave device has been lost due to the effects of noise, etc., the output of a stop condition is not possible. The additional SCL output function can be used to output additional clock pulses one by one to make the slave device release the SDA_n line from being held low, thus recovering the bus from an unusable state. Release of the SDA_n line by the slave device can be monitored by reading the ICCR1.SDAI bit. After confirming release of the SDA_n line by the slave device, complete communications by regenerating a stop condition.

Use this function with the ICFER.MALE bit set to 0 (master arbitration-lost detection disabled).

Conditions for using the ICCR1.CLO bit

- When the bus is free (ICCR2.BBSY flag is 0) or in master mode (ICCR2.MST bit is 1 and ICCR2.BBSY flag is 1)
- When the communication device does not hold the SCL_n line low

Figure 42.39 shows the operation timing of the additional SCL output function (CLO bit).

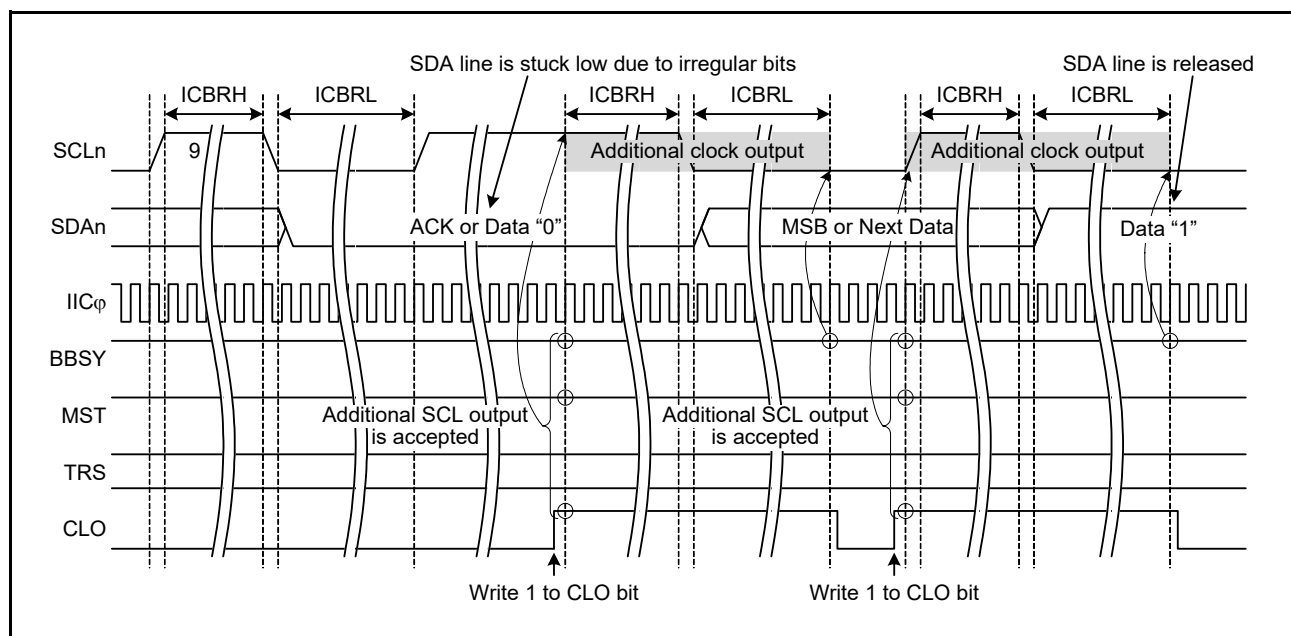


Figure 42.39 Additional SCL Output Function (CLO Bit)

42.11.3 RIIC Reset and Internal Reset

The RIIC module incorporates a function for resetting itself. There are two types of reset. One is referred to as an RIIC reset; this initializes all registers including the ICCR2.BBSY flag. The other is referred to as an internal reset; this releases the RIIC from the slave-address matched state and initializes the internal counter while retaining other settings. After applying a reset, be sure to set the ICCR1.IICRST bit to 0.

Both types of reset are effective for release from bus-hung states because both restore the output state of the SCLn and SDAn pins to the high-impedance state.

Applying a reset during slave operation may lead to a loss of synchronization between the master device clock and the slave device clock, so avoided this where possible. Note that monitoring of the bus state, such as for the presence of a start condition, is not possible during an RIIC reset (bits ICE and IICRST in the ICCR1 register are 01b).

For a detailed description of the RIIC and internal resets, refer to **section 42.14, Initialization of Registers and Functions When a Reset is Applied or a Condition is Detected**.

42.12 SMBus Operation

The RIIC is available for data communication conforming to the SMBus (Version 2.0). To perform SMBus communication, set the ICMR3.SMBS bit to 1. To use the transfer rate within a range of 10 kbps to 100 kbps of the SMBus specification, set the ICMR1.CKS[2:0] bits, the ICBRH register, and the ICBRL register. In addition, determine the values of the ICMR2.DLCS bit and the ICMR2.SDDL[2:0] bits to meet the data hold time (300 ns (min.)). If the RIIC is used only as a slave device, the transfer rate setting is not necessary, whereas the ICBRL register needs to be set to a value longer than the data setup time (250 ns).

For the SMBus device default address (1100 001b), use one of the slave address registers L0 to L2 (registers SARL0, SARL1, and SARL2), and set the corresponding SARUy.FS bit (7-bit/10-bit address format select) (y = 0 to 2) to 0 (7-bit address format).

When transmitting the UDID (Unique Device Identifier), set the ICFER.SALE bit to 1 to enable the slave arbitration-lost detection function.

42.12.1 SMBus Timeout Measurement

(1) Measuring timeout of slave device

The following period (timeout interval: $T_{\text{LOW:SEXT}}$) must be measured for slave devices in SMBus communication.

- From start condition to stop condition

To measure timeout for slave devices, measure the period from start condition detection to stop condition detection with the MTU or TMR timer using a start condition detection interrupt (STI) and stop condition detection interrupt (SPI) of the RIIC. The measured timeout period must be within the cumulative clock low extend time (slave device) ($T_{\text{LOW:SEXT}}$: 25 ms (max.)) of the SMBus specification.

If the time measured with the MTU or TMR exceeds the detect clock low timeout (T_{TIMEOUT} : 25 ms (min.)) of the SMBus specification, the slave device must release the bus by writing 1 to the ICCR1.IICRST bit to apply an internal reset of the RIIC. When an internal reset is applied to the RIIC, it stops driving the SCLn and SDAn pins of the bus and makes the SCLn/SDAn pin outputs high-impedance, thus releasing the bus.

(2) Measuring timeout of master device

The following periods (timeout interval: $T_{\text{LOW:MEXT}}$) must be measured for master devices in SMBus communication.

- From start condition to acknowledgment bit
- Between acknowledgment bits
- From acknowledgment bit to stop condition

To measure timeout for master devices, measure these periods with the MTU or TMR timer using a start condition detection interrupt (STI), stop condition detection interrupt (SPI), and transmission end interrupt (TEI) or receive data full interrupt (RXI) of the RIIC. The measured timeout period must be within the cumulative clock low extend time (master device) ($T_{\text{LOW:MEXT}}$: 10 ms (max.)) of the SMBus specification, and the total of all $T_{\text{LOW:MEXT}}$ from start condition to stop condition must be within $T_{\text{LOW:SEXT}}$: 25 ms (max.).

For the ACK receive timing (rising edge of the ninth SCL), monitor the ICSR2.TEND flag in master transmit mode (master transmitter) and the ICSR2.RDRF flag in master receive mode (master receiver). For this reason, perform bitwise transmit operation in master transmit mode, and hold the ICMR3.RDRFS bit 0 until the byte just before reception of the final byte in master receive mode. While the RDRFS bit is 0, the RDRF flag is set to 1 at the rising edge of the ninth SCL.

If the period measured with the MTU or TMR exceeds the cumulative clock low extend time (master device) ($T_{\text{LOW:MEXT}}$: 10 ms (max.)) of the SMBus specification or the total of measured periods exceeds the detect clock low timeout (T_{TIMEOUT} : 25 ms (min.)) of the SMBus specification, the master device must stop the transaction by generating a stop condition. In master transmit mode, immediately stop the transmit operation (writing data to the ICDRT register).

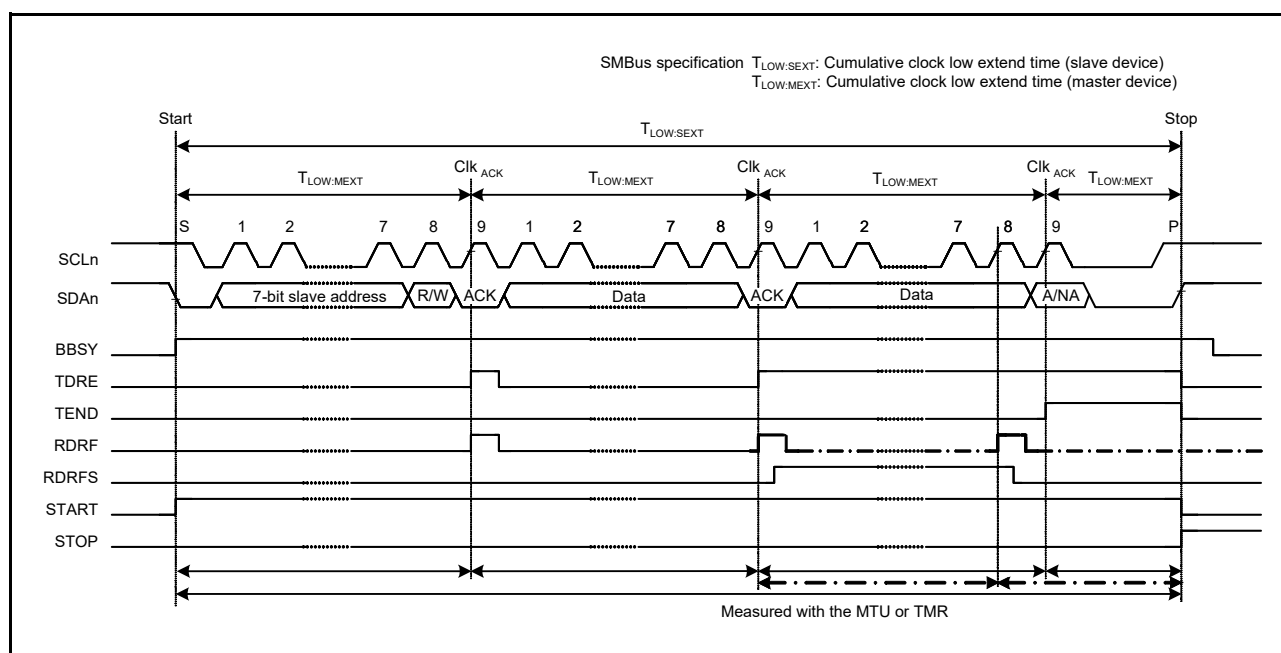


Figure 42.40 SMBus Timeout Measurement

42.12.2 Packet Error Code (PEC)

This MCU incorporates a CRC calculator. The CRC calculator enables transmission of a packet error code (PEC) or checking the received data of the SMBus in data communication of the RIIC. For the CRC generating polynomials of the CRC calculator, refer to [section 46, CRC Calculator \(CRCA\)](#).

The PEC data in master transmit mode can be generated by writing all transmit data to the CRC data input register (CRCDIR) in the CRC calculator.

The PEC data in master receive mode can be checked by writing all receive data to CRCDIR in the CRC calculator and comparing the obtained value in the CRC data output register (CRCDOR) with the received PEC data.

To send ACK or NACK according to the match or mismatch result when the final byte is received as a result of the PEC code check, set the ICMR3.RDRFS bit to 1 before the rising edge of the eighth SCL during reception of the final byte, and hold the SCLn line low at the falling edge of the eighth clock pulse.

42.12.3 SMBus Host Notification Protocol (Notify ARP Master Command)

In communications over an SMBus, a slave device can temporarily act as a master device to notify the SMBus host (or ARP master) of its own slave address or to request its own slave address from the SMBus host.

For a product of this MCU to operate as an SMBus host (or ARP master), the host address (0001 000b) sent from the slave device must be detected as a slave address, so the RIIC has a function for detecting the host address. To detect the host address as a slave address, set the ICMR3.SMBS bit and the ICSER.HOAE bit to 1. Operation after the host address has been detected is the same as normal slave operation.

42.13 Interrupt Sources

The RIIC generates four types of interrupt requests: communication error or communication event (arbitration-lost detection, NACK detection, timeout detection, start condition detection, and stop condition detection), receive data full, transmit data empty, and transmission end.

Table 42.6 lists details of the several interrupt requests. The receive data full and transmit data empty interrupt requests allow the DTC or DMAC to start data transfer.

Table 42.6 Interrupt Sources

Symbol	Interrupt Source	Interrupt Flag	Start DTC/DMA Transfer	Interrupt Generation Condition
EEI	Communication error/ communication event	AL	Not possible	AL = 1 and ALIE = 1
		NACKF		NACKF = 1 and NAKIE = 1
		TMOF		TMOF = 1 and TMOIE = 1
		START		START = 1 and STIE = 1
		STOP		STOP = 1 and SPIE = 1
RXI*2	Receive data full	RDRF	Possible	RDRF = 1 and RIE = 1
TXI*1	Transmit data empty	TDRE	Possible	TDRE = 1 and TIE = 1
TEI*3	Transmission end	TEND	Not possible	TEND = 1 and TEIE = 1

Note: There is a delay time between the execution of a write instruction for a peripheral module by the CPU and actual writing to the module. Thus, when an interrupt flag has been cleared or an interrupt request has been masked, read the relevant flag again to check whether clearing or masking has been completed, and then return from interrupt handling. Returning from interrupt handling without checking that writing to the module has been completed creates a possibility of repeated processing of the same interrupt.

Note 1. Because TXI is an edge-detected interrupt, it does not require clearing. Furthermore, the ICSR2.TDRE flag (a condition for TXI) is automatically set to 0 when data for transmission are written to the ICDRT register or a stop condition is detected (ICSR2.STOP flag is 1).

Note 2. Because RXI is an edge-detected interrupt, it does not require clearing. Furthermore, the ICSR2.RDRF flag (a condition for RXI) is automatically set to 0 when data are read from the ICDRR register.

Note 3. When using the TEI interrupt, clear the ICSR2.TEND flag in the TEI interrupt handling.
Note that the ICSR2.TEND flag is automatically set to 0 when data for transmission are written to the ICDRT register or a stop condition is detected (ICSR2.STOP flag is 1).

Clear the each flag or mask the interrupt request during interrupt handling.

42.13.1 Buffer Operation for TXI and RXI Interrupts

If the conditions for generating a TXI and RXI interrupt are satisfied while the corresponding ICU.IRn.IR flag is 1, the interrupt request is not output for the ICU but retained internally (the capacity for internal retention is one request per source).

An interrupt request that was being retained internally is output to the ICU when the value of the IR flag becomes 0. Internally retained interrupt requests are automatically cleared under normal conditions of usage.

Internally retained interrupt requests can also be cleared by writing 0 to the corresponding interrupt enable bit in the ICIER register.

42.14 Initialization of Registers and Functions When a Reset is Applied or a Condition is Detected

The RIIC can be reset by MCU reset, RIIC reset, and internal reset functions. Table 42.7 lists the reset states of registers and functions when a reset is applied or a condition is detected.

Table 42.7 Reset States of Registers and Functions When a Reset is Applied or a Condition is Detected

		MCU Reset	RIIC Reset (ICE = 0, IICRST = 1)	Internal Reset (ICE = 1, IICRST = 1)	Start Condition/ Restart Condition Detection	Stop Condition Detection
ICCR1	SDAO, SCLO	To be reset	To be reset	To be reset	Retained	Retained
	IICRST, ICE		Retained	Retained		
	Others		To be reset			
ICCR2	ST, RS	To be reset	To be reset	To be reset	To be reset	Retained
	SP					To be reset
	TRS				See note 1	
	MST				See note 1	
	BBSY			Retained	Becomes 1	
ICMR1	BC[2:0]	To be reset	To be reset	To be reset	To be reset	Retained
	Others			Retained	Retained	
ICMR2		To be reset	To be reset	Retained	Retained	Retained
ICMR3	ACKBT	To be reset	To be reset	Retained	Retained	To be reset
	Others					Retained
ICFER		To be reset	To be reset	Retained	Retained	Retained
ICSER		To be reset	To be reset	Retained	Retained	Retained
ICIER		To be reset	To be reset	Retained	Retained	Retained
ICSR1		To be reset	To be reset	To be reset	Retained	To be reset
ICSR2	START	To be reset	To be reset	To be reset	Becomes 1	To be reset
	STOP				Retained	Becomes 1
	TEND					To be reset
	TDRE				See note 1	
	Others				Retained	Retained
SARL0, SARL1, SARL2, SARU0, SARU1, SARU2		To be reset	To be reset	Retained	Retained	Retained
ICBRH, ICBRL		To be reset	To be reset	Retained	Retained	Retained
ICDRT		To be reset	To be reset	Retained	Retained	Retained
ICDRR		To be reset	To be reset	Retained	Retained	Retained
ICDRS		To be reset	To be reset	To be reset	Retained	Retained
Timeout function		To be reset	To be reset	To be reset	Operation	Operation
Bus free time measurement		To be reset	To be reset	Operation	Operation	Operation

Note 1. This bit is not reset. This bit becomes 0 or 1 in accordance with the conditions.

42.15 Event Link Function (Output)

The RIIC0 handles event output for the event link controller (ELC) corresponding to the following sources.

- Communication error/communication event
- Receive data full
- Transmit data empty
- Transmission end

42.15.1 Interrupt Handling and Event Linking

The RIIC has four types of interrupts: communication error or communication event (arbitration-lost detection, NACK detection, timeout detection, start condition detection, or stop condition detection), receive data full, transmit data empty, and transmission end. Each of these has an enable bit to control enabling and disabling of the interrupt signal. An interrupt request signal is output for the ICU when an interrupt source condition is satisfied while the setting of the corresponding enable bit is enabled.

The corresponding event signals are sent to other modules via the ELC when the interrupt source conditions are satisfied, regardless of the settings of the interrupt enable bits.

For details on interrupt sources, see Table 42.6.

42.16 Usage Notes

42.16.1 Setting Module Stop Function

Module stop state can be entered or released using module stop control register B (MSTPCRB) or module stop control register C (MSTPCRC). The initial setting is for operation of the RIIC to be stopped. RIIC register access is enabled by releasing the module stop state.

For details on module stop control registers B and C, refer to **section 11, Low Power Consumption**.

42.16.2 Notes on Starting Transfer

If the IR flag corresponding to the RIIC interrupt is 1 when transfer is started (ICCR1.ICE bit is 1), follow the procedure below to clear interrupts before enabling operations. Starting transfer with the IR flag set to 1 while the ICCR1.ICE bit is 1 leads to an interrupt request being internally retained after transfer starts, and this can lead to unanticipated behavior of the IR flag.

1. Confirm that the ICCR1.ICE bit is 0.
2. Set the relevant interrupt enable bits (ICIER.TIE, etc.) on the peripheral function side to 0.
3. Read the relevant interrupt enable bits (ICIER.TIE, etc.) on the peripheral function side and confirm that its value is 0.
4. Set the IR flag to 0.